

Real Analysis
Modern Techniques and Their Applications
Solutions

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Chapter 1 : Measures

Problem 1.1. A family of sets $\mathcal{R} \subset \mathcal{P}(X)$ is called a **ring** if it is closed under finite unions and differences (i.e., if $E_1, \dots, E_n \in \mathcal{R}$, then $\cup_1^n E_j \in \mathcal{R}$, and if $E, F \in \mathcal{R}$, then $E \setminus F \in \mathcal{R}$). A ring that is closed under countable unions is called a **σ -ring**.

- a. Rings (resp. σ -rings) are closed under finite (resp. countable) intersections.
- b. If $\mathcal{R}$ is a ring (resp. σ -ring), then $\mathcal{R}$ is an algebra (resp. σ -algebra) iff $X \in \mathcal{R}$.
- c. If $\mathcal{R}$ is a σ -ring, then $\{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$ is a σ -algebra.
- d. If $\mathcal{R}$ is a σ -ring, then $\{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$ is a σ -algebra.

Solution.

- a. If $A, B \in \mathcal{R}$ then, $A \setminus B \in \mathcal{R}$ and again, $A \setminus (A \setminus B) = A \cap B \in \mathcal{R}$. We can induct on the number of sets and conclude that $\mathcal{R}$ is closed under finite intersections. If $\mathcal{R}$ was a σ -ring and $\{A_n\} \subseteq \mathcal{R}$ then $A_1 \setminus A_1, A_1 \setminus A_2, A_1 \setminus A_3, \dots \in \mathcal{R}$ and $\cup(A_1 \setminus A_n) = A_1 \setminus (\cap A_n) \in \mathcal{R}$ by definition. Thus, $A_1 \setminus (A_1 \setminus \cap A_n) = \cap A_n \in \mathcal{R}$.
- b. The existence of $X \in \mathcal{R}$ implies that its closed under complements and we can conclude using part (a). Now if it was a σ -algebra to begin with then, $X \in \mathcal{R}$ follows from definition.
- c. Let this set be $\mathcal{S}$. As, $\phi \in \mathcal{R} \subseteq \mathcal{S}$ we must also have $X \in \mathcal{S}$. If A is in this set then $A \in \mathcal{R}$ or, $A^c \in \mathcal{R}$, in the first case $X \setminus A = A^c \in \mathcal{R}$ and in the second case $A^c \in \mathcal{R}$ already, proving the closure of this set under complements. Now let $\{E_n\}_{n=1}^\infty \subset \mathcal{S}$. If $E_n \in \mathcal{R}$ for all n , then $\cup E_n \in \mathcal{R} \subset \mathcal{S}$ because $\mathcal{R}$ is a σ -ring. If there exists at least one k such that $E_k \notin \mathcal{R}$, then necessarily $E_k^c \in \mathcal{R}$. We examine the complement of the union,

$$E_k^c \supseteq \left(\bigcup_{n=1}^\infty E_n \right)^c = \bigcap_{n=1}^\infty E_n^c = \bigcap_{n=1}^\infty (E_n^c \cap E_k^c)$$

For any n , if $E_n \in \mathcal{R}$, then $E_n^c \cap E_k^c = E_k^c \setminus E_n \in \mathcal{R}$. If $E_n^c \in \mathcal{R}$, then $E_n^c \cap E_k^c \in \mathcal{R}$ by closure under intersections. Thus, the sequence $\{E_n^c \cap E_k^c\}_n$ lies entirely in $\mathcal{R}$. Since σ -rings are closed under countable intersections by part (a), we conclude $(\cup E_n)^c \in \mathcal{R}$, which implies $\cup E_n \in \mathcal{S}$.

- d. Clearly, ϕ, X are included in this set. If A is in this then $\forall F \in \mathcal{R}, A \cap F \in \mathcal{R}$ and by definition $F \setminus (F \cap A) = F \cap A^c \in \mathcal{R}$ so A^c is in it, thus its closed under complements. If $A_1, A_2 \dots$ are in it then $\forall n \in \mathbb{N}, \forall F \in \mathcal{R}, A_n \cap F \in \mathcal{R}$ and a $\mathcal{R}$ is a σ -ring we must have $\cup(A_n \cap F) = F \cap (\cup A_n) \in \mathcal{R}$ which is enough to show that its closed under countable unions, thus its a σ -algebra.

□

Problem 1.2. Complete the proof of Proposition 1.2.

Solution. Trivial.

□

Problem 1.3. Let $\mathcal{M}$ be an infinite σ -algebra.

- a. $\mathcal{M}$ contains an infinite sequence of disjoint sets.
b. $\text{card}(\mathcal{M}) \geq \mathfrak{c}$.

Solution. Let $\{E_n\}_{n \in \mathbb{N}}$ be a countably infinite collection of distinct sets in $\mathcal{M}$ which we can choose by the virtue of it being infinite. Consider the atoms formed by this collection,

$$\left\{ \bigcap_{n \in \mathbb{N}} E_n^{f(n)} \mid f : \mathbb{N} \rightarrow \{1, c\} \right\}$$

This is a subset of $\mathcal{M}$ as $\mathcal{M}$ is a sigma algebra. Clearly for any two different f, g the intersections they produce are disjoint. Also, every set in our collection is an union of some sets from these atoms,

$$E_n = \bigcup_{\substack{f \in \{1, c\}^{\mathbb{N}} \\ f(n)=1}} \bigcap_{n \in \mathbb{N}} E_n^{f(n)}$$

So if there are finitely many nonempty atoms, say n then its easy to show that there are atmost 2^n sets to begin with which neccesitates that there be infinite atoms. Thus there exists some countably infinite sequence of pairwise disjoint nonempty sets in $\mathcal{M}$ (choose some atoms) proving part (a). Let $\{F_n\}_{n \in \mathbb{N}}$ be the collection of some pairwise disjoint sets in $\mathcal{M}$. Identifying A^0 with ϕ , consider the collection

$$\mathcal{G} := \left\{ \bigcup_{n \in \mathbb{N}} F_n^{f(n)} \mid f : \mathbb{N} \rightarrow \{0, 1\} \right\}$$

Clearly $\mathcal{G} \subseteq \mathcal{M}$ and all the sets in this are distinct as F_n are all pairwise disjoint. Also this is in bijection to $\{0, 1\}^{\mathbb{N}}$. Thus, $|\mathcal{M}| \geq |\mathcal{G}| = |\{0, 1\}^{\mathbb{N}}| = \mathfrak{c}$. $\square$

Problem 1.4. An algebra $\mathcal{A}$ is a σ -algebra iff $\mathcal{A}$ is closed under countable increasing unions (i.e., if $\{E_j\}_1^\infty \subset \mathcal{A}$ and $E_1 \subset E_2 \subset \dots$, then $\cup_1^\infty E_j \in \mathcal{A}$).

Solution. If its a σ -algebra then its tirvially an algebra closed under all sorts of intersections. If its an algebra closed under increasing countable unions then for $A_1, \dots \in \mathcal{A}$ consider $B_n := \cup_{m \leq n} A_m \in \mathcal{A}$. Clearly B_n is increasing and thus, $\cup B_n = \cup A_n \in \mathcal{A}$. $\square$

Problem 1.5. If $\mathcal{M}$ is the σ -algebra generated by $\mathcal{E}$, then $\mathcal{M}$ is the union of the σ -algebras generated by $\mathcal{F}$ as $\mathcal{F}$ ranges over all countable subsets of $\mathcal{E}$. (Hint: Show that the latter object is a σ -algebra.)

Solution. We wish to prove that,

$$\mathcal{M}(\mathcal{E}) = \bigcup_{\substack{A \subseteq \mathcal{E} \\ |A| \leq |\mathbb{N}|}} \mathcal{M}(A)$$

Izt suffices to show that the set on the right is a σ -algebra as $\mathcal{E}$ is already a subset and we can conclude using the definition of generated σ -algebra. The set is clearly closed under complements and also ϕ, X are in it. Now, say $\{A_n\}$ is subset, then we can find $\mathcal{E}_n \subseteq \mathcal{E}$ such that $A_n \in \mathcal{M}(\mathcal{E}_n)$. Then, $|\cup \mathcal{E}_n| \leq |\mathbb{N}|$ and

$$\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{M} \left(\bigcup_{n \in \mathbb{N}} \mathcal{M}(\mathcal{E}_n) \right) \subseteq \mathcal{M} \left(\bigcup_{n \in \mathbb{N}} \mathcal{E}_n \right) \subseteq \bigcup_{\substack{A \subseteq \mathcal{E} \\ |A| \leq |\mathbb{N}|}} \mathcal{M}(A)$$

proving that its a sigma algebra. $\square$

Problem 1.6. Complete the proof of Theorem 1.9.

Solution. First we show that $\overline{\mathcal{M}}$ is a sigma algebra. It clearly has ϕ, X and is closed under countable unions, all thats left to show is that its closed under complements. Now, if $E \cup F \in \overline{\mathcal{M}}$ with $E \in \mathcal{M}$ and $F \subseteq N \in mN$, write it as $E \cup F = E \cup (E \setminus F) \in \overline{\mathcal{M}}$ with $E \subseteq \mathcal{M}$ and $F \setminus E \subseteq N \setminus E \in \mathcal{N}$. As, E and $N \setminus E$ are disjoint we see that it suffices to assume $E \cap N = \phi$ from the start and we do so. Write $E \cup F = (E \cup N) \cap (F \cup N^c) = (E \cup N) \cap (N \setminus F)^c$ and $(E \cup F)^c = (E \cup N)^c \cup (N \setminus F)$ with $E \cup N \in \mathcal{M}$ and $N \setminus F \subseteq N \in \mathcal{N}$ so $(E \cup F) \in \overline{\mathcal{M}}$ proving that its a σ -algebra. Set $\bar{\mu}(E \cup F) := \mu(E)$ when $E \in \mathcal{M}, F \subseteq N \in \mathcal{N}$, this is well defined because if $E_1 \cup F_1 = E_2 \cup F_2$ with $E_1, E_2 \in \mathcal{M}$ and $F_1, F_2 \subseteq N_1, N_2$ resp. in $\mathcal{N}$. Then, $\mu(E_1) = \bar{\mu}(E_1 \cup F_1) = \bar{\mu}(E_1 \cup N_1) = \mu(E_1 \cup N_1) \geq \mu(E_2)$ as $E_2 \subseteq E_1 \cup F_1 \subseteq E_1 \cup N_1$. Similarly we get that $\mu(E_1) \leq \mu(E_2)$, thus $\bar{\mu}(E_2 \cup E_2) = \mu(E_2) = \mu(E_1) = \bar{\mu}(E_1 \cup F_1)$. $\bar{\mu}$ is measure as trivially, $\bar{\mu}(\phi) = 0$ and for any disjoint countable collection of sets $\{E_n \cup F_n\}_{n \in \mathbb{N}} \subseteq \overline{\mathcal{M}}$ with $E_n \in \mathcal{M}$ and $F_n \subseteq N_n \in \mathcal{N}$ we have,

$$\bar{\mu}\left(\bigcup_{n \in \mathbb{N}} (E_n \cup F_n)\right) = \bar{\mu}\left(\left(\bigcup_{n \in \mathbb{N}} E_n\right) \cup \left(\bigcup_{n \in \mathbb{N}} F_n\right)\right) = \mu\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \sum_{n=1}^{\infty} \mu(E_n) = \sum_{n=1}^{\infty} \bar{\mu}(E_n \cup F_n)$$

where $\cup E_n \in \mathcal{M}$ and $\cup F_n \subseteq \cup N_n \in \mathcal{N}$ by sigma additivity of μ . Now we prove uniqueness, suppose μ_* is another such measure which extends μ . Then, by definition $\mu_*|_{\mathcal{M}} = \mu$ so, for $E \cup F \in \overline{\mathcal{M}}$ as before,

$$\bar{\mu}(E \cup F) = \mu(E) = \mu_*(E) \leq \mu_*(E \cup F) \leq \mu_*(E \cup N) = \mu(E \cup N) = \mu(E) + 0 = \bar{\mu}(E \cup F)$$

by monotonicity, which proves that $\mu_* = \bar{\mu}$ and we are done. $\square$

Problem 1.7. If $\mu_1, \dots, \mu_n$ are measures on $(X, \mathcal{M})$ and $a_1, \dots, a_n \in [0, \infty)$, then $\sum_1^n a_j \mu_j$ is a measure on $(X, \mathcal{M})$.

Solution. $(\sum_i a_i \mu_i)(\phi) = \sum_i a_i \mu_i(\phi) = \sum_i a_i \cdot 0 = 0$ and for any disjoint countable collection of sets $\{E_n\}$ we find that, as $a_i \cdot \mu_i(E_n) \geq 0$ for all valid i, n we can interchange the order of summation as we are working in the extended reals and deduce,

$$\begin{aligned} \left(\sum_i a_i \mu_i\right)\left(\bigcup_{n \in \mathbb{N}} E_n\right) &= \sum_i a_i \mu_i\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \sum_i \sum_{n \in \mathbb{N}} a_i \mu_i(E_n) \\ &= \sum_{n \in \mathbb{N}} \sum_i a_i \mu_i(E_n) = \sum_{n \in \mathbb{N}} \left(\sum_i a_i \mu_i\right)(E_n) \end{aligned}$$

proving that $\sum_i a_i \mu_i$ is a measure. $\square$

Problem 1.8. If $(X, \mathcal{M}, \mu)$ is a measure space and $\{E_j\}_1^\infty \subset \mathcal{M}$, then $\mu(\liminf E_j) \leq \liminf \mu(E_j)$. Also, $\mu(\limsup E_j) \geq \limsup \mu(E_j)$ provided that $\mu(\cup_1^\infty E_j) < \infty$.

Solution.

$$\mu(\liminf E_j) = \mu\left(\bigcup_{k \geq 1} \bigcap_{n \geq k} E_n\right) = \lim_{k \rightarrow \infty} \mu\left(\bigcap_{n \geq k} E_n\right)$$

as the sequence of sets $\{\cap_{n \geq k} E_n\}_{k \in \mathbb{N}}$ is increasing. Now, $\mu(E_n) \geq \mu(\cap_{n \geq k} E_n)$ for all $k \geq n$ by monotonicity and using the definition of infimums, $\inf_{n \geq k} \mu(E_n) \geq \mu(\cap_{n \geq k} E_n)$ and taking the limits on both sides as k tends to infinity we have,

$$\liminf \mu(E_j) = \lim_{k \rightarrow \infty} \inf_{n \geq k} \mu(E_n) \geq \lim_{k \rightarrow \infty} \mu \left(\bigcap_{n \geq k} E_n \right) = \mu(\liminf E_j)$$

Now assume that $\mu(\cup E_n) < \infty$, then similarly

$$\mu(\limsup E_j) = \mu \left(\bigcap_{k \geq 1} \bigcup_{n \geq k} E_n \right) = \lim_{k \rightarrow \infty} \mu \left(\bigcup_{n \geq k} E_n \right)$$

as the sequence of sets $\{\cup_{n \geq k} E_n\}_{k \in \mathbb{N}}$ is decreasing and the set at index 1 has finite measure. Now, clearly $\mu(\cup_{n \geq k} E_n) \geq \sup_{n \geq k} \mu(E_n)$ from the definition of supremums and taking the limits on both sides as $k \rightarrow \infty$ we have,

$$\mu(\limsup E_j) = \lim_{k \rightarrow \infty} \mu \left(\bigcup_{n \geq k} E_n \right) \geq \lim_{k \rightarrow \infty} \sup_{n \geq k} \mu(E_n) = \limsup \mu(E_j)$$

□

Problem 1.9. If $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$, then $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$.

Solution. If atleast one of $\mu(E), \mu(F)$ is infinite then the equality holds trivially. So assume that both of the sets E, F have finite measure, in which case both $\mu(E \cap F) \leq \mu(E)$ and $\mu(E \cup F) \leq \mu(E) + \mu(F)$ also have finite measure. As $(E \cup F) \setminus F = E \setminus F = E \setminus (E \cap F)$, using monotonicity we can write

$$\mu(E \cup F) - \mu(F) = \mu((E \cup F) \setminus F) = \mu(E \setminus (E \cap F)) = \mu(E) - \mu(E \cap F)$$

rearranging (all the terms are finite so we can do this) which gives us the required equality. □

Problem 1.10. Given a measure space $(X, \mathcal{M}, \mu)$ and $E \in \mathcal{M}$, define $\mu_E(A) = \mu(A \cap E)$ for $A \in \mathcal{M}$. Then μ_E is a measure.

Solution. Clearly $\mu_E(\emptyset) = \mu(\emptyset \cap E) = \mu(\emptyset) = 0$ and if $A_1, A_2, \dots$ are disjoint then,

$$\mu_E \left(\bigcup_{n \in \mathbb{N}} E_n \right) = \mu \left(A \cap \bigcap_{n \in \mathbb{N}} E_n \right) = \mu \left(\bigcap_{n \in \mathbb{N}} (A \cap E_n) \right) = \sum_{n=1}^{\infty} \mu(A \cap E_n) = \sum_{n=1}^{\infty} \mu_E(A_n)$$

as $A_n \cap E$ are also disjoint, hence proving that μ_E is a measure. □

Problem 1.11. A finitely additive measure μ is a measure iff it is continuous from below as in Theorem 1.8c. If $\mu(X) < \infty$, μ is a measure iff it is continuous from above as in Theorem 1.8d.

Solution. This is trivial. □

Problem 1.12. Let $(X, \mathcal{M}, \mu)$ be a finite measure space.

- a. If $E, F \in \mathcal{M}$ and $\mu(E \Delta F) = 0$, then $\mu(E) = \mu(F)$.
- b. Say that $E \sim F$ if $\mu(E \Delta F) = 0$; then $\sim$ is an equivalence relation on $\mathcal{M}$.
- c. For $E, F \in \mathcal{M}$, define $\rho(E, F) = \mu(E \Delta F)$. Then $\rho(E, G) \leq \rho(E, F) + \rho(F, G)$, and hence ρ defines a metric on the space $\mathcal{M}/\sim$ of equivalence classes.

Solution.

1. We have $E \Delta F = (E \cup F) \setminus (E \cap F)$. If $\mu(E \cap F) = \infty$ then clearly $\mu(E) = \infty = \mu(F)$ and we are done. Otherwise using monotonicity and rearranging we can write, $\mu(E \cup F) = \mu(E \cap F)$. Again, using monotonicity we have, $\mu(E) \leq \mu(E \cup F) = \mu(E \cap F) \leq \mu(E) \implies \mu(E) = \mu(E \cup F)$. Similarly, $\mu(F) = \mu(E \cup F)$ and we have $\mu(E) = \mu(F)$.
2. This is clearly a equivalence relation as its reflexive ($\mu(A \Delta A) = \mu(\emptyset) = 0$), symmetric as the symmetric difference operator is symmetric in its arguments, transitive as $A \sim B, B \sim C \implies \mu(A \Delta C) \leq \mu(A \Delta B) + \mu(B \Delta C) = 0 + 0$, this is proved in the next part.
3. We have the following inclusion for $A, B, C \in \mathcal{M}$

$$(A \setminus B) \cup (B \setminus C) \supseteq A \setminus C$$

Using this we get, $\mu(A \setminus B) + \mu(B \setminus C) \geq \mu(A \setminus C)$. Similarly, $\mu(B \setminus A) + \mu(C \setminus B) \geq \mu(C \setminus A)$ and adding the two gives,

$$(\mu(A \setminus B) + \mu(B \setminus A)) + (\mu(B \setminus C) + \mu(C \setminus B)) \geq (\mu(A \setminus C) + \mu(C \setminus A))$$

And as $\mu(A \Delta B) = \mu(A \setminus B) + \mu(B \setminus A)$ we get the desired inequality. Thus $\rho(E, F) := \mu(E \Delta F)$ is a metric on $\mathcal{M}$. If $E' \in [E] \in \mathcal{M}/\sim, F' \in [F] \in \mathcal{M}/\sim$ then we wish to prove that $\rho(E, F) = \rho(E', F')$. This is true as $\rho(E, F) \leq \rho(E, E') + \rho(E', F) = \rho(E', F) \leq \rho(E', F') + \rho(F', F) = \rho(E', F')$ and similarly $\rho(E', F') \leq \rho(E, F)$ as well, so we have equality. We showed that the choice of the representative from an equivalence class of $\sim$ does not matter so we can define ρ to be a metric on $\mathcal{M}/\sim$ by letting $\rho([E], [F]) := \rho(E', F')$ where $E' \in [E], F' \in [F]$ are some representatives.

□

Problem 1.13. Every σ -finite measure is semifinite.

Solution. Let $E_1, E_2, \dots$ be such that $X = \cup E_n$ and $\mu(E_n) < \infty$, then for any $E \in \mathcal{M}$ with $\mu(E) = \infty$ we see that, $\mu(E) = \mu(\cup(E \cap E_n)) \leq \sum_{n=1}^{\infty} \mu(E_n \cap E)$. As the sum is positive, not all terms can be zero, let $\mu(E \cap E_n) > 0$, we also see that $\mu(E \cap E_n) \leq \mu(E_n) < \infty$. This proves that μ is semifinite as we found $E \cap E_n \subseteq E$ with positive finite measure.

□

Problem 1.14. If μ is a semifinite measure and $\mu(E) = \infty$, for any $C > 0$ there exists $F \subset E$ with $C < \mu(F) < \infty$.

Solution. For the sake of contradiction assume that $s = \sup\{\mu(F) | \mathcal{M} \ni F \subseteq E, \mu(F) < \infty\} < \infty$. Then there exist $F_n \subseteq E$ with $s - 1/n \leq \mu(F_n) \leq s$ and thus, $K := \cup F_n$

has measure atleast s . There are two cases, either $\mu(\cup F_n)$ is finite or its infinite. If its finite, as $\mu(E) = \infty$, we must have $\mu(E \setminus K) = \infty$ and as μ is semifinite there exists some $G \subseteq E \setminus K$ with finite positive measure, but this leads to a contradiction as $\mu(E \cup K) = \mu(E) + \mu(K)$ is finite and stricly larger than s , here we used the fact that, by construction $G \cap K = \phi$. In the other case, if $\mu(K) = \infty$ then consider the increasing sequence of sets $G_n := \cup_{k \leq n} F_k$, we thus have that $\mu(\cup G_n) = \lim \mu(G_n) = \infty$ so there exists some $n \in \mathbb{N}$ such that $\mu(G_n) > S$ and also by construction we find that $\mu(G_n) \leq n \cdot s < \infty$ contradicting the maximality of s . So s must have been ∞ to begin with which implies that there subsets of E of arbitrarily large finite measure. $\square$

Problem 1.15. Given a measure μ on $(X, \mathcal{M})$, define μ_0 on $\mathcal{M}$ by $\mu_0(E) = \sup\{\mu(F) : F \subset E \text{ and } \mu(F) < \infty\}$.

- μ_0 is a semifinite measure. It is called the **semifinite part** of μ .
- If μ is semifinite, then $\mu = \mu_0$. (Use Exercise 14.)
- There is a measure ν on $\mathcal{M}$ (in general, not unique) which assumes only the values 0 and ∞ such that $\mu = \mu_0 + \nu$.

Solution.

- Clearly, $\mu_0(\phi) = 0$ and for disjoint $\{E_n\} \subseteq \mathcal{M}$,

$$\begin{aligned} \mu_0\left(\bigcup_{n \in \mathbb{N}} E_n\right) &:= \sup\{\mu(E) < \infty : E \in \mathcal{M}, E \subseteq \bigcup_{n \in \mathbb{N}} E_n\} \\ &= \sup\left\{\sum_{n=1}^{\infty} \mu(E_n \cap E) < \infty : E \in \mathcal{M}, E \subseteq \bigcup_{n \in \mathbb{N}} E_n\right\} \\ &\leq \sum_{n=1}^{\infty} \sup\{\mu(F) < \infty : F \in \mathcal{M}, F \subseteq E_n\} = \sum_{n=1}^{\infty} \mu_0(E_n) \end{aligned}$$

as, for all $n \in \mathbb{N}$, $\mu(E \cap E_n) \leq \mu(E) < \infty$ and $\mu(E \cap E_n) \leq \mu_0(E_n)$. It suffices to prove the other direction of the equality as we have already established sub additivity. There are two cases:

- If $\sum_{n=1}^{\infty} \mu_0(E_n) < \infty$ then for arbitrary $\epsilon > 0$, for all $n \in \mathbb{N}$ let $\mathcal{M} \ni F_n \subseteq E_n$ be such that $\mu_0(E_n) - \epsilon 2^{-n} \leq \mu(F_n) \leq \mu_0(E_n)$ as $\mu_0(E_n) < \infty$ is a supremum. Summing both sides as $n \in \mathbb{N}$ we get,

$$\sum_{n=1}^{\infty} \mu_0(E_n) - \epsilon \leq \sum_{n=1}^{\infty} \mu(F_n) \leq \sum_{n=1}^{\infty} \mu_0(E_n)$$

we see that $\cup F_n$ is considered in the set $\{\mu(E) < \infty : \mathcal{M} \ni E \subseteq \cup E_n\}$ as $\mu(\cup F_n) \leq \sum \mu_0(E_n) < \infty$ and thus $\sum \mu_0(E_n) - \epsilon \leq \mu_0(\cup E_n)$. As ϵ was arbitrary, we must have that $\sum \mu_0(E_n) \leq \mu_0(\cup E_n)$ and we are done.

- If $\sum_{n=1}^{\infty} \mu_0(E_n) = \infty$, then its trivial if $\mu_0(E_n) = \infty$ for some $n \in \mathbb{N}$ so assume thats not the case. For any $M > 0$ there must exist some $n \in \mathbb{N}$ such that $\sum_{k \leq n} \mu_0(E_k) \geq M$. Using arguements similar to those in the first case, for any $\epsilon > 0$ we can find $\mathcal{M} \ni F \subseteq \cup_{k \leq n} E_k \subseteq \cup E_k$ such that $M - \epsilon \leq$

$\sum_{n \leq k} \mu_0(E) - \epsilon \leq \mu(F) \leq \sum_{k \leq n} \mu_0(E_k) < \infty$. Again using a similar line of reasoning we find that $M - \epsilon \leq \mu_0(\cup E_k)$ and as M was arbitrarily large and ϵ arbitrarily small we must have $\mu_0(\cup E_n) = \infty = \sum \mu_0(E_n)$. we have hence proven that μ_0 is a measure

we have hence proven that μ_0 is a measure. Now suppose $\mu_0(E) = \infty$, then there must exist $\mathcal{M} \ni F \subseteq E$ with finite positive measure, and for these F , we trivially have $\mu_0(F) = \mu(F)$ so μ_0 is semifinite.

- b. Clearly, $\mu_0 \leq \mu$ always holds and we have equality when $\mu < \infty$, we will show that equality still holds when $\mu = \infty$ if μ is semifinite. Say $\mu(E) = \infty$, then there exists $\mathcal{M} \ni F \subseteq E$ with arbitrarily large $\mu(F) < \infty$ by [Problem 1.14](#) which implies that $\mu_0(E) = \infty$ and we are done.
- c. Let $\nu(E)$ be 0 if E is μ - σ finite and ∞ otherwise. To prove that ν is a measure it suffices to prove that if $\cup E_n$ is not μ - σ finite for disjoint $\{E_n\}_{n \in \mathbb{N}}$ then not all E_n can be μ - σ finite. This is true because if all E_n are μ - σ finite, we can write $E_n = \cup_j F_{n,j}$ with $\mu(F_{n,j}) < \infty$ and thus, $\cup E_n = \cup_{n,j} F_{n,j}$ which is a countable union, making $\cup E_n$ μ - σ finite which is not true. This is sufficient as, if $\{E_n\}$ are disjoint sets and $\cup E_n$ is not μ - σ finite then $\nu(\cup E_n) = \infty$ by definition and $\sum \nu(E_n) = \infty$ too as some E_n is not μ - σ finite, making $\nu(E_n) = \infty$ and when $\cup E_n$ is μ - σ finite we see that all of E_n are trivially also μ - σ finite, thus $\nu(\cup E_n) = 0 = \sum \nu(E_n)$ as $\nu(E_n) = 0$ for all $n \in \mathbb{N}$. As $\nu(\phi) = 0$ clearly and ν has sigma additivity, its a measure. Now $\mu = \mu_0 + \nu$ as $\mu = m\mu_0$ on μ - σ finite sets (if $E = \cup E_n$ with $\mu(E_n) < \infty$ then consider, for $n \in \mathbb{N}$ $G_n := E_n \setminus \cup_{k < n} E_k \in \mathcal{M}$, these are of finite measure too by monotonicity and partition E , thus $\mu(E) = \mu(\cup G_n) = \sum \mu(G_n) = \sum \mu_0(G_n) = \mu_0(\cup G_n) = \mu_0(E)$ and when E is not sigma finite we clearly have $\mu(E) = \infty$ and $\mu_0(E) + \nu(E) = \mu_0(E) = \infty = \mu(E)$ so we are done.

□

Problem 1.16. Let $(X, \mathcal{M}, \mu)$ be a measure space. A set $E \subset X$ is called **locally measurable** if $E \cap A \in \mathcal{M}$ for all $A \in \mathcal{M}$ such that $\mu(A) < \infty$. Let $\tilde{\mathcal{M}}$ be the collection of all locally measurable sets. Clearly $\mathcal{M} \subset \tilde{\mathcal{M}}$; if $\mathcal{M} = \tilde{\mathcal{M}}$, then μ is called **saturated**.

- a. If μ is σ -finite, then μ is saturated.
- b. $\tilde{\mathcal{M}}$ is a σ -algebra.
- c. Define $\tilde{\mu}$ on $\tilde{\mathcal{M}}$ by $\tilde{\mu}(E) = \mu(E)$ if $E \in \mathcal{M}$ and $\tilde{\mu}(E) = \infty$ otherwise. Then $\tilde{\mu}$ is a saturated measure on $\tilde{\mathcal{M}}$, called the **saturation** of μ .
- d. If μ is complete, so is $\tilde{\mu}$.
- e. Suppose that μ is semifinite. For $E \in \tilde{\mathcal{M}}$, define $\underline{\mu}(E) = \sup\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E\}$. Then $\underline{\mu}$ is a saturated measure on $\tilde{\mathcal{M}}$ that extends μ .
- f. Let X_1, X_2 be disjoint uncountable sets, $X = X_1 \cup X_2$, and $\mathcal{M}$ the σ -algebra of countable or co-countable sets in X . Let μ_0 be counting measure on $\mathcal{P}(X_1)$, and define μ on $\mathcal{M}$ by $\mu(E) = \mu_0(E \cap X_1)$. Then μ is a measure on $\mathcal{M}$, $\tilde{\mathcal{M}} = \mathcal{P}(X)$, and in the notation of parts (c) and (e), $\tilde{\mu} \neq \underline{\mu}$.

Solution.

- a. Let, $X = \cup E_n$ with $\mu(E_n) < \infty$ then, $A \in \tilde{\mathcal{M}} \implies \forall n \in \mathbb{N}, A \cap E_n \in \mathcal{M}$ and thus $\cup(E_n \cap A) = A \cap X = A \in \mathcal{M}$ proving $\tilde{\mathcal{M}} \subseteq \mathcal{M}$ making μ saturated.
- b. Clearly $\phi, X \in \mathcal{M}$ and if $A \in \tilde{\mathcal{M}}$ then for all $E \in \mu^{-1}([0, \infty))$, $A \cap E \in \mathcal{M}$ and thus, $E \setminus (A \cap E) = E \cap A^c \in \mathcal{M} \implies A^c \in \mathcal{M}$ so $\tilde{\mathcal{M}}$ is closed under complements. If $\{A_n\}_{n \in \mathbb{N}} \subseteq \tilde{\mathcal{M}}$ then, $\mu(E) < \infty \implies \forall n \in \mathbb{N}, A_n \cap E \in \mathcal{M} \implies \cup(A_n \cap E) = E \cap (\cup A_n) \in \mathcal{M} \implies \cup A_n \in \tilde{\mathcal{M}}$ and hence $\tilde{\mathcal{M}}$ is a σ -algebra.
- c. Clearly $\tilde{\mu}(\phi) = 0$ and if $\{E_n\}_{n \in \mathbb{N}} \subseteq \tilde{\mathcal{M}}$ are disjoint then when $\tilde{\mu}(\cup E_n) = \infty$ there are two cases : either $\cup E_n \notin \mathcal{M}$, in which case there's some $E_n \notin \mathcal{M}$ as well, making $\sum \tilde{\mu}(E_n) = \infty$ as well, otherwise $\cup E_n \in \mathcal{M}$, now if $A = \{n \in \mathbb{N} : E_n \in \mathcal{M}\}$ then $\cup_{n \in \mathbb{N} \setminus A} E_n \in \mathcal{M}$ and thus $\tilde{\mu}(\cup E_n) = \mu(\cup_{n \in A} E_n) + \mu(\cup_{n \in \mathbb{N} \setminus A} E_n)$. If $\mu(\cup_{n \in A} E_n) = \infty$ then clearly we have sigma additivity for $\{E_n\}$ and otherwise, we must have $\mu(\cup_{n \in \mathbb{N} \setminus A} E_n) = \infty$ and we again have sigma additivity as $\tilde{\mu}(E_k) = \infty$ for $k \in \mathbb{N} \setminus A$. The other case is that of $\tilde{\mu}(\cup E_n) < \infty$ in which case $\cup E_n \in \tilde{\mathcal{M}} \implies E_n \cap (\cup_m E_m) = E_n \in \mathcal{M}$ for all $n \in \mathbb{N}$ by definition so we have sigma additivity. We have thus shown that $\tilde{\mu}$ is sigma additive and maps ϕ to 0 making it a measure. To show saturation we have to show that $\tilde{\mathcal{M}} \subseteq \tilde{\tilde{\mathcal{M}}}$. If $A \in \tilde{\mathcal{M}}$ we have that $E \in \tilde{\mathcal{M}}$ and $\tilde{\mu}(E) < \infty \implies A \cap E \in \tilde{\mathcal{M}}$ and as $E \in \mathcal{M}$ in this case we have that $E \in \mathcal{M}$ and $\tilde{\mu}(E) = \mu(E) < \infty \implies A \cap E \in \tilde{\mathcal{M}}$ and again, $A \cap E \in \tilde{\mathcal{M}} \implies (A \cap E) \cap E = A \cap E \in \mathcal{M} \implies A \in \tilde{\mathcal{M}}$ thus $\tilde{\mathcal{M}} \subseteq \tilde{\tilde{\mathcal{M}}}$ and we are done.
- d. If $A \in \tilde{\mathcal{M}}$ and $\tilde{\mu}(A) = 0 \implies A \in \mathcal{M}$ and thus $\tilde{\mu}(A) = \mu(A) = 0$. Now if $F \subseteq A$ then as μ is complete we have $F \in \mathcal{M} \subseteq \tilde{\mathcal{M}} \ni F$ and thus, $\tilde{\mu}(F) = \mu(F) = 0$ so $\tilde{\mu}$ is complete as well.
- e. Clearly $\underline{\mu}|_{\mathcal{M}} = \mu$ by monotonicity, we will prove sigma additivity now. First we prove the claim : if μ is semifinite then $\underline{\mu}(E) = \sup\{\mu(F) : F \in \mathcal{M}, F \subseteq E, \mu(F) < \infty\}$. If $\mu(E) = \infty$ then that means that there's some $\mathcal{M} \ni F \subseteq E$ with $\mu(F) = \infty$ or, arbitrarily large $\mu(F)$, in the later case we are done but in the prior case, as μ is semifinite we can find arbitrarily large $\mu(K)$ for $\mathcal{M} \ni K \subseteq F \subseteq E$ by using [Problem 1.14](#) which implies $\sup\{\mu(F) : F \in \mathcal{M}, F \subseteq E, \mu(F) < \infty\} = \infty = \underline{\mu}(E)$. In the other case where $\underline{\mu}(E) < \infty$ its trivial to see that our claim is true as, using similar definitions, $\mu(F) < \infty$ is forced already. We will use this now, let $\{E_n\}_{n \in \mathbb{N}} \subseteq \tilde{\mathcal{M}}$ be disjoint, then

$$\begin{aligned}
\underline{\mu}\left(\bigcup_{n \in \mathbb{N}} E_n\right) &= \sup \left\{ \mu(E) : E \in \mathcal{M}, E \subseteq \bigcup_{n \in \mathbb{N}} E_n, \mu(E) < \infty \right\} \\
&= \sup \left\{ \sum_{n=1}^{\infty} \underbrace{\mu(E \cap E_n)}_{\leq \underline{\mu}(E_n), \text{ as } E \cap E_n \in \mathcal{M}} : E \in \mathcal{M}, E \subseteq \bigcup_{n \in \mathbb{N}} E_n, \mu(E) < \infty \right\} \leq \sum_{n=1}^{\infty} \underline{\mu}(E_n) \\
&= \sum_{n=1}^{\infty} \sup \{ \mu(F_n) : F_n \in \mathcal{M}, F_n \subseteq E_n, \mu(F_n) < \infty \} \\
&= \sup \left\{ \sum_{n=1}^{\infty} \mu(F_n) : n \in \mathbb{N}, F_n \in \mathcal{M}, F_n \subseteq E_n, \mu(F_n) < \infty \right\}
\end{aligned}$$

$$\begin{aligned}
&= \sup \left\{ \mu \left(\bigcup_{n \in \mathbb{N}} F_n \right) : \mathcal{M} \ni \bigcup_{n \in \mathbb{N}} F_n \subseteq \bigcup_{n \in \mathbb{N}} E_n \right\} \\
&\leq \sup \{ \mu(F) : \mathcal{M} \ni F \subseteq \bigcup E_n \} = \underline{\mu}(\bigcup E_n)
\end{aligned}$$

and thus we have sigma additivity making $\underline{\mu}$ a measure. Now its saturated as $A \in \tilde{M}$ implies that for all $B \in \mathcal{M} \subseteq \tilde{M}$ with $\mu(B) < \infty$ we have $\mu(B) = \underline{\mu}(B) < \infty$ and thus, $B \cap A \in \tilde{M}$ and $(B \cap A) \cap B = B \cap A \in \mathcal{M} \implies A \in \mathcal{M} \implies \tilde{M} \subseteq \tilde{M}$ and we are done.

- f. Write subsets S as $S_1 \cup S_2$ where $S_i \subseteq X_i; i = 1, 2$ from now on. μ is a measure by [Problem 1.10](#). We see that a set in $\mathcal{M}$ has finite measure iff it has finite intersection with X_1 , thus these are countable or cocountable sets of form $Y_1 \cup Y_2$ with Y_1 finite and when these are countable we clearly have that Y_2 is countable or otherwise we need them to be cocountable in which case $(Y_1 \cup Y_2)^c = (X_1 \setminus Y_1) \cup (X_2 \setminus Y_2)$ is countable which is impossible as $|X_1 \setminus Y_1| = \mathfrak{c} > \omega$. Thus the sets of finite measure are exactly those of form $Y_1 \cup Y_2$ with Y_1 finite and Y_2 countable. Given any $S_1 \cup S_2 \subseteq X$ we see that $(S_1 \cup S_2) \cap (Y_1 \cup Y_2) = (S_1 \cap Y_1) \cup (S_2 \cap Y_2)$ is countable and as its a subset of a countable set $Y_1 \cup Y_2$ making $\tilde{M} = \mathcal{P}(X)$. Now, $\tilde{\mu}(X_2 \cup \{x_1\})$ where x_1 is some element in X_1 is ∞ as $X_2 \cup \{x_1\}$ not cocountable nor countable whereas $\underline{\mu}(X_2 \cup \{x_1\}) = \sup \{ \mu(Y_1) = \mu(Y_1 \cup Y_2) : Y_1 \in \{\{x_1\}, \emptyset\}, |Y_2| \leq \omega, Y_2 \subseteq X_2 \} = 1$ and thus $\underline{\mu} \neq \tilde{\mu}$.

□

Problem 1.17. If μ^* is an outer measure on X and $\{A_j\}_1^\infty$ is a sequence of disjoint μ^* -measurable sets, then $\mu^*(E \cap (\bigcup_1^\infty A_j)) = \sum_1^\infty \mu^*(E \cap A_j)$ for any $E \subset X$.

Solution. As outer measures are subadditive, it suffices to show $\mu^*(E \cap (\bigcup_n A_n)) \geq \sum \mu^*(E \cap A_n)$. Let, $B_n = \bigcup_{k \leq n} A_k$ and $B = \bigcup_n A_n = \bigcup B_n$. Then as A_n are all measurable and hence we can say,

$$\begin{aligned}
\mu^*(E \cap B) &\geq \mu^*(E \cap B_n) = \mu^*(E \cap B_n \cap A_n) + \mu^*(E \cap B_n \cap A_n^c) \\
&= \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1}) = \dots = \sum_{m \leq n} \mu^*(E \cap A_m)
\end{aligned}$$

by monotonicity and the definition of measurability for outer measures. As $n \rightarrow \infty$ we get our desired inequality. □

Problem 1.18. Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra, $\mathcal{A}_\sigma$ the collection of countable unions of sets in $\mathcal{A}$, and $\mathcal{A}_{\sigma\delta}$ the collection of countable intersections of sets in $\mathcal{A}_\sigma$. Let μ_0 be a premeasure on $\mathcal{A}$ and μ^* the induced outer measure.

- For any $E \subset X$ and $\epsilon > 0$ there exists $A \in \mathcal{A}_\sigma$ with $E \subset A$ and $\mu^*(A) \leq \mu^*(E) + \epsilon$.
- If $\mu^*(E) < \infty$, then E is μ^* -measurable iff there exists $B \in \mathcal{A}_{\sigma\delta}$ with $E \subset B$ and $\mu^*(B \setminus E) = 0$.
- If μ_0 is σ -finite, the restriction $\mu^*(E) < \infty$ in (b) is superfluous.

Solution.

- a. By the definition of the outer measure as an infimum, for any $\epsilon > 0$ there exists (wlog we can consider these to be disjoint) $\{K_n\} \subseteq \mathcal{A}$ such that $\sum \mu_0(K_n) \leq \mu^*(E) + \epsilon$. Now we see that,

$$\sum_{n=1}^{\infty} \mu_0(K_n) = \lim_{m \rightarrow \infty} \sum_{n < m} \mu_0(K_n) = \lim_{m \rightarrow \infty} \mu_0\left(\underbrace{\bigcup_{n < m} K_n}_{\in \mathcal{A}}\right) = \lim_{m \rightarrow \infty} \mu^*\left(\underbrace{\bigcup_{n < m} K_n}_{\in \mathcal{M}(\mathcal{A})}\right) = \mu^*(\cup K_n)$$

using continuity from below as μ^* is a measure on the sigma algebra of μ^* -measurable sets which contains $\mathcal{A}$ and thus $\mathcal{M}(\mathcal{A})$ as well. As $\cup K_n \in \mathcal{A}_\sigma$ we are done.

- b. Assume that E is measurable. By part (a) we can find $E \subseteq E_n \in \mathcal{A}_\sigma$ with $\mu^*(E) + 1/n > \mu^*(E_n) \geq \mu^*(E)$. Then using continuity from above ($\mu^*(E) + 1/n < \infty$ as per the statement) we get that $\mu^*(\cap E_n) = \mu^*(E)$ and $E \subseteq K := \cap E_n \in \mathcal{A}_{\sigma\delta}$. Using the definition of measurability and finitude of $\mu^*(E)$ we get that $\mu^*(K \setminus E) = \mu^*(K) - \mu^*(E) = 0$. For the other direction, let B, E be as in the statement, then as μ^* is complete and $\mathcal{A}_{\sigma\delta} \subseteq \mathcal{M}(\mathcal{A})$, $B \setminus E$ and B are measurable here, this makes their difference measurable too i.e. $B \setminus (B \setminus E) = E$ is also measurable.
- c. Assuming σ -finiteness, let $\{H_n\}_{n \in \mathbb{N}} \subseteq \mathcal{A}$ be disjoint and of finite μ_0 -measure such that $X \subseteq \cup H_n$. Assume that E is measurable now. H_n being measurable forces $E \cap H_n$ to be measurable as well. Using part (a), fixing $\epsilon > 0$ we can find $B_n \in \mathcal{A}_\sigma$ containing $E \cap H_n$ such that $\mu^*(B_n) \leq \mu^*(E \cap H_n) + \epsilon 2^{-n}$ which, by the finitude of $\mu^*(E \cap H_n)$ (its clearly less than $\mu_0(H_n) < \infty$) and definition of μ^* -measurability, after moving a term to another side of the equality implies $\mu^*(B_n \setminus (E \cap H_n)) = \mu^*(B_n) - \mu^*(E \cap H_n) < \epsilon 2^{-n}$. Thus summing these over $n \in \mathbb{N}$ and using subadditivity of outer measures we have

$$\mu^*\left(\bigcup_{n \in \mathbb{N}} (B_n \setminus (E \cap H_n))\right) \leq \sum_{n \in \mathbb{N}} \mu^*(B_n \setminus (H_n \cap E)) \leq \epsilon$$

Now using monotonicity and the fact that $(\cup B_n) \setminus E = (\cup B_n) \setminus (\cup (H_n \cap E)) \subseteq \cup (B_n \setminus (H_n \cap E))$ we see that $\mu^*((\cup B_n) \setminus E) < \epsilon$ where $E \subseteq \cup B_n \in \mathcal{A}_{\sigma\sigma} = \mathcal{A}_\sigma$. As ϵ was arbitrarily small we can find $V_n \in \mathcal{A}_\sigma$ containing E for which $\mu^*(V_n \setminus E) < 1/n$. And again by monotonicity we see that $\mu^*((\cap V_n) \setminus E) = 0$ with $\cap V_n \in \mathcal{A}_{\sigma\delta}$. The proof for another direction does not depend on the finitude of $\mu^*(E)$ and is exactly the same to that in part (b) so we are done. □

Problem 1.19. Let μ^* be an outer measure on X induced from a finite premeasure μ_0 . If $E \subset X$, define the inner measure of E to be $\mu_*(E) = \mu_0(X) - \mu^*(E^c)$. Then E is μ^* -measurable iff $\mu^*(E) = \mu_*(E)$. (Use Exercise 18.)

Solution. μ_0 being finite clearly makes μ^* finite as well so we can rearrange terms freely. The only if direction is trivial, we will prove the if direction so assume that $\mu^*(E) = \mu_*(E)$, equivalently $\mu^*(X) = \mu^*(E) + \mu^*(E^c)$. For any $S \subseteq X$ and any $\epsilon > 0$, from [Problem 1.18](#) we can find μ^* -measurable K containing S with $\mu^*(S) + \epsilon > \mu^*(K)$. As K is μ^* -measurable, by definition we have

$$(\mu^*(S) + \epsilon) + \mu^*(K^c) > \mu^*(K) + \mu^*(K^c) = \mu^*(X) = \mu^*(E) + \mu^*(E^c)$$

$$\begin{aligned}
&= \mu^*(E \cap K) + \mu^*(E \cap K^c) + \mu^*(E^c \cap K) + \mu^*(E^c \cap K^c) \geq \mu^*(E \cap K) + \mu^*(E^c \cap K) + \mu^*(K^c) \\
&\geq \mu^*(E \cap S) + \mu^*(E^c \cap S) + \mu^*(K^c)
\end{aligned}$$

by using subadditivity and monotonicity. Subtracting $\mu^*(K^c)$ we get, $\mu^*(S) + \epsilon \geq \mu^*(E \cap S) + \mu^*(E^c \cap S)$ and as ϵ was arbitrarily small we can remove it from the inequality and it will still be valid, which is enough to prove equality as we already have subadditivity by the virtue of μ^* being an outer measure. $\square$

Problem 1.20. Let μ^* be an outer measure on X , $\mathcal{M}^*$ the σ -algebra of μ^* -measurable sets, $\bar{\mu} = \mu^*|_{\mathcal{M}^*}$, and μ^+ the outer measure induced by $\bar{\mu}$ as in (1.12) (with $\bar{\mu}$ and $\mathcal{M}^*$ replacing μ_0 and $\mathcal{A}$).

- If $E \subset X$, we have $\mu^*(E) \leq \mu^+(E)$, with equality iff there exists $A \in \mathcal{M}^*$ with $A \supset E$ and $\mu^*(A) = \mu^*(E)$.
- If μ^* is induced from a premeasure, then $\mu^* = \mu^+$. (Use Exercise 18a.)
- If $X = \{0, 1\}$, there exists an outer measure μ^* on X such that $\mu^* \neq \mu^+$.

Solution.

- $\mu^* \leq \mu^+$ follows from subadditivity, monotonicity and the definition of an infimum. Now suppose we have $\mu^*(E) = \mu^+(E)$, by the definition of μ^+ , for all $n \in \mathbb{N}$ we can find $A_n = \cup_m K_{m,n} \supseteq E$ with $K_{m,n} \in \mathcal{M}^*$ such that

$$\mu^+(E) + 1/n \geq \sum_m \mu^*(K_{m,n}) \geq \mu^*(A_n)$$

using monotonicity. As $\mathcal{M}^*$ forms a σ -algebra we see that $\cap A_n \in \mathcal{M}^*$ and by monotonicity, $\mu^*(\cap A_n) \leq \mu^*(E)$ and we have equality as $E \subseteq \cap A_n$. The other direction is trivial.

- Let μ^* be induced from a premeasure μ_0 on $\mathcal{A}$. Then using part (a) of [Problem 1.18](#), for any $E \subseteq X$ we can find $A_n \in \mathcal{A}_\sigma \subseteq \mathcal{M}^*$ containing E such that $\mu^*(E) + 1/n \geq \mu^*(A_n)$, now as $\mathcal{M}^*$ is a sigma algebra we have $E \subseteq \cap A_n \in \mathcal{M}^*$ and also $\mu^*(E) \geq \mu^*(\cap A_n)$, which implies equality by monotonicity. Thus we have found a set $\cap A_n \in \mathcal{M}^*$ containing E with $\mu^*(\cap A_n) = \mu^*(E)$ and thus by part (a) we must have $\mu^*(E) = \mu^+(E)$. As E was arbitrary we have $\mu^* = \mu^+$.
- Define $\mu^* : \emptyset \mapsto 0, \{0\} \mapsto 2, \{1\} \mapsto 3, \{0, 1\} \mapsto 4$. This meets all the criterions for being an outer measure i.e. monotonicity and σ -subadditivity and being 0 on the emptyset. It can be shown that $\mathcal{M}^* = \{\emptyset, \{0, 1\}\}$ and that $\mu^+(\{1\}) = 4 \neq \mu^*(\{1\})$ so we are done.

$\square$

Problem 1.21. Let μ^* be an outer measure induced from a premeasure and $\bar{\mu}$ the restriction of μ^* to the μ^* -measurable sets. Then $\bar{\mu}$ is saturated. (Use Exercise 18.)

Solution. If $A \subseteq X$ be locally $\bar{\mu}$ -measurable. If $F \subseteq X$ is such that $\mu^*(F) = \infty$ then by subadditivity atleast one of $\mu^*(F \cap A), \mu^*(F \cap A^c)$ is infinite and thus we must have $\mu^*(F) = \infty = \mu^*(F \cap A) + \mu^*(F \cap A^c)$. Now suppose we had $\mu^*(F) < \infty$, using part

(a) of [Problem 1.18](#) and the arguments in part (a) of [Problem 1.20](#) we can find a μ^* -measurable $C \supset F$ such that $\mu^*(F) = \mu^*(C)$. As $\mu^*(C) < \infty$ we must have that $C \cap F$ is measurable and thus so is $C \setminus (C \cap A) = C \cap A^c$ as the measurable sets form a σ -algebra here. Thus we have,

$$\begin{aligned}\mu^*(F) &= \mu^*(C) = \bar{\mu}(C) = \bar{\mu}(C \cap A) + \bar{\mu}(C \cap A^c) = \mu^*(C \cap A) + \mu^*(C \cap A^c) \\ &\geq \mu^*(F \cap A) + \mu^*(F \cap A^c)\end{aligned}$$

using additivity of $\bar{\mu}$ and monotonicity of μ^* . This implies, by subadditivity of μ^* that $\mu^*(F) = \mu^*(F \cap A) + \mu^*(F \cap A^c)$ and we have proven that this holds for all $F \subseteq X$, hence proving measurability of A by definition. This shows that all locally measurable sets are measurable w.r.t $\bar{\mu}$, making it saturated. $\square$

Problem 1.22. Let $(X, \mathcal{M}, \mu)$ be a measure space, μ^* the outer measure induced by μ according to (1.12), $\mathcal{M}^*$ the σ -algebra of μ^* -measurable sets, and $\bar{\mu} = \mu^*|_{\mathcal{M}^*}$.

- If μ is σ -finite, then $\bar{\mu}$ is the completion of μ . (Use Exercise 18.)
- In general, $\bar{\mu}$ is the saturation of the completion of μ . (See Exercises 16 and 21.)

Solution.

- σ -algebras are algebras and measures are also premeasures, this allows us to utilise parts of [Problem 1.18](#). The completion of μ is denoted by $\hat{\mu}$. We will show that

$$\mathcal{M}^* = \overline{\mathcal{M}} := \{E \cup F | E \in \mathcal{M}, \exists N \in \mathcal{M} : \mu(N) = 0 \text{ and } F \subseteq N\}$$

this is sufficient as by Carathéodory's theorem $\bar{\mu}$ is complete and we also know that $\mu^*|_{\mathcal{M}} = \mu$ (from proposition 1.13), so it extends μ . As this complete extension is unique, $\bar{\mu}$ must be the completion here. If $S \in \mathcal{M}^*$ then using part (c) of problem 18 we can find $B \in \mathcal{M}_{\sigma\delta} = \mathcal{M}$ such that $S \subseteq B$ and $\mu^*(B \setminus S) = 0$. By part (a) of problem 18, with some work we can show that there exists some $N \in \mathcal{M}_\sigma = \mathcal{M}$ such that $B \setminus S \subseteq N$ and $\mu^*(B \setminus S) = \mu^*(N) = \mu(N) = 0$. Now, to show that $E \in \mathcal{M}$, it suffices to show that $E^c \in \mathcal{M}$ as $\mathcal{M}$ is a σ -algebra. Rewriting E as $B \setminus (B \setminus E)$ we have $E^c = B^c \cup (B \setminus E)$ with $B^c \in \mathcal{M}$ and $B \setminus E \subseteq N \in \mathcal{M}$ of zero measure and thus $E^c \in \overline{\mathcal{M}}$ implying $\mathcal{M}^* \subseteq \overline{\mathcal{M}}$. For the other direction of the set inclusion, any $E \cup F$ with $E \in \mathcal{M}, F \subseteq N \in \mathcal{M}$ with $\mu(N) = 0$. Also, $E \cup N \in \mathcal{M}$ and $(E \cup N) \setminus (E \cup F) \subseteq N \setminus F$ so $\mu^*((E \cup N) \setminus (E \cup F)) \leq \mu^*(N \setminus F) \leq \mu^*(N) = \mu(N) = 0$, making $E \cup F$ measurable by part (b) of Problem 18.

- We need to first show that $\mathcal{M}^* = \tilde{\overline{\mathcal{M}}}$. Let $E \in \mathcal{M}^*$, consider $K \in \overline{\mathcal{M}} \subseteq \mathcal{M}^*$ of finite μ -measure, then we see that $E \cap K \in \mathcal{M}^*$ and they have finite μ^* -measure, using part (a,b) of problem 18 as we did in part (a) of this problem we can prove that $E \cap K \in \tilde{\overline{\mathcal{M}}}$. This shows one direction of the equality $\mathcal{M}^* \subseteq \tilde{\overline{\mathcal{M}}}$. Now we show the other direction $\tilde{\overline{\mathcal{M}}} \subseteq \mathcal{M}^*$. Suppose $G \in \tilde{\overline{\mathcal{M}}}$ then for all $K \in \overline{\mathcal{M}}$, of finite $\hat{\mu}$ -measure we have $G \cap K \in \overline{\mathcal{M}}$. It suffices to show that $G \in \tilde{\mathcal{M}}^*$ as μ^* is saturated. If $K \in \mathcal{M}^*$ has finite μ^* -measure then we have shown earlier that $K \in \overline{\mathcal{M}}$. We also have that $\mu^* = \hat{\mu}$ on $\overline{\mathcal{M}}$ as any $E \cup F$ with $E \in \mathcal{M}$ and $F \subseteq N \in \mu^{-1}(\{0\})$, thus $\mu^*(E \cup F) \leq \mu^*(E \cup N) \leq \mu^*(E) + \mu^*(N) = \mu(E) + \mu(N) = \mu(E)$ and we have equality by monotonicity. Now we see that $\mu^*(K) < \infty \implies \hat{\mu}(K) < \infty$ and thus

$E \cap K \in \overline{\mathcal{M}} \subseteq \mathcal{M}^*$ proving $\overline{\mathcal{M}} \subseteq \mathcal{M}^*$, finally resulting in equality. Now to show that both of these measures agree, we already know that they agree on $\overline{\mathcal{M}}$, outside which $\hat{\mu} = \infty$ and thus we need to show that $\mu^* = \infty$ too. However this is true as $\mu^*(E) < \infty$ for $E \in \mathcal{M}^*$ implies $E \in \overline{\mathcal{M}}$ so we are done.

□

Problem 1.23. Let $\mathcal{A}$ be the collection of finite unions of sets of the form $(a, b] \cap \mathbb{Q}$ where $-\infty \leq a < b \leq \infty$.

- $\mathcal{A}$ is an algebra on $\mathbb{Q}$. (Use Proposition 1.7.)
- The σ -algebra generated by $\mathcal{A}$ is $\mathcal{P}(\mathbb{Q})$.
- Define μ_0 on $\mathcal{A}$ by $\mu_0(\emptyset) = 0$ and $\mu_0(A) = \infty$ for $A \neq \emptyset$. Then μ_0 is a premeasure on $\mathcal{A}$, and there is more than one measure on $\mathcal{P}(\mathbb{Q})$ whose restriction to $\mathcal{A}$ is μ_0 .

Solution.

- This is trivial.
- Given any nonempty $X \subseteq \mathbb{Q}$ we see that it must be countable, thus

$$X = \bigcup_{x \in X} \{x\} = \bigcup_{x \in X} \bigcap_{n \in \mathbb{N}} \left(x - \frac{1}{n}, x\right] \in \mathcal{A}_{\delta\sigma} \subseteq \mathcal{M}(\mathcal{A})$$

thus we must have equality as $\mathcal{M}(\mathcal{A}) \subseteq \mathcal{P}(\mathbb{Q})$.

- All elements of $\mathcal{A}$ can be easily shown to have nonempty interior. Let $\mu_1 = \infty$ except on $\emptyset$ and $\mu_2 = \infty$ except on $\{\emptyset, \{0\}\}$. It can be easily shown that $\mu_1|_{\mathcal{A}} = \mu_2|_{\mathcal{A}} = \mu_0$ with $\mu_1 \neq \mu_2$

□

Problem 1.24. Let μ be a finite measure on $(X, \mathcal{M})$, and let μ^* be the outer measure induced by μ . Suppose that $E \subset X$ satisfies $\mu^*(E) = \mu^*(X)$ (but not that $E \in \mathcal{M}$).

- If $A, B \in \mathcal{M}$ and $A \cap E = B \cap E$, then $\mu(A) = \mu(B)$.
- Let $\mathcal{M}_E = \{A \cap E : A \in \mathcal{M}\}$, and define the function ν on $\mathcal{M}_E$ defined by $\nu(A \cap E) = \mu(A)$ (which makes sense by (a)). Then $\mathcal{M}_E$ is a σ -algebra on E and ν is a measure on $\mathcal{M}_E$.

Solution.

- We have $A \setminus B = ((A \cap E) \setminus (B \cap E)) \cup ((A \cap E^c) \setminus (B \cap E^c)) = (A \cap E^c) \setminus (B \cap E^c)$ and also $(A \setminus B)^c = (B \cap E^c) \cup (A \cap E^c)^c = (B \cap E^c) \cup (A^c \cap E) \supseteq E$. This implies $\mu((A \setminus B)^c) = \mu^*((A \setminus B)^c) \geq \mu^*(E) = \mu(X)$ and thus we must have $\mu(A \setminus B) = \mu(X) - \mu((A \setminus B)^c) = \mu(X) - \mu(X) = 0$. Thus, $\mu(A) = \mu(A \setminus B) + \mu(B \cap A) = \mu(B \cap A)$ by additivity and similarly $\mu(B) = \mu(A \cap B)$ too, so we have $\mu(A) = \mu(B)$.

- b. $\mathcal{M}_E$ clearly contains $E, \emptyset$ and is closed under countable unions. If $E \cap K \in \mathcal{M}_E$ with $K \in \mathcal{M}$, we see that $K^c \in \mathcal{M}$ and thus $E \cap K^c = E \setminus (E \cap K)$ i.e. the complement of $E \cap K$ in E is also in $\mathcal{M}_E$ so its closed under complements as well, making it a σ -algebra. Clearly $\nu(\emptyset) = 0$ and if $\{A_n \cap E\}_{n \in \mathbb{N}} \subseteq \mathcal{M}_E$ are disjoint, then using the fact that $S \cap E = (S \cap E) \cap E$ for all $S \in \mathcal{M}$ and thus $\mu(S) = \mu(S \cap E)$, we have

$$\begin{aligned} \nu \left(\bigcup_{n \in \mathbb{N}} (A_n \cap E) \right) &= \nu \left(E \cap \bigcup_{n \in \mathbb{N}} A_n \right) = \underbrace{\mu \left(\bigcup_{n \in \mathbb{N}} A_n \right)}_{\in \mathcal{M}} = \mu \left(E \cap \bigcup_{n \in \mathbb{N}} A_n \right) \\ &= \mu \left(\bigcup_{n \in \mathbb{N}} (E \cap A_n) \right) = \sum_{n \in \mathbb{N}} \mu(E \cap A_n) = \sum_{n \in \mathbb{N}} \mu(A_n) = \sum_{n \in \mathbb{N}} \nu(E \cap A_n) \end{aligned}$$

□

Problem 1.25. Complete the proof of Theorem 1.19.

Solution. We start with the following lemma: If $F_n \subseteq (n, n+1]$ are closed subsets for $n \in \mathbb{Z}$ then so is their union. For a proof, consider any sequence with terms in the union that converges. If the limit is in $(n, n+1]$ then we see that the sequence is eventually in $F_n \cup F_{n+1}$ which is indeed closed and the sequence converges in $F_n \cup F_{n+1}$, and thus also in $\cup F_n$ making it closed. Clearly (b,c) imply (a), we prove the other directions. As the measure μ defined here is sigma finite, we can consider $\{H_n = (n, n+1]\}_{n \in \mathbb{Z}} \subseteq \mathcal{M}_\mu$ of finite measure which partitions $\mathbb{R}$. Now, for every $n \in \mathbb{N}$ let $U_{n,k}$ and $F_{n,k}$ be open and closed sets respectively such that $F_{n,k} \subseteq H_n \cap E \subseteq U_{n,k}$ and

$$\mu(U_{n,k}) - 2^{-|n|}/3k \leq \mu(E) \leq \mu(F_{n,k}) + 2^{-|n|}/3k$$

as every term is finite here we can rearrange them to get

$$\mu(U_{n,k} \setminus E \cap H_n), \mu(E \cap H_n \setminus F_{n,k}) \leq 2^{-|n|}/3k$$

We have that $U_k^* := \cup_n U_{n,k}$ is open, contains E and $F_k^* := \cup_n F_{n,k}$ (this is a disjoint union too) is closed and contained in E such that

$$\mu(U_k^* \setminus E) \leq \mu \left(\bigcup_{n \in \mathbb{Z}} (U_{n,k} \setminus H_n \cap E) \right) \leq \sum_{n \in \mathbb{Z}} \mu(U_{n,k} \setminus H_n \cap E) \leq \sum_{n \in \mathbb{Z}} 2^{-|n|}/3k = 1/k$$

using monotonicity as $(\cup U_{n,k}) \setminus (\cup E \cap H_n) \subseteq \cup (U_{n,k} \setminus E \cap H_n)$. Similarly,

$$\mu(E \setminus F_k^*) \leq \mu \left(\bigcup_{n \in \mathbb{Z}} (E \cap H_n \setminus F_{n,k}) \right) = \sum_{n \in \mathbb{Z}} \mu(E \cap H_n \setminus F_{n,k}) \leq \sum_{n \in \mathbb{Z}} 2^{-|n|}/3k = 1/k$$

Let, $U := \cap_k U_k^* \in G_\delta$ and $F := \cup_k F_k^* \in F_\sigma$ then $F \subseteq E \subseteq U$ and we have $\mu(E \setminus F) = \mu(U \setminus E) = 0$. Now, using additivity of μ we have $\mu(U) = \mu(U \setminus E) + \mu(E) = \mu(E)$ and $\mu(E) = \mu(F) + \mu(E \setminus F) = \mu(F)$. Now we can write $E = U \setminus (U \setminus E) = F \cup (E \setminus F)$ proving that (a) implies (b,c) too and we are done.

□

Problem 1.26. Prove Proposition 1.20. (Use Theorem 1.18.)

Solution. Using theorem 1.18, given any $\epsilon > 0$ we can find open $U \supset E$ with $\mu(U) \leq \mu(E) + \epsilon$ rearranging which gives $\mu(E \Delta U) < \epsilon$. Suppose $U = \cup I_k$ where $\{I_k\}$ are disjoint open intervals, then the sum $\sum \mu(I_k) = \mu(U) < \infty$. Truncating this we can find $M \in \mathbb{N}$ such that $\mu(\cup_{1 \leq k \leq M} I_k) + \epsilon \geq \mu(U)$ and thus $\mu((\cup_{1 \leq k \leq M} I_k) \Delta U) < \epsilon$. As $\mu(\cup_{1 \leq k \leq M} I_k), \mu(E), \mu(U)$ are all finite, we have

$$\mu\left(E \Delta \left(\bigcup_{1 \leq k \leq M} I_k\right)\right) \leq \mu(E \Delta U) + \mu\left(U \Delta \left(\bigcup_{1 \leq k \leq M} I_k\right)\right) < 2\epsilon$$

proving the proposition. $\square$

Problem 1.27. Prove Proposition 1.22a. (Show that if $x, y \in C$ and $x < y$, there exists $z \notin C$ such that $x < z < y$.)

Solution. This is trivial by considering ternary expansions. $\square$

Problem 1.28. Let F be increasing and right continuous, and let μ_F be the associated measure. Then $\mu_F(\{a\}) = F(a) - F(a-)$, $\mu_F([a, b)) = F(b-) - F(a-)$, $\mu_F([a, b]) = F(b) - F(a-)$, and $\mu_F((a, b)) = F(b-) - F(a)$.

Solution. All of these follows from upper and lower continuity theorems of measures. $\square$

Problem 1.29. Let E be a Lebesgue measurable set.

- If $E \subset N$ where N is the nonmeasurable set described in §1.1, then $m(E) = 0$.
- If $m(E) > 0$, then E contains a nonmeasurable set. (It suffices to assume $E \subset [0, 1]$. In the notation of §1.1, $E = \cup_{r \in R} (E \cap N_r)$.)

Solution.

- If E is a measurable subset of N then we find that, for $r \in \mathbb{Q} \cap [0, 1)$, $E_r := (r + E \cap [0, 1 - r)) \cup (-1 + r + E \cap [1 - r, 1)) \subseteq N_r$ is measurable and $m(E) = m(E_r)$. As N_r are disjoint, E_r must also be disjoint. Thus,

$$\sum_{r \in \mathbb{Q} \cap [0, 1)} m(E) = \sum_{r \in [0, 1) \cap \mathbb{Q}} m(E_r) = m\left(\bigcup_{r \in \mathbb{Q} \cap [0, 1)} E_r\right) \leq m([0, 1)) = 1$$

this is possible only when $\mu(E) = 0$ so that must be the case here.

- After shifting taking a finite measured subset of E , we can assume $E \subseteq [0, 1)$. If all of $E \cap N_r$ are measurable, then as any measurable subset of the nonmeasurable sets N_r has measure 0 from the previous part (This can be proven using the same arguments we used to prove it for N i.e. when $r = 0$) we have,

$$m(E) = m\left(\bigcup_{r \in \mathbb{Q} \cap [0, 1)} N_r \cap E\right) = \sum_{r \in \mathbb{Q} \cap [0, 1)} m(E \cap N_r) = \sum_{r \in \mathbb{Q} \cap [0, 1)} 0 = 0$$

which is false as $m(E) > 0$.

□

Problem 1.30. If $E \in \mathcal{L}$ and $m(E) > 0$, for any $\alpha < 1$ there is an open interval I such that $m(E \cap I) > \alpha m(I)$.

Solution. Unless $m(E) < \infty$, we can consider some finite positive measured subset of E so wlog it is safe to assume $m(E) < \infty$ from the start and we do so. First we define the following supremum,

$$C := \sup \left\{ \frac{m(E \cap I)}{m(I)} : I \text{ is a nonempty open interval} \right\}$$

which exists because $m(E \cap I)/m(I) \leq 1$ by monotonicity. Thus we must have that $m(E \cap I) \leq Cm(I)$ for all nonempty open I . By [Problem 1.18](#), for any $\epsilon > 0$ we can find open $U \supset E$ with $m(E) + \epsilon > m(U)$. Suppose $U = \cup I_k$ with I_k being mutually disjoint open intervals. Then as $E = E \cap U = \cup (E \cap I_k)$ we can write,

$$m(E) = m\left(\bigcup_k (E \cap I_k)\right) = \sum_k m(E \cap I_k) \leq \sum_k Cm(I_k) = Cm(U)$$

and thus, for all $\epsilon > 0$ we must have $Cm(U) + \epsilon \geq m(E) + \epsilon > m(U)$ implying that

$$\epsilon > (1 - C)m(U) \geq (1 - C)m(E)$$

Now as $m(E) > 0$ this is only possible when $1 - C = 0$ and we are done. □

Problem 1.31. If $E \in \mathcal{L}$ and $m(E) > 0$, the set $E - E = \{x - y : x, y \in E\}$ contains an interval centered at 0. (If I is as in Exercise 30 with $\alpha > 3/4$, then $E - E$ contains $(-1/2m(I), 1/2m(I))$).

Solution. Consider the setup in the hint, then for $r \in (-0.5m(I), 0.5m(I))$ we see that $r \in E - E$ iff $(|r| + E) \cap E$ is nonempty. As it contains $\supset (|r| + E \cap I) \cap (E \cap I)$, proving this to be nonempty suffices. Now by the inclusion exclusion principle for measures, for our case we can write

$$\begin{aligned} m((|r| + E \cap I) \cap E \cap I) &= m(|r| + E \cap I) + m(E \cap I) - m((|r| + E \cap I) \cup E \cap I) \\ &\geq 2m(E \cap I) - m(I \cup (|r| + I)) \geq 1.5m(I) - (1 + |r|)m(I) = (0.5 - |r|)m(I) > 0 \end{aligned}$$

so it definitely can not be empty and we are done. □

Problem 1.32. Suppose $\{\alpha_j\}_1^\infty \subset (0, 1)$.

- $\prod_1^\infty (1 - \alpha_j) > 0$ iff $\sum_1^\infty \alpha_j < \infty$. (Compare $\sum \log(1 - \alpha_j)$ to $\sum \alpha_j$.)
- Given $\beta \in (0, 1)$, exhibit a sequence $\{\alpha_j\}$ such that $\prod_1^\infty (1 - \alpha_j) = \beta$.

Solution. The first one follows from continuity of the logarithm on its domain and comparison principle for sums using the fact that $\log(1 - x) = -x + O(x^2)$ for small x . For the second one consider $\alpha_j := 1 - \beta^{2^{-j}}$. □

Problem 1.33. There exists a Borel set $A \subset [0, 1]$ such that $0 < m(A \cap I) < m(I)$ for every subinterval I of $[0, 1]$. (Hint: Every subinterval of $[0, 1]$ contains Cantor-type sets of positive measure.)

Solution. By the previous problem we easily notice that there are cantor-like measurable sets $\subseteq [0, 1]$ which are closed, nowhere dense and have positive measure. Let $\{I_n\}$ be an enumeration of the nonempty open (this is clearly sufficient) subintervals with rational endpoints in $[0, 1]$. Let $K_1 \subseteq I_1$ be such a set as discussed before, as K_1 is nowhere dense there is an open interval in $I_1 \setminus K_1$ and we can choose a positive measured nowhere dense set $C_1 \subseteq I_1 \setminus K_1$ after shifting and dilating sets like K . Inductively choose $K_1, C_1, \dots, K_n, C_n$. As the union of finitely many nowhere dense sets is still nowhere dense, there is an open interval in $I_{n+1} \setminus (\cup_{k \leq n} K_k \cup C_k)$ and we can choose a nowhere dense set K_{n+1} in it of positive measure. Again we can choose a similar $C_{n+1} \subseteq I_{n+1} \setminus (K_{n+1} \cup (\cup_{k \leq n} K_k \cup C_k))$. Continuing ad infinitum we get two collections of sets $\{K_n\}$ and $\{C_n\}$, and clearly $K := \cup K_n$ is disjoint with $C := \cup C_n$. We claim that $A = K$ works. Now, given any subinterval I of $[0, 1]$ having positive measure we can find some $I_n \subseteq I$ and thus, because $K_n \subseteq I_n$ we see that $m(I \cap K) \geq m(I_n \cap K_n) = m(K_n) > 0$ and we also have that $m(I \cap K^c) \geq m(I_n \cap C_n) = m(C_n) > 0$ as $C \subseteq K^c$. Thus we have that $0 < m(A \cap I) = m(I) - m(A^c \cap I) < m(I)$ so this A satisfies the criterion, As all the K_n were closed, $A \in F_\sigma$ so its clearly borel and we are done. $\square$

Chapter 2 : Integration

Problem 2.1. Let $f : X \rightarrow \overline{\mathbb{R}}$ and $Y = f^{-1}(\mathbb{R})$. Then f is measurable iff $f^{-1}(\{-\infty\}) \in \mathcal{M}$, $f^{-1}(\{\infty\}) \in \mathcal{M}$, and $f|_Y$ is measurable on Y .

Solution. For the only if direction, $\{\infty\}, \{-\infty\}, \mathbb{R} = Y \in \mathcal{B}_{\overline{\mathbb{R}}}$ making f^{-1} map them into $\mathcal{M}$ as f is measurable and as $\mathcal{M}$ is a sigma algebra we see that $f|_Y : Y \rightarrow \mathbb{R}$ is $\mathcal{M}_Y$ -measurable as $f^{-1}(E) \cap Y \in \mathcal{M}_Y$ for all $E \in \mathcal{B}_{\mathbb{R}}$. Now for the if direction, as $\mathcal{B}_{\overline{\mathbb{R}}} = \{E \cup I : E \in \mathcal{B}_{\mathbb{R}}, I \subseteq \{\infty, -\infty\}\}$ we see that for any such $E \cup I \in \mathcal{B}_{\overline{\mathbb{R}}}$ we have $f^{-1}(E \cup I) = f^{-1}(E) \cup f^{-1}(I)$ and as $f^{-1}(E) \cap Y = f^{-1}(E) \in \mathcal{M}_Y \subseteq \mathcal{M}$ and $f^{-1}(I) \in \mathcal{M}$ so f is measurable on $\mathcal{M}$. $\square$

Problem 2.2. Suppose $f, g : X \rightarrow \overline{\mathbb{R}}$ are measurable.

- fg is measurable (where $0 \cdot (\pm\infty) = 0$).
- Fix $\alpha \in \overline{\mathbb{R}}$ and define $h(x) = \alpha$ if $f(x) = -g(x) = \pm\infty$ and $h(x) = f(x) + g(x)$ otherwise. Then h is measurable.

Solution.

- Consider the two maps,

$$X \ni x \longrightarrow (f(x), g(x)) \text{ and } \overline{\mathbb{R}}^2 \ni (x, y) \mapsto xy$$

the first is a measurable map from X (with sigma algebra $\mathcal{M}$) to $R \times R$ (with sigma algebra $\otimes_1^2 \mathcal{B}_{\overline{\mathbb{R}}}$) and the second from this to $\overline{\mathbb{R}}$ (with sigma algebra $\mathcal{B}_{\overline{\mathbb{R}}}$) and as the composition of measurable maps are measurable we are done. Here, the measurability of $(x, y) \rightarrow xy$ follows from its continuity (note: the metric on $\overline{\mathbb{R}}$ considered throughout the chapter is usually $d(x, y) := |\arctan(x) - \arctan(y)|$ with $\arctan(\pm\infty) = \pm\pi/2$).

b. Similarly again consider $x \mapsto (f(x), g(x))$ but this time with $(x, y) \rightarrow J(x, y)$ where,

$$J(x, y) := \begin{cases} \alpha & \text{if } x = -y = \pm\infty \\ x + y & \text{otherwise} \end{cases}$$

We will use [Problem 1.5](#) here. Let $A = \{(+\infty, -\infty), (-\infty, +\infty)\}$ and $B = \mathbb{R}^2 \setminus A$, it is clear that J is continuous on B and hence measurable on it and as it is constant on A it is trivially measurable on it so we are done. As compositions of measurable functions are measurable we are done. □

Problem 2.3. If $\{f_n\}$ is a sequence of measurable functions on X , then $\{x : \lim f_n(x) \text{ exists}\}$ is a measurable set.

Solution. We know that $g_1 = \limsup f_n$ and $g_2 = -\liminf f_n$ are measurable functions, define

$$h(x) := \begin{cases} 2026 & \text{if } g_1(x) = -g_2(x) = \pm\infty \\ g_1(x) + g_2(x) & \text{otherwise} \end{cases}$$

h is a measurable function by [Problem 1.2](#) and f_n converges iff $g_1 = -g_2$ and we get that

$$\{x : \lim f_n(x) \text{ exists}\} = h^{-1}(\{0, 2026\})$$

which implies that the set in fact is measurable as all the sets we performed unions and intersections on are in $\mathcal{M}$. □

Problem 2.4. If $f : X \rightarrow \mathbb{R}$ and $f^{-1}((r, \infty]) \in \mathcal{M}$ for each $r \in \mathbb{Q}$, then f is measurable.

Solution. It can easily be shown that these sets being in $\mathcal{M}$ implies that $f^{-1}((k, \infty]) \in \mathcal{M}$ for any $k \in \mathbb{R}$ easily so we are done. □

Problem 2.5. If $X = A \cup B$ where $A, B \in \mathcal{M}$, a function f on X is measurable iff $f|_A$ is measurable on A and $f|_B$ is measurable on B .

Solution. If f is measurable then measurability on $A, B \in \mathcal{M}$ are trivial. For the other direction, suppose $f : X \rightarrow Y$ with $\mathcal{N}$ being the sigma algebra on Y , for any $E \in \mathcal{N}$ we have,

$$f^{-1}(E) = (f^{-1}(E) \cap A) \cup (f^{-1}(E) \cap B) = (f|_A^{-1}(E)) \cup (f|_B^{-1}(E)) \in \mathcal{M}$$

as these two sets are in $\mathcal{M}_A, \mathcal{M}_B \subseteq \mathcal{M}$ respectively anyways. □

Problem 2.6. The supremum of an uncountable family of measurable $\mathbb{R}$ -valued functions on X can fail to be measurable (unless the σ -algebra $\mathcal{M}$ is very special).

Solution. Suppose $X = \mathbb{R}$ with some sigma algebra containing singletons (most of the ones we care about do have them, for example the borel sigma algebra), then for any nonempty set $E \subseteq \mathbb{R}$ consider the family $\{\chi_{\{r\}}\}_{r \in E}$, these are all measurable as singletons are in the sigma algebra whereas $\sup_{r \in E} \chi_{\{r\}} = \chi_E$ always being measurable would imply that the sigma algebra was the whole powerset stemming from E being arbitrary. □

Problem 2.7. Suppose that for each $\alpha \in \mathbb{R}$ we are given a set $E_\alpha \in \mathcal{M}$ such that $E_\alpha \subset E_\beta$ whenever $\alpha < \beta$, $\bigcup_{\alpha \in \mathbb{R}} E_\alpha = X$, and $\bigcap_{\alpha \in \mathbb{R}} E_\alpha = \emptyset$. Then there is a measurable function $f : X \rightarrow \mathbb{R}$ such that $f(x) \leq \alpha$ on E_α and $f(x) \geq \alpha$ on E_α^c for every α . (Use Exercise 4.)

Solution. We claim that the following function works

$$f(x) := \sup\{r \in \mathbb{Q} : x \notin E_r\}$$

For $x \in E_\alpha$, from the definition we have that $x \in E_\beta$ for all $\mathbb{Q} \ni \beta \geq \alpha$ and thus $f(x) = \sup\{r \in \mathbb{Q} : r < \alpha, x \notin E_r\} \leq \alpha$. Now, for $x \in E_\alpha^c$ its clearly $\geq \alpha$. We claim that, for $r \in \mathbb{Q}$ we have

$$f^{-1}((r, \infty)) = \bigcup_{\mathbb{Q} \ni \alpha > r} E_\alpha^c$$

If $f(x) > r$, then the set of β for which $x \in E_\beta^c$ (its an open ray unbounded below) must contain r , i.e its of form $(-\infty, v)$ with $v > r$ and we can find a rational $q \in (r, v)$ such that $x \in E_q$ so we have one direction of the equality, the other direction is trivial and we are done as the set on the right is in $\mathcal{M}$ being a countable union of sets from $\mathcal{M}$ which is sufficient to prove measurability via [Problem 2.4](#) $\square$

Problem 2.8. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is monotone, then f is Borel measurable.

Solution. trivial. $\square$

Problem 2.9. Let $f : [0, 1] \rightarrow [0, 1]$ be the Cantor function (§1.5), and let $g(x) = f(x) + x$.

- g is a bijection from $[0, 1]$ to $[0, 2]$, and $h = g^{-1}$ is continuous from $[0, 2]$ to $[0, 1]$.
- If C is the Cantor set, $m(g(C)) = 1$.
- By Exercise 29 of Chapter 1, $g(C)$ contains a Lebesgue nonmeasurable set A . Let $B = g^{-1}(A)$. Then B is Lebesgue measurable but not Borel.
- There exist a Lebesgue measurable function F and a continuous function G on $\mathbb{R}$ such that $F \circ G$ is not Lebesgue measurable.

Solution.

- It is known that the cantor function is continuous on $[0, 1]$ as well as being increasing. This makes $g(x) = f(x) + x$ strictly increasing and continuous, with this we can say using the intermediate value theorem that g is is a continuous bijection from $[0, 1]$ to $[0, 2]$ and $h = g^{-1}$ is also continuous.
- As the cantor set is borel and h is continuous, it must be borel measurable. Thus, $h^{-1}(C) = g(C)$ is borel (as well as being lebesgue measurable which is weaker). We can write $m(g([0, 1])) = m(g(C)) + m(g([0, 1] \setminus C))$ as the sets here are disjoint (the second one is clearly measurable). From the previous parts we see that $m(g(C)) = 2 - m(g([0, 1] \setminus C))$. This set is the disjoint union of the middle third intervals removed in the process of constructing the cantor set, on which, by defintion the cantor function is constant. These removed open intervals have measures summing to 1 and as g is a translation on these we see that $m(g([0, 1] \setminus C)) = 1$ showing that $m(g(C)) = 1$.

- c. As the lebesgue measure is complete we see that B being the subset of the cantor set which has measure zero, must be lebesgue measurable too. Now, if B was borel then so would be $h^{-1}(B) = A$ as g^{-1} is continuous, so it is not borel.
- d. Consider $\chi_B \circ h$, we see that $(\chi_B \circ h)^{-1}(\{1\}) = h^{-1}(\chi_B^{-1}(\{1\})) = h^{-1}(B) = A$ which is not lebesgue measurable while we had h be continuous (a homeomorphism even) and χ_B lebesgue measurable so we are done as $\{1\}$ is borel.

□

Problem 2.10. Prove that the following implications are valid iff the measure μ is complete:

- a. If f is measurable and $f = g$ μ -a.e. then g is measurable.
- b. If f_n is measurable for $n \in \mathbb{N}$ and $f_n \rightarrow f$ μ -a.e. then f is measurable.

Solution.

- a. First assume that μ is complete and that f and g disagree on some $N \in \mathcal{M}$ with $\mu(N) = 0$, then for any measurable E in the range we must have that

$$\begin{aligned} g^{-1}(E) &= (g^{-1}(E) \cap N^c) \cup (g^{-1}(E) \cap N) = (g|_{N^c}^{-1}(E)) \cup (g^{-1}(E) \cap N) \\ &= \underbrace{(f|_{N^c \in \mathcal{M}}^{-1}(E))}_{\in \mathcal{M}} \cup \underbrace{(g^{-1}(E) \cap N)}_{\subseteq N \Rightarrow \in \mathcal{M}} \end{aligned}$$

and thus this set must be in $\mathcal{M}$ itself. For the other direction, suppose $\mu(N) = 0$ for some $N \in \mathcal{M}$ and $K \subseteq N$. Let, $f = \chi_N$ and $g = \chi_K$, we see that $f = g$ on N^c and thus $f = g$ μ -a.e. which makes g measurable and thus $g^{-1}(\{1\}) = K$ must also be measurable making μ complete.

- b. For one direction assume that μ is complete. By [Problem 2.3](#) we know that the set where we have convergence, is measurable so let it be E . Further, a.e. convergence implies that $\mu(X \setminus E) = 0$. Let $g = \lim \chi_E f_n(x)$, we see that g is measurable on E as well as on E^c and is thus measurable on X , being the limit (it always exists) of measurable functions. Suppose, N is the set where $f_n \not\rightarrow f$, by our hypothesis this is of measure zero. For $x \in N^c$ however $f_n(x) \rightarrow f(x)$ and thus $f(x) = g(x)$ so $f(x) \neq g(x)$ can only happen on N implying that $f = g$ a.e after which part a proves measurability of f and we are done. For the other direction, suppose E is a set of measure zero and $K \subseteq E$. Let, $f = \chi_K$ and $f_n = 0$, with this we see that $f_n \rightarrow f$ on E^c which shows that $f_n \rightarrow f$ a.e. and thus f is measurable so we are done.

□

Problem 2.11. Suppose that f is a function on $\mathbb{R} \times \mathbb{R}^k$ such that $f(x, \cdot)$ is Borel measurable for each $x \in \mathbb{R}$ and $f(\cdot, y)$ is continuous for each $y \in \mathbb{R}^k$. For $n \in \mathbb{N}$, define f_n as follows. For $i \in \mathbb{Z}$ let $a_i = i/n$, and for $a_i \leq x \leq a_{i+1}$ let

$$f_n(x, y) = \frac{f(a_{i+1}, y)(x - a_i) - f(a_i, y)(x - a_{i+1})}{a_{i+1} - a_i}$$

Then f_n is Borel measurable on $\mathbb{R} \times \mathbb{R}^k$ and $f_n \rightarrow f$ pointwise; hence f is Borel measurable on $\mathbb{R} \times \mathbb{R}^k$. Conclude by induction that every function on $\mathbb{R}^n$ that is continuous in each variable separately is Borel measurable.

Solution. We can write $\mathbb{R} \times \mathbb{R}^k$ as a countable union of the borel sets $\{[a_i, a_{i+1}] \times \mathbb{R}^k\}_{i \in \mathbb{Z}}$ and proving borel measurability of f_n on these suffices to prove borel measurability globally. It being borel on $[a_i, a_{i+1}] \times \mathbb{R}^k$ follows from these maps being clearly borel

$$(x, y) \mapsto (x - a_i), (x, y) \mapsto (x - a_{i+1}), (x, y) \mapsto f(a_i, y), (x, y) \mapsto f(a_{i+1}, y)$$

as the borel measurability is preserved under products and sums. To prove pointwise convergence just notice that for $(x, y) \in [a_i, a_{i+1}] \times \mathbb{R}^k$, $f_n(x, y)$ is a convex combination of $f(a_i, y)$ and $f(a_{i+1}, y)$ which converge to $f(x, y)$ as $n \rightarrow \infty$ by continuity of $f(\cdot, y)$. This shows that if $f(\cdot, y)$ is continuous and $f(x, \cdot)$ borel, for all $(x, y) \in \mathbb{R} \times \mathbb{R}^k$ then f is borel. Now, suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous in each variable, suppose that the proposition holds for $n - 1$, then for all $x \in \mathbb{R}$, $f(x, \cdot) : y \in \mathbb{R}^{n-1} \mapsto f(x, y)$ is borel because this is continuous in each variable as well and for all $y \in \mathbb{R}^{n-1}$, $f(\cdot, y) : x \in \mathbb{R} \mapsto f(x, y)$ is continuous too, which implies that f is borel. We can induct on n from the base case of $n = 1$ where the proposition is true due to continuity in each variable coinciding with continuity (which is stronger than borel measurability), hence proving the proposition. $\square$

Problem 2.12. Prove Proposition 2.20. (See Proposition 0.20, where a special case is proved.)

Solution. If $f^{-1}(\{\infty\})$ is not null then the integral will trivially be infinite and thus it must be a null set. Now, consider the sets $H_n = f^{-1}(1/n, \infty]$ and we find that $\{x : f(x) > 0\} = \cup H_n$. Here, we see that $f \geq n^{-1} \chi_{H_n}$ for all $n \geq 1$ and thus $\int f \geq n^{-1} \mu(H_n)$ forces all the H_n to have finite measure. $\square$

Problem 2.13. Suppose $\{f_n\} \subset L^+$, $f_n \rightarrow f$ pointwise, and $\int f = \lim \int f_n < \infty$. Then $\int_E f = \lim \int_E f_n$ for all $E \in \mathcal{M}$. However, this need not be true if $\int f = \lim \int f_n = \infty$.

Solution. For any $E \in \mathcal{M}$ we can see that $f_n \chi_E \rightarrow f \chi_E$ pointwise as well and thus, by fatou's lemma and the additivity of the integral we have,

$$\begin{aligned} \int f \chi_E &\leq \liminf \int f_n \chi_E \text{ and } \int f \chi_{X \setminus E} \leq \liminf \int f_n \chi_{X \setminus E} \\ \implies \int f &= \int f \chi_E + \int f \chi_{X \setminus E} \leq \liminf \int f_n \chi_E + \liminf \int f_n \chi_{X \setminus E} \end{aligned}$$

Using the fact that the limit infimum is subadditive we see that the above is

$$\leq \liminf \left(\int f_n \chi_E + f_n \chi_{X \setminus E} \right) = \liminf \int f_n = \int f$$

as all the terms involved in this string of inequalities are finite (this follows from the fact that $\int f < \infty$ and that $\int f_n$ is close to $\int f$ eventually) we must have that,

$$\int f \chi_E = \liminf \int f_n \chi_E$$

Assume wlog $\int f_n$ is always finite, then by the superadditivity of the limit supremum and the equality we developed earlier, we have

$$\begin{aligned}\limsup \int f_n \chi_E &= \limsup \left(\int f_n - \int f_n \chi_{E^c} \right) \leq \limsup \int f_n + \limsup \left(- \int f_n \chi_{E^c} \right) \\ &= \limsup \int f_n - \liminf \int f_n \chi_{E^c} = \int f - \int f \chi_{E^c} = \int f \chi_E \\ \limsup \int f_n \chi_E &\leq \int f \chi_E = \liminf \int f_n \chi_E \implies \lim \int f_n \chi_E = \int f \chi_E\end{aligned}$$

For a counterexample we can consider a function on $\mathbb{R}$ given by $f_n = n\chi_{(0,1/n)} + \chi_{[1,\infty)}$ which converges pointwise to $\chi_{[1,\infty)}$ and $\int f = \lim \int f_n = \infty$ but if we focus on $(0,1)$ then $\lim \int f_n \chi_{(0,1)} = 1$ whereas $\int f \chi_{(0,1)} = 0$. Here the area escaped to infinity in the unit open interval and hence the contradiction. $\square$

Problem 2.14. If $f \in L^+$, let $\lambda(E) = \int_E f d\mu$ for $E \in \mathcal{M}$. Then λ is a measure on $\mathcal{M}$, and for any $g \in L^+$, $\int g d\lambda = \int gf d\mu$. (First suppose that g is simple.)

Solution. For any disjoint collection $\{E_n\}_{n \in \mathbb{N}} \subseteq \mathcal{M}$ with union E then by the sigma additivity of the integral we have,

$$\lambda(E) = \int f \chi_E d\mu = \int \sum_{n \in \mathbb{N}} f \chi_{E_n} d\mu = \sum_{n \in \mathbb{N}} \int f \chi_{E_n} d\mu = \sum_{n \in \mathbb{N}} \lambda(E_n)$$

Now, suppose $\phi = \sum a_j \chi_{E_j}$ is a simple function, then we see that

$$\int \phi d\lambda = \sum a_j \lambda(E_j) = \sum a_j \int f \chi_{E_j} d\mu = \int \sum a_j f \chi_{E_j} d\mu = \int f \phi d\mu$$

Now, if $g \in L^+$, we can find a sequence of simple functions $\{\phi_n\}$ that increase to g , for these we also see that $f\phi_n$ increases to fg and by the monotone convergence theorem and the fact that λ and μ are defined on the same sigma algebra $\mathcal{M}$ we have that,

$$\int fg d\mu = \int \lim f \phi_n d\mu \longleftarrow \int f \phi_n d\mu = \int \phi_n \lambda \longrightarrow \int \lim \phi_n \lambda = \int g \lambda$$

where we used the the monotone convergence theorem on both sides. $\square$

Problem 2.15. If $\{f_n\} \subset L^+$, f_n decreases pointwise to f , and $\int f_1 < \infty$, then $\int f = \lim \int f_n$.

Solution. Clearly we have $\int f \leq \lim \int f_n < \infty$. For the other direction, consider the measurable set $E = \{f > 0\}$. Fix $\alpha > 1$ and define $F_n := \{x \in E : f_n(x) \leq \alpha f(x)\}$, then $\{F_n\}$ is an increasing sequence of measurable sets and $\cup F_n = E$. From the definition we see that $\int_{F_n} f_n \leq \alpha \int_{F_n} f$ and $\int_E f_n - \int_{F_n} f_n = \int_{E \setminus F_n} f_n \leq \int_{E \setminus F_n} f_1$. As $A \mapsto \int_A f_1$ is a measure on $\mathcal{M}$ and $E \setminus F_n$ is a decreasing sequence with $\int_{E \setminus F_1} f_1 \leq \int f_1 < \infty$ we see that

$$\lim_{n \rightarrow \infty} \int_{F \setminus F_n} f_1 = \int_{\cap E \setminus F_n} f_1 = \int_{E \setminus E} f_1 = 0$$

this implies that $\lim \int_{F_n} f_n = \lim \int_E f_n$. As $A \mapsto \int_A f$ is also a measure, we have

$$\lim \int_{F_n} f_n = \lim \int_E f_n \implies \lim \int_E f_n \leq \lim \alpha \int_{F_n} f = \alpha \lim \int_{F_n} f = \alpha \int_{\cup F_n} f = \int_E f$$

As this is true for any $\alpha > 1$ we have $\lim \int_E f_n \leq \int_E f$. Now, fix $\alpha > 0$ and consider E^c on which f is zero and thus f_n decreases to 0 here. Let $G_n := \{x \in E^c : f_n(x) \leq \alpha f_1(x)\}$ see that $\{G_n\}$ is an increasing sequence of measurable sets with $\cup G_n = E^c$. It can also be shown that $\lim \int_{G_n} f_n = \lim \int_{E^c} f_n$ using similar arguments as we did when working over E and from the definition we see that $\int_{G_n} f_n \leq \alpha \int_{G_n} f_1 \leq \alpha \int f_1$ and thus, $\lim \int_{E^c} f_n = 0 = \int_{E^c} f$ as $\alpha > 0$ was arbitrary. Combining these two deductions, namely $\lim \int_E f_n \leq \int_E f$ and $\lim \int_{E^c} f_n = 0 = \int_{E^c} f$ we see that $\lim \int f_n \leq \int f$ as the integral is additive. Having shown both directions of the inequality we must have $\int f = \lim \int f_n$. $\square$

Problem 2.16. If $f \in L^+$ and $\int f < \infty$, for every $\epsilon > 0$ there exists $E \in \mathcal{M}$ such that $\mu(E) < \infty$ and $\int_E f > (\int f) - \epsilon$.

Solution. From [Problem 2.12](#) we see that the set $E = \{x : f(x) > 0\}$ is sigma finite so we write it as a disjoint(standard technique) union of countably many sets of finite measure as $E = \cup F_n$, now from the additivity of the integral we have,

$$\int f = \int f \chi_E + \underbrace{\int f \chi_{E^c}}_{=0} = \int_E f = \sum_{n \in \mathbb{N}} \int_{F_n} f = \lim_n \sum_{k=1}^n \int_{F_n} f = \lim_n \int_{\cup_{k \leq n} F_k} f$$

Now, for any $\epsilon > 0$ we can find a N such that $\int f < \epsilon + \int_{\cup_{k \leq N} F_k} f$ and as $\mu(\cup_{k \leq N} F_k) = \sum_{k \leq N} \mu(F_k) < \infty$ we are done. $\square$

Problem 2.17. Assume Fatou's lemma and deduce the monotone convergence theorem from it.

Solution. Fatou's lemma implies that $\lim \int f_n = \liminf \int f_n \geq \int \liminf f_n = \int f$ while the other direction of the inequality is trivial. $\square$

Problem 2.18. Fatou's lemma remains valid if the hypothesis that $f_n \in L^+$ is replaced by the hypothesis that f_n is measurable and $f_n \geq -g$ where $g \in L^+ \cap L^1$. What is the analogue of Fatou's lemma for nonpositive functions?

Solution. Clearly $f_n + g \in L^+$ so we can apply fatou's lemma on it and get

$$\begin{aligned} \int g + \liminf \int f_n &= \liminf \left(\int f_n + \int g \right) = \liminf \int g + f_n \\ &\geq \int \liminf (f_n + g) = \int g + \liminf f = \int g + \int \liminf f_n \end{aligned}$$

and now we can just cancel out the $\int g$ terms. For nonpositive functions f ($-f \in L^+$) we will have that

$$\limsup \int f_n \leq \int \limsup f_n$$

$\square$

Problem 2.19. Suppose $\{f_n\} \subset L^1(\mu)$ and $f_n \rightarrow f$ uniformly.

- If $\mu(X) < \infty$, then $f \in L^1(\mu)$ and $\int f_n \rightarrow \int f$.
- If $\mu(X) = \infty$, the conclusions of (a) can fail. (Find examples on $\mathbb{R}$ with Lebesgue measure.)

Solution.

- This is trivial.
- consider $f_n = n^{-1}\chi_{[0,n^2]}$ on $\mathbb{R}$ with respect to the lebesgue measure.

□

Problem 2.20. (A generalized Dominated Convergence Theorem) If $f_n, g_n, f, g \in L^1$, $f_n \rightarrow f$ and $g_n \rightarrow g$ a.e., $|f_n| \leq g_n$, and $\int g_n \rightarrow \int g$, then $\int f_n \rightarrow \int f$. (Rework the proof of the dominated convergence theorem.)

Solution. We see that $g_n + f_n, g_n - f_n \in L^1$ and also as $f_n \rightarrow f, g_n \rightarrow g$ a.e. we can restrict the integrals to a set E where convergence is global such that $\int = \int_E$ (as operators) and we work inside E now without mention. By fatou's lemma

$$\begin{aligned} \int g + \liminf \int f_n &= \liminf \int f_n + \lim \int g_n = \liminf \left(\int f_n + \int g_n \right) = \liminf \int f_n + g_n \\ &\geq \int \liminf (g_n + f_n) \geq \int \liminf g + \liminf f_n = \int f + g = \int f + \int g \end{aligned}$$

cancelling out the $\int g$ terms we get $\liminf \int f_n \geq \int f$ and doing the same with $g_n - f_n$ we get $-\limsup \int f_n = \liminf \int -f_n \geq \int -f = -\int f$ which implies that $\int f \geq \limsup \int f$ and thus finally we have $\limsup \int f_n \leq \int f \leq \liminf \int f_n \implies \lim \int f_n = \int f$. □

Problem 2.21. Suppose $f_n, f \in L^1$ and $f_n \rightarrow f$ a.e. Then $\int |f_n - f| \rightarrow 0$ iff $\int |f_n| \rightarrow \int |f|$. (Use Exercise 20.)

Solution. For the if direction we can use generalised dominated convergence (Problem 2.20) on $|f_n - f| \rightarrow 0$ a.e. where it is dominated by $|f_n| + |f| \rightarrow 2|f|$ a.e. as $\int |f_n| + |f| = \int |f| + \int |f_n| \rightarrow 2 \int |f| = \int 2|f|$ and for the only if direction we can proceed by the reverse triangle inequality as follows

$$\left| \int |f_n| - \int |f| \right| \leq \int ||f_n| - |f|| \leq \int |f_n - f| \rightarrow 0$$

□

Problem 2.22. Let μ be counting measure on $\mathbb{N}$. Interpret Fatou's lemma and the monotone and dominated convergence theorems as statements about infinite series.

Solution. A sequence of functions $\{f_k\}$ on $\mathbb{N}$ corresponds to a double sequence $a_{k,n} = f_k(n)$. Pointwise convergence $f_k \rightarrow f$ a.e. means that for each fixed n , $\lim_{k \rightarrow \infty} a_{k,n} = a_n$ (since the only set of measure zero for the counting measure is the empty set). With this we can interpret the three theorems as follows:

Fatou's Lemma: If $f_k \in L^+$, then $\int \liminf f_k \leq \liminf \int f_k$. In series notation: If $a_{k,n} \geq 0$ for all $k, n \in \mathbb{N}$, then

$$\sum_{n=1}^{\infty} \left(\liminf_{k \rightarrow \infty} a_{k,n} \right) \leq \liminf_{k \rightarrow \infty} \sum_{n=1}^{\infty} a_{k,n}$$

Monotone Convergence Theorem: If $\{f_k\} \subset L^+$, $f_k \leq f_{k+1}$, and $f_k \rightarrow f$, then $\lim \int f_k = \int f$. In series notation: If $0 \leq a_{1,n} \leq a_{2,n} \leq \dots$ for all n , and $\lim_{k \rightarrow \infty} a_{k,n} = a_n$, then

$$\lim_{k \rightarrow \infty} \sum_{n=1}^{\infty} a_{k,n} = \sum_{n=1}^{\infty} a_n$$

Dominated Convergence Theorem: If $f_k \rightarrow f$ pointwise, and $|f_k| \leq g$ where $g \in L^1(\mu)$, then $\lim \int f_k = \int f$. In series notation: If $\lim_{k \rightarrow \infty} a_{k,n} = a_n$ for each n , and there exists a bounding sequence $\{b_n\}$ such that $|a_{k,n}| \leq b_n$ for all k, n , and $\sum_{n=1}^{\infty} b_n < \infty$, then

$$\lim_{k \rightarrow \infty} \sum_{n=1}^{\infty} a_{k,n} = \sum_{n=1}^{\infty} a_n$$

□

Problem 2.23. Given a bounded function $f : [a, b] \rightarrow \mathbb{R}$, let

$$H(x) = \lim_{\delta \rightarrow 0} \sup_{|y-x| \leq \delta} f(y), \quad h(x) = \lim_{\delta \rightarrow 0} \inf_{|y-x| \leq \delta} f(y).$$

Prove Theorem 2.28b by establishing the following lemmas:

- $H(x) = h(x)$ iff f is continuous at x .
- In the notation of the proof of Theorem 2.28a, $H = G$ a.e. and $h = g$ a.e. Hence H and h are Lebesgue measurable, and $\int_{[a,b]} H \, dm = \bar{I}_a^b(f)$ and $\int_{[a,b]} h \, dm = \underline{I}_a^b(f)$.

Solution.

- This is trivial.
- If it was continuous outside a null set then we can show riemann integrability by considering partitions of form $P = \{a_1 < b_1 < \dots < a_n < b_n\} \cup P_\delta$ where, by our assumptions $m(E = \cup(a_i, b_i)) = \epsilon$ is small such that f is continuous on the compact set $[a, b] \setminus E$ and P_δ is of small enough mesh such as to make $U(f|_{[a,b] \setminus E}, P_\delta) - L(f|_{[a,b] \setminus E}) < \delta$. Now, as f is bounded there is some M such that $|f| \leq M$ then

$$U(f, P) - L(f, P) \leq 2M\epsilon + \delta$$

which can be made arbitrarily small so f must be integrable. Now we will prove the converse. If f is integrable then there exists an increasing sequence of finite partitions P_k such that $\|P_k\| \rightarrow 0$ and $\lim \int G_{P_k} dm = \int f = \lim \int g_{P_k} dm$. It was already shown in the book that $g = G$ a.e. when f is riemann integrable so if we show that $H = G$ a.e. and $h = g$ a.e. then we would be done using part (a) as this implies $H = h$ a.e. which is equivalent to continuity a.e. by part (a). We will show that $H = G$ a.e. and the rest will follow similarly. Let K be the set of all endpoints of some interval in P_k for some k , this is a null set and we will show that $H = G$ outside it. Say $x \in [a, b] \setminus K$ is any point then we first show that $H \geq G$.

First define $H_\delta(x) := \sup_{|x-y| \leq \delta} f(y)$ then we see that $H = \lim_{\delta \rightarrow 0} H_\delta$ by definition and $H_\delta \leq H_{\delta'}$ for all $0 \leq \delta \leq \delta'$. Now, for any $\delta > 0$ we can find some k such that $x \in (a_k, b_k) \subseteq (x - \delta, x + \delta)$ where (a_k, b_k) is an interval in P_k as $\|P_k\| \rightarrow 0$. Then we must have that $G_{P_k}(x) \leq H_\delta(x)$ and as G_{P_k} is decreasing we have that $G(x) \leq H_\delta(x)$ and then taking the limit as $\delta \searrow 0$ we have $G(x) \leq H(x)$. To show the other direction of the equality we notice that for any P_k with the interval (a_k, b_k) containing x within it we can find some $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq (a_k, b_k)$ and this implies that $G_{P_k}(x) \geq H_\delta(x)$ and as $H_\delta(x)$ is decreasing in delta, taking the limit we see that $G_{P_k}(x) \geq H(x)$ and again taking the limit as $k \rightarrow \infty$ we see that $G \geq H$ so we are done. $\square$

Problem 2.24. Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) < \infty$, and let $(X, \overline{\mathcal{M}}, \overline{\mu})$ be its completion. Suppose $f : X \rightarrow \mathbb{R}$ is bounded. Then f is $\overline{\mathcal{M}}$ -measurable (and hence in $L^1(\overline{\mu})$) iff there exist sequences $\{\phi_n\}$ and $\{\psi_n\}$ of $\mathcal{M}$ -measurable simple functions such that $\phi_n \leq f \leq \psi_n$ and $\int(\psi_n - \phi_n) d\mu < n^{-1}$. In this case, $\lim \int \phi_n d\mu = \lim \int \psi_n d\mu = \int f d\overline{\mu}$.

Solution. To show the if direction we can consider the functions $g := \sup_n \phi_n$ and $h := \inf_n \psi_n$ then these are $\mathcal{M}$ -measurable and we have that $\int g - h < 1/n$ for all n and thus it is zero which makes $h = g$ a.e. and thus $f = g = h$ μ -a.e. Now, If h, g are $\mathcal{M}$ measurable then they are clearly $\overline{\mathcal{M}}$ measurable as well and as $\mathcal{M} \subseteq \overline{\mathcal{M}}$ we see that $\overline{\mu}$ -a.e. too so we must have that $h = f = g$ $\overline{\mu}$ -a.e as well. which implies that f is $\overline{\mathcal{M}}$ measurable. For the converse, we first deal with the case where $f \geq 0$ consider the following simple functions which are $\overline{\mathcal{M}}$ measurable,

$$\phi_n^* := \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \cdot \chi_{f^{-1}([k2^{-n}, (k+1)2^{-n}))}$$

$$\psi_n^* := \sum_{k=0}^{2^{2n}-1} \frac{k+1}{2^n} \cdot \chi_{f^{-1}([k2^{-n}, (k+1)2^{-n}))}$$

Then, for all large enough n we have that $\psi_n^* \geq f \geq \phi_n^*$ and $0 \leq f - \phi_n^*, \psi_n^* - f \leq 2^{-n}$. As f is bounded we can find some constant M with $f \leq M$. For brevity write

$$\phi_n^* = \sum_{k=1}^m a_k \chi_{E_k \cup F_k} \text{ where } E_i \cup F_i \text{ are disjoint } E_i \in \mathcal{M}, F_i \subseteq N_i \in \mathcal{M} \text{ with } \mu(N_i) = 0$$

Then define, $\phi_n := \sum_{k=1}^m a_k \chi_{E_k}$ and we have $\phi_n \leq \phi_n^*$ with equality $\overline{\mu}$ a.e. Again writing ψ_n^* in the exact same notation with $N = \cup N_i \in \mathcal{M}$ we see that

$$\psi_n := \chi_{N^c} \psi_n^* + (M+1) \chi_N = (M+1) \chi_N + \sum_{k=1}^m a_k \chi_{E_k \cap N^c}$$

is such that $\psi_n \geq \psi_n^*$ with equality with equality $\overline{\mu}$ a.e. (We use $M+1$ instead of M to guarantee the inequality, we need to do this due to how $\psi_n^* \leq f + 2^{-n} \leq M+1$ for large enough n). We now have $\mathcal{M}$ measurable simple functions ψ_n, ϕ_n such that $\psi_n \geq f \geq \phi_n$ and also $\psi_n = \psi_n^*$ and $\phi_n = \phi_n^*$ $\overline{\mu}$ -a.e. (both together), from this we can say that $\psi_n - f, f - \phi_n \leq 2^{-n}$ together $\overline{\mu}$ -a.e. implying $\psi_n - \phi_n \leq 2^{-n+1}$ $\overline{\mu}$ -a.e. and thus,

$$\int \psi_n - \phi_n d\overline{\mu} = \int \psi_n - \phi_n d\mu \leq \int 2^{-n+1} d\mu = \frac{\mu(X)}{2^{-n+1}} \rightarrow 0$$

and we can start from a larger index to have $\int \psi_n - \phi_n < n^{-1}$. In the previous line we assumed the following: if g is some $\mathcal{M}$ measurable function then $\int g d\mu = \int g d\bar{\mu}$, this is easy to prove and we omit the proof. We also see that $\phi_n \leq f \leq \phi_n + 2^{-n}$ $\bar{\mu}$ -a.e. for large enough n and thus for large n we have $\int \phi_n d\mu \leq \int f d\bar{\mu} \leq \mu(X) \cdot 2^{-n} + \int \phi_n d\mu$ using the assumption from earlier along the way which implies that $\lim \int \phi_n d\mu = \int f d\bar{\mu}$. The same can be proved for ψ_n similarly so we are done with the proof for $f \geq 0$. Now, for general such f we first split it into its measurable (and bounded in this case positive and negative parts, in usual notation $f = f_+ - f_-$. Let ϕ_n^+, ψ_n^+ be as in the previous case, for f_+ and ϕ_n^-, ψ_n^- be those for f_- , then we see that $\phi_n^+ - \psi_n^- \leq f_+ - f_- \leq \psi_n^+ - \phi_n^-$. As $|\phi_n^+ - \psi_n^- - (f_+ - f_-)| \leq |\phi_n^+ - f_+| + |\psi_n^- - f_-| \leq 2^{-n} + 2^{-n}$ $\bar{\mu}$ -a.e for large enough n implying

$$\phi_n^+ - \psi_n^- \leq f_+ - f_- \leq \phi_n^+ - \psi_n^- + 2^{-n+1}$$

and we can get the corresponding bounds for $\psi_n^+ - \phi_n^-$ as well and finish like in the previous case using the $\mathcal{M}$ measurable simple functions $\zeta_n := \phi_n^+ - \psi_n^-$ and $\gamma_n := \psi_n^+ - \phi_n^-$ with $\zeta_n \leq f \leq \gamma_n$ and $\gamma_n - \zeta_n \leq 2^{-n+2}$ holding $\bar{\mu}$ -a.e. □

Problem 2.25. Let $f(x) = x^{-1/2}$ if $0 < x < 1$, $f(x) = 0$ otherwise. Let $\{r_n\}_1^\infty$ be an enumeration of the rationals, and set $g(x) = \sum_1^\infty 2^{-n} f(x - r_n)$.

- a. $g \in L^1(m)$, and in particular $g < \infty$ a.e.
- b. g is discontinuous at every point and unbounded on every interval, and it remains so after any modification on a Lebesgue null set.
- c. $g^2 < \infty$ a.e., but g^2 is not integrable on any interval.

Solution.

- a. It follows from the monotone convergence theorem and the fact that the lebesgue integral and riemann integral agree that $\int g dm = \sum_{n=1}^\infty \int 2^{-n} f(x - r_n) dm = \sum_{n=1}^\infty 2^{-n} \cdot \int_0^1 x^{-1/2} dx = 2$ implying $g \in L^1(m)$ which again forces $\{g \geq \infty\}$ to be null.
- b. given any real y , if $r_n < y < 1 + r_n$ then we see that $g(y) > 2^{-n}(y - r_n)^{-1/2}$. Now, given any $x \in \mathbb{R}$ and small enough $\delta > 0$ we can find some r_n such that $x - \delta < r_n < x < x + \delta < r_n + 1$ and thus, for all small enough $\epsilon > 0$ with $r_n < r_n + \epsilon < x$ we see that $g(r_n + \epsilon) > 2^{-n}\epsilon^{-2}$ which can be made arbitrarily large, this shows that g is unbounded in any open neighbourhood around x from which the claim follows.
- c. As $g < \infty$ a.e. we clearly have $g^2 < \infty$ a.e. as well. With the same setup as the previous part, we see that $g(y)^2 > 4^{-n}(y - r_n)^{-1}$ and thus

$$\int_{x-\delta}^{x+\delta} g^2 dm \geq \int_{r_n}^x \frac{1}{2^n(y - r_n)} dm(y) = \infty$$

thus, its not integrable on any interval of form $(x - \delta, x + \delta)$ proving the claim after noticing how all intervals have an interval of this form. □

Problem 2.26. If $f \in L^1(m)$ and $F(x) = \int_{-\infty}^x f(t) dt$, then F is continuous on $\mathbb{R}$.

Solution. First we prove right continuity using the sequential criterion. Say $x_n \searrow x$, then as $A \mapsto \int \chi_A |f|$ is a finite measure on $\mathcal{L}$ we see that

$$\lim |F(x_n) - F(x)| = \lim \left| \int_{[x, x_n]} f dm \right| \leq \lim \int_{[x, x_n]} |f| dm = \int_{\cap [x, x_n]} |f| dm = 0$$

as $\cap [x, x_n] = \{x\}$ is a lebesgue null set and we follow the convention of $0 \cdot \infty = 0$. Right continuity is proven similarly and hence we have continuity. $\square$

Problem 2.27. Let $f_n(x) = ae^{-nax} - be^{-nbx}$ where $0 < a < b$.

- a. $\sum_1^\infty \int_0^\infty |f_n(x)| dx = \infty$.
- b. $\sum_1^\infty \int_0^\infty f_n(x) dx = 0$.
- c. $\sum_1^\infty f_n \in L^1([0, \infty), m)$, and $\int_0^\infty \sum_1^\infty f_n(x) dx = \log(b/a)$.

Solution. This is just a bunch of calculations. However it furnishes an example to how how $\int \sum |f_n| \notin L^1$ might result in $\int \sum f_n \neq \sum \int f_n$ even whe both are finite and $\sum f_n \in L^1$ $\square$

Problem 2.28. Compute the following limits and justify the calculations:

- a. $\lim_{n \rightarrow \infty} \int_0^\infty (1 + (x/n))^{-n} \sin(x/n) dx$.
- b. $\lim_{n \rightarrow \infty} \int_0^1 (1 + nx^2)(1 + x^2)^{-n} dx$.
- c. $\lim_{n \rightarrow \infty} \int_0^\infty n \sin(x/n)[x(1 + x^2)]^{-1} dx$.
- d. $\lim_{n \rightarrow \infty} \int_a^\infty n(1 + n^2 x^2)^{-1} dx$. (The answer depends on whether $a > 0$, $a = 0$, or $a < 0$. How does this accord with the various convergence theorems?)

Solution.

- a. Let, $f_n(x) = (1 + x/n)^{-n} \sin(x/n)$ then it converges $e^{-x} \cdot \sin(0) = 0$ pointwise and it can be checked that $|f_n| \leq (1 + x/2)^{-2}$ for all $n \geq 2$ and as this is L^1 , it allows us to conclude that the limit is 0 by DCT.
- b. Let $f_n(x) = (1 + nx^2)/(1 + x^2)^n$ then it converges to $\chi_{\{1\}}$ pointwise and as $|f_n| \leq 1$ in the interval $[0, 1]$ we can apply DCT to conclude that the limit is 0
- c. In a similar fashion we see that the integrands are dominated by $(1 + x^2)^{-1}$ and converge pointwise to it as well after which we can conclude with DCT that the limit is $\pi/2$ using basic calculus.
- d. If $a > 0$ then we see that the integrands are eventually dominated by $(1 + x^2)^{-1}$ as $1/n + nx^2 \geq 1 + x^2 \iff x^2 > 1/n$ which is always true in our case for large enough n allowing us to apply DCT and conclude that the limit is zero as the integrands converge to 0 pointwise. However, if $a \leq 0$ then the integrands converge to $\infty \cdot \chi_{\{0\}}$ pointwise which integrates to 0 whereas the limit is that of the terms $\pi/2 - \arctan(na)$ which has a strictly positive limit here for the $a \leq 0$ we

consider. This shows us that DCT *can't be applied* here, in fact it is easily proven without the DCT that any function which dominates this sequence can not be L^1 by considering the values on intervals of form $[1/n, 1/(n-1)]$ which is left to the reader as an exercise.

□

Problem 2.29. Show that $\int_0^\infty x^n e^{-x} dx = n!$ by differentiating the equation $\int_0^\infty e^{-tx} dx = 1/t$. Similarly, show that $\int_{-\infty}^\infty x^{2n} e^{-x^2} dx = (2n)! \sqrt{\pi}/4^n n!$ by differentiating the equation $\int_{-\infty}^\infty e^{-tx^2} dx = \sqrt{\pi/t}$ (see Proposition 2.53).

Solution. Let $f : [1, \infty) \times [0, \infty) \rightarrow \mathbb{R}$ be defined as $f(t, x) = e^{-tx}$, we see that for all t, x we have

$$\left| \frac{\partial}{\partial t} f(x, t) \right| = x e^{-tx} \leq x e^{-x}$$

which is L^1 allowing us to use Theorem 2.27 (page 56) to deduce that

$$\frac{-1}{t^2} = \frac{d}{dt} \int_0^\infty e^{-tx} dx = \int_0^\infty \frac{\partial}{\partial t} f(x, t) dx = \int_0^\infty -x e^{-tx} dx$$

repeating this a total of n times with the dominating functions being $x^n e^{-x}$ we get the desired conclusion after setting $t = 1$ at the end. For the second parts we work with t over the same interval but this time x varies over $\mathbb{R}$ with $f(t, x) = e^{-tx^2}$ and following the exact same steps with dominating functions of form $x^{2n} e^{-x^2}$ we arrive at a proof. □

Problem 2.30. Show that $\lim_{k \rightarrow \infty} \int_0^k x^n (1 - k^{-1}x)^k dx = n!$.

Solution. The integrands $\chi_{[0, k]} x^n (1 - x/k)^k$ are dominated by $x^n e^{-x}$ which is L^1 and also the pointwise limit of the integrands. Applying DCT and using the previous problem we arrive at the required conclusion. □

Problem 2.31. Derive the following formulas by expanding part of the integrand into an infinite series and justifying the term-by-term integration. Exercise 29 may be useful. (Note: In (d) and (e), term-by-term integration works, and the resulting series converges, only for $a > 1$, but the formulas as stated are actually valid for all $a > 0$.)

- For $a > 0$, $\int_0^\infty e^{-x^2} \cos ax dx = \sqrt{\pi} e^{-a^2/4}$.
- For $a > -1$, $\int_0^1 x^a (1-x)^{-1} \log x dx = \sum_1^\infty (a+k)^{-2}$.
- For $a > 1$, $\int_0^\infty x^{a-1} (e^x - 1)^{-1} dx = \Gamma(a) \zeta(a)$, where $\zeta(a) = \sum_1^\infty n^{-a}$.
- For $a > 1$, $\int_0^\infty e^{-ax} x^{-1} \sin x dx = \arctan(a^{-1})$.
- For $a > 1$, $\int_0^\infty e^{-ax} J_0(x) dx = (a^2 + 1)^{-1/2}$, where $J_0(x) = \sum_0^\infty (-1)^n x^{2n} / 4^n (n!)^2$ is the Bessel function of order zero.

Solution. All of these are very easily proven by expanding some function into its power series representation and then using DCT (with the dominating function being of form $\sum |f_n|$ for functions of form $\sum f_n$) to switch the summation and integration so we state what we expand for each.

- a. expand the cosine term and use the previous problem (we set $t = 1$ at the end, here we require the result without that substitution.)
- b. expand the denominator.
- c. Rewrite the integrand as

$$\frac{x^{a-1}}{e^x - 1} = \frac{x^{a-1}e^{-x}}{1 - e^{-x}} = \sum_{n=0}^{\infty} x^{a-1}e^{-x}(e^{-x})^n$$

- d. expand the sine term and use the previous problem.
- e. this is already given as a power series

□

Problem 2.32. Suppose $\mu(X) < \infty$. If f and g are complex-valued measurable functions on X , define

$$\rho(f, g) = \int \frac{|f - g|}{1 + |f - g|} d\mu.$$

Then ρ is a metric on the space of measurable functions if we identify functions that are equal a.e., and $f_n \rightarrow f$ with respect to this metric iff $f_n \rightarrow f$ in measure.

Solution. For the if part, first notice that $x \mapsto x/(1+x)$ is an increasing map on the nonnegatives and thus we have for any arbitrary $\epsilon > 0$ that,

$$\begin{aligned} \int_X \frac{|f - f_n|}{1 + |f - f_n|} d\mu &= \int_{|f - f_n| \geq \epsilon} \frac{|f - f_n|}{1 + |f - f_n|} d\mu + \int_{|f - f_n| < \epsilon} \frac{|f - f_n|}{1 + |f - f_n|} d\mu \\ &\leq 1 \cdot \mu(\{|f_n - f| \geq \epsilon\}) + \frac{\epsilon}{1 + \epsilon} \cdot \mu(\{|f_n - f| < \epsilon\}) \leq \mu(\{|f_n - f| \geq \epsilon\}) + \epsilon \cdot \mu(X) \end{aligned}$$

And as $f_n \rightarrow f$ in measure we see that $\lim \rho(f_n, f) \leq \epsilon \mu(X)$ which can be made arbitrarily small proving this direction. For the other direction we similarly have that

$$\rho(f_n, f) \geq \frac{\epsilon}{1 + \epsilon} \cdot \mu(\{|f_n - f| \geq \epsilon\})$$

As $\rho(f_n, f) \rightarrow 0$ we must have the right side of the inequality tend to 0 proving the claim. □

Problem 2.33. If $f_n \geq 0$ and $f_n \rightarrow f$ in measure, then $\int f \leq \liminf \int f_n$.

Solution. By the definition of the limit infimum we can find a subsequence f_{n_k} such that $\liminf \int f_n = \lim \int f_{n_k}$. As f_n converges in measure to f so does f_{n_k} and we can find a subsequence $f_{n_{k_m}}$ that converges a.e. to f . Now, we see that

$$\liminf_n \int f_n = \lim_k \int f_{n_k} \limsup_k \int f_{n_k} \geq \limsup_m \int f_{n_{k_m}} \geq \liminf_m \int f_{n_{k_m}} \geq \int \liminf f_{n_{k_m}}$$

where we use fatou's lemma for the last inequality. As $f_{n_{k_m}} \rightarrow f$ a.e., the last expression is just $\int f$ so we are done. □

Problem 2.34. Suppose $|f_n| \leq g \in L^1$ and $f_n \rightarrow f$ in measure.

- a. $\int f = \lim \int f_n$.
- b. $f_n \rightarrow f$ in L^1 .

Solution.

- a. By [Problem 1.33](#) we can use fatou's lemma just like we did in the proof for DCT here as well by considering the sequences $g - f_n, g + f_n$ which are nonnegative and converge to $g - f, g + f$ respectively in measure.
- b. By part (a) and the assumption we clearly see that f_n, f are L^1 and as $|f_n| \rightarrow |f|$ in measure too while being dominated by g we can apply part (a) again to see that $\int |f| = \lim \int |f_n|$ which is equivalent to the claim by [Problem 1.21](#).

enumerate □

Problem 2.35. $f_n \rightarrow f$ in measure iff for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\mu(\{x : |f_n(x) - f(x)| \geq \epsilon\}) < \epsilon$ for $n \geq N$.

Solution. The only if direction follows directly from the definition and for the if direction, given $\epsilon > 0$ and $\delta > 0$ let, $\nu = \min\{\epsilon, \delta\}$ we see that $\{|f_n - f| \geq \epsilon\} \subseteq \{|f_n - f| \geq \nu\}$ and we can find some $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\mu(\{|f_n - f| \geq \epsilon\}) \leq \mu(\{|f_n - f| \geq \nu\}) < \nu \leq \delta$$

and we are done. □

Problem 2.36. If $\mu(E_n) < \infty$ for $n \in \mathbb{N}$ and $\chi_{E_n} \rightarrow f$ in L^1 , then f is (a.e. equal to) the characteristic function of a measurable set.

Solution. As the convergence is L^1 we can find a subsequence $\chi_{E_{n_k}}$ that converges to f outside a set of measure 0. Outside this set we clearly have that f is a characteristic function and its also measurable restricted to this set being the limit of measurable functions, thus it is a.e. equal to a characteristic function of a measurable set. □

Problem 2.37. Suppose that f_n and f are measurable complex valued functions and $\phi : \mathbb{C} \rightarrow \mathbb{C}$.

- a. If ϕ is continuous and $f_n \rightarrow f$ a.e., then $\phi \circ f_n \rightarrow \phi \circ f$ a.e.
- b. If ϕ is uniformly continuous and $f_n \rightarrow f$ uniformly, almost uniformly, or in measure, then $\phi \circ f_n \rightarrow \phi \circ f$ uniformly, almost uniformly, or in measure, respectively.
- c. There are counter examples when the continuity assumptions on ϕ are not satisfied

Solution.

- a. This is obvious from the fact that composition by continuous functions conserve pointwise convergence which is clear from the definition of continuity.
- b. It is a fact from basic real analysis that composition with uniformly continuous functions conserves uniform convergence and in the case of almost uniform convergence we just restrict ourselves to a suitable set. In the case of convergence in

measure, first choose $\delta > 0$ such that $|x - y| < \delta \implies |\phi(x) - \phi(y)| < \epsilon$, then we see that,

$$\{x \in X : |f_n(x) - f(x)| < \delta\} \subseteq \{x \in X : |\phi(f_n(x)) - \phi(f(x))| < \epsilon\}$$

and taking complements,

$$\{x \in X : |f_n(x) - f(x)| \geq \delta\} \supseteq \{x \in X : |\phi(f_n(x)) - \phi(f(x))| \geq \epsilon\}$$

Thus, we see that,

$$0 \longleftarrow \mu(\{x \in X : |f_n(x) - f(x)| \geq \delta\}) \geq \mu(\{x \in X : |\phi(f_n(x)) - \phi(f(x))| \geq \epsilon\})$$

proving this claim.

- c. A counterexample for part (a) if continuity of ϕ is not assumed is: $X = \mathbb{R}$ with the lebesgue measure, $\phi(x) = 1 - \chi_{\{0\}}$ and $f_n(x) = 1/n \rightarrow 0$. For part (b) in the case of uniform and almost uniform convergence is, in the same measurable space as the previous is: $f_n(x) = x + 1/n \rightarrow x$ and $\phi(x) = x^2$, this is also a counterexample for the case of convergence in measure.

□

Problem 2.38. If $f_n \rightarrow f$ in measure and $g_n \rightarrow g$ in measure, then $f_n + g_n \rightarrow f + g$ in measure, and $f_n g_n \rightarrow f g$ in measure if $\mu(X) < \infty$, but not necessarily if $\mu(X) = \infty$.

Solution. The first part follows from the fact that

$$\{x \in X : |f_n + g_n - (f + g)| \geq \epsilon\} \subseteq \{x \in X : |f_n - f| \geq \epsilon\} \cup \{x \in X : |g_n - g| \geq \epsilon\}$$

from the triangle inequality. For the second part it suffices to prove that if $\psi_n \rightarrow 0$ in measure then $\psi_n \phi \rightarrow 0$ in measure for any measurable ϕ as,

$$|f_n g_n - f g| \leq |f_n - f| |g_n| + |f| |g_n - g| \leq |g| |f_n - f| + |g - g_n| |f - f_n| + |f| |g_n - g|$$

due to the middle term converging to 0 in measure, which follows from the set inclusion below. For any $\epsilon > 0$

$$\{|f_n - f| |g_n - g| \geq \epsilon\} \subseteq \{|f_n - f| \geq \sqrt{\epsilon}\} \cup \{|g_n - f| \geq \sqrt{\epsilon}\}$$

After which we can apply the previous result that convergence in measure is preserved under addition. Now consider $S_n := \{x \in X : n \leq |\phi| < n + 1\}$. Then $\{S_n\}_{n \geq 0}$ is a partition of X . We see that

$$\sum_{n \geq 0} \mu(S_n) = \mu(X) < \infty$$

and thus for any $\epsilon > 0$ there exists some m_ϵ such that

$$\sum_{n > m_\epsilon} \mu(S_n) < \epsilon/2$$

Implied that,

$$\mu(\{x \in X : |\phi \psi_k| \geq \epsilon\}) < \sum_{n=0}^{m_\epsilon} \mu(\{x \in S_n : |\psi_k \phi| \geq \epsilon\}) + \epsilon/2$$

From construction, for $k = 0, 1, \dots, m_\epsilon$ we have that

$$\{x \in S_n : |\phi\psi_k| \geq \epsilon\} \subseteq \{x \in X : |\psi_k| \geq \frac{\epsilon}{n+1}\} \subseteq \{|\psi_k| \geq \frac{\epsilon}{m_\epsilon+1}\}$$

Let N be such that $k \geq N \implies \mu(\{|\psi_k| \geq \epsilon/(m_\epsilon+1)\}) < \epsilon/2(m_\epsilon+1)$. Then from the previous work we see that

$$\mu(\{|\phi\psi_k| \geq \epsilon\}) < \sum_{n=0}^{m_\epsilon} \mu(\{x \in S_n : |\psi_k\phi| \geq \epsilon\}) + \epsilon/2 < \frac{(m_\epsilon+1)\epsilon}{2(m_\epsilon+1)} + \frac{\epsilon}{2} = \epsilon$$

for all $k \geq N$ implying convergence in measure (by [Problem 1.35](#)) and we are done. For a counter example in the case of $\mu(X) = \infty$ consider $g_n(x) = x^2$ and $f_n(x) = 1/n$ on $\mathbb{R}$ with the lebesgue measure. $\square$

Problem 2.39. If $f_n \rightarrow f$ almost uniformly, then $f_n \rightarrow f$ a.e. and $f_n \rightarrow f$ in measure.

Solution. Suppose E_n are measurable sets such that $\mu(E_n^c) < 1/n$ then we have that $\mu((\cup E_n)^c) = \cap E_n^c < 1/n$ for all n and hence 0. If $x \in \cup E_n$ then $x \in E_n$ for some n and thus $f_n(x) \rightarrow f(x)$ as we have uniform convergence which implies pointwise convergence on E_n . Thus, we have convergence a.e. due to $(\cup E_n)^c$ being null. Given $\delta > 0$, almost uniform convergence gives a set E with $\mu(E^c) < \delta$ such that $f_n \rightarrow f$ uniformly on E . Because of uniform convergence, for large enough n , $|f_n(x) - f(x)| < \epsilon$ for all $x \in E$. This means that the set $\{|f_n - f| \geq \epsilon\}$ is entirely contained within E^c . Therefore, its measure is less than δ . Since δ was arbitrary, the measure tends to 0. $\square$

Problem 2.40. In Egoroff's theorem, the hypothesis " $\mu(X) < \infty$ " can be replaced by " $|f_n| \leq g$ for all n , where $g \in L^1(\mu)$."

Solution. We first restrict ourselves to a set with a null complement such that we have pointwise convergence on it and assume this to be the whole space here on which causes no problems due to the complement being null. It would suffice for the sets $E_1(k)$ to have finite measure, which they do as

$$\bigcup_{m \geq n} \left\{x \in X : |f_m(x) - f(x)| \geq \frac{1}{k}\right\} \subseteq \left\{x \in X : 2g(x) \geq \frac{1}{k}\right\}$$

with the set on the right being of finite measure due to g being L^1 (more precisely, it is easily shown that it has measure at most $2k \int_X g$) $\square$

Problem 2.41. If μ is σ -finite and $f_n \rightarrow f$ a.e., there exist measurable $E_1, E_2, \dots \subset X$ such that $\mu((\cup_1^\infty E_j)^c) = 0$ and $f_n \rightarrow f$ uniformly on each E_j .

Solution. Using sigma finiteness write $X = \cup E_n$ with E_n being finite. We can apply egoroff's on each of the E_n 's to get $E_n^{(m)} \subseteq E_n$ on which we have uniform convergence, with $\mu(E_n \setminus E_n^{(m)}) < 1/m$, then we see that $\mu(E_n \setminus \cup_m E_n^{(m)}) = 0$. As,

$$X \setminus \bigcup_{m,n} E_n^{(m)} = \left(\bigcup_n E_n\right) \setminus \bigcup_n \bigcup_m E_n^{(m)} \subseteq \bigcup_n \left(E_n \setminus \bigcup_m E_n^{(m)}\right)$$

we see that the collection $\{E_n^{(m)}\}_{m,n}$ satisfies the condition in the problem statement as a countable union of null sets (which are $E_n \setminus \cup_m E_n^{(m)}$ here) is still null. $\square$

Problem 2.42. Let μ be counting measure on $\mathbb{N}$. Then $f_n \rightarrow f$ in measure iff $f_n \rightarrow f$ uniformly.

Solution. For the only if direction notice that, for any $\epsilon > 0$ if $\mu(\{|f_n - f| \geq \epsilon\}) < 1/2$ for all $n \geq N$ then this set is empty implying that $|f_n(x) - f(x)| < \epsilon$ for all $x \in \mathbb{N}$, for all $n \geq N$ proving uniform convergence. The only if direction is always true regardless of the domain measurable space as, if $|f_n(x) - f(x)| < \epsilon$ for all n large enough, then $\{|f_n - f| \geq \epsilon\}$ is eventually always empty and thus has measure 0 for large enough n making the measure tend to 0 as $n \rightarrow \infty$ proving convergence in measure. $\square$

Problem 2.43. Suppose that $\mu(X) < \infty$ and $f : X \times [0, 1] \rightarrow \mathbb{C}$ is a function such that $f(\cdot, y)$ is measurable for each $y \in [0, 1]$ and $f(x, \cdot)$ is continuous for each $x \in X$.

- If $0 < \epsilon, \delta < 1$ then $E_{\epsilon, \delta} = \{x : |f(x, y) - f(x, 0)| \leq \epsilon \text{ for all } y < \delta\}$ is measurable.
- For any $\epsilon > 0$ there is a set $E \subset X$ such that $\mu(E) < \epsilon$ and $f(\cdot, y) \rightarrow f(\cdot, 0)$ uniformly on E^c as $y \rightarrow 0$.

Solution.

- By continuity we have that,

$$E_{\epsilon, \delta} = \bigcap_{r \in \mathbb{Q} \cap (0, \delta)} \{x \in X : |f(x, r) - f(x, 0)| \leq \epsilon\}$$

as the sets we take the intersection of are measurable, so is the set itself.

- This is almost like a continuous version (as in, the variable we judge uniform convergence over is continuous) of egoroff's theorem. Consider the sets

$$F_n(k) = \bigcup_{m \geq n} E_{1/k, 1/m}^c$$

$$= \left\{ x \in X : \exists m (m \geq n) \text{ such that } |f(x, y) - f(x, 0)| \geq \frac{1}{k} \text{ for some } y < \frac{1}{m} \right\}$$

By continuity we know that for any fixed $x \in X$ we have $f(x, y) \rightarrow f(x, 0)$ as $y \rightarrow 0$, thus for any fixed k , we see that $\bigcap_n F_n(k)$ is empty as if some x were to be in it then for any n we could find some $m \geq n$ such that there exists some y with $0 < y < 1/m \leq 1/n$ and $|f(x, y) - f(x, 0)| \geq 1/k$ which would contradict continuity as there is some n such that $y < 1/n$ implies $|f(x, y) - f(x, 0)| < 1/k$. As we have $\mu(X) < \infty$ and $F_n(k)$ decreases in n we see that $\lim_n \mu(F_n(k)) = \mu(\bigcap_n F_n(k)) = 0$. For any $\epsilon > 0$, for all k we can find some n_k with $\mu(F_{n_k}(k)) < \epsilon 2^{-k}$. Thus, $\mu(\bigcup_k F_{n_k}(k)) < \epsilon$ and as,

$$\left(\bigcup_k F_{n_k}(k) \right)^c = \bigcap_k (F_{n_k}(k))^c$$

$$= \left\{ x \in X : \forall k, m \geq n_k \implies |f(x, y) - f(x, 0)| < \frac{1}{k} \text{ for all } y < \frac{1}{m} \right\}$$

Let, $E = \bigcup_k F_{n_k}(k)$, then for all $x \in E^c$, if $y < 1/n_k$ then $|f(x, y) - f(x, 0)| < 1/k$ and thus we have uniform convergence on this set. As $\mu(E) < \epsilon$ we are done.

□

Problem 2.44. (Lusin's Theorem) If $f : [a, b] \rightarrow \mathbb{C}$ is Lebesgue measurable and $\epsilon > 0$, there is a compact set $E \subset [a, b]$ such that $m(E^c) < \epsilon$ and $f|_E$ is continuous. (Use Egoroff's theorem and Theorem 2.26.)

Solution. Fix an $\epsilon > 0$, then following a similar argument as we did in [Problem 1.38](#) we can find some M such that $m(\{x \in [a, b] : |f(x)| \geq M\}) < \epsilon$, let the complement of this set be E . Then, as f is bounded on E , it is L^1 and using Theorem 2.26 we can find continuous functions arbitrarily close to this on E in the L^1 metric. We can hence choose a sequence of continuous functions that converge to f on E in the L^1 metric, due to which we can choose a subsequence of this sequence that converges in measure to our function by Corollary 2.32, after that using Egoroff's theorem we can find some measurable $F \subseteq E$ with $m(E \setminus F) < \epsilon$ on which convergence is uniform. As the limit of a uniformly convergent sequence of continuous functions is continuous we have that f is continuous on F . Now, as $F^c \subseteq (E \setminus F) \cup E^c$ we see that $m(F^c) = m(E^c) + m(E \setminus F) < \epsilon + \epsilon = 2\epsilon$. As the Lebesgue measure is inner regular (Theorem 1.18 on page 36) we can find some compact $K \subseteq F$ such that $\mu(F \setminus K) < \epsilon$ and thus similarly, $m(K^c) = m(F \setminus K) + m(F^c) < \epsilon + 2\epsilon = 3\epsilon$. Thus, we have found a compact set $K \subseteq [a, b]$ with $m(K^c) < 3\epsilon$ on which f is continuous. □

Problem 2.45. If $(X_j, \mathcal{M}_j)$ is a measurable space for $j = 1, 2, 3$, then $\bigotimes_1^3 \mathcal{M}_j = (\mathcal{M}_1 \otimes \mathcal{M}_2) \otimes \mathcal{M}_3$. Moreover, if μ_j is a σ -finite measure on $(X_j, \mathcal{M}_j)$, then $\mu_1 \times \mu_2 \times \mu_3 = (\mu_1 \times \mu_2) \times \mu_3$.

Solution. Having identified $A \times B \times C$ with $(A \times B) \times C$ in the text, we find that by definition we see that $\bigotimes_1^3 \mathcal{M}_j = (\mathcal{M}_1 \otimes \mathcal{M}_2) \otimes \mathcal{M}_3$ as each of these clearly contains the generators of the other. These two measures also clearly agree on the measurable rectangles and thus we can say that they are equal on the algebra formed by the finite disjoint unions of these rectangles, hence we can say that they are equal on the whole sigma algebra by the uniqueness criterion in Carathéodory's extension theorem from chapter 1 as the premeasures in this case are clearly σ -finite (the reasoning behind this is given for the case of two measurable spaces at the bottom of page 64 and we thus omit the extremely similar argument needed here) □

Problem 2.46. Let $X = Y = [0, 1]$, $\mathcal{M} = \mathcal{N} = \mathcal{B}_{[0,1]}$, μ = Lebesgue measure, and ν = counting measure. If $D = \{(x, x) : x \in [0, 1]\}$ is the diagonal in $X \times Y$, then $\iint \chi_D d\mu d\nu$, $\iint \chi_D d\nu d\mu$, and $\int \chi_D d(\mu \times \nu)$ are all unequal. (To compute $\int \chi_D d(\mu \times \nu)$, go back to the definition of $\mu \times \nu$.)

Solution. D is clearly measurable and thus $\chi_D \in L^+$. We have that $D_x = \{x\}$ and $D^y = \{y\}$, we use this to compute the iterated integrals combined with the fact that for any E we have $(\chi_E)_x = \chi_{E_x}$ and similarly for y as well.

$$\int \int \chi_D d\mu(x) d\nu(y) = \int \int \chi_{D^y}(x) d\mu(x) d\nu(y) = \int \int \chi_{\{y\}}(x) d\mu(x) d\nu(y) = \int \mu(\{y\}) d\nu(y) = 0$$

as $\mu(\{y\})$ is always 0.

$$\int \int \chi_D d\nu(y) d\mu(x) = \int \nu(\{x\}) d\mu(x) = \int 1 d\mu(x) = \int_{[0,1]} 1 d\mu(x) = 1$$

We will now show that $\int \chi_D d(\mu \times \nu) = \mu \times \nu(D) = \infty$. If $D \subseteq \cup I_n$ with $I_n \in \mathcal{A}$ then, we can break these I_n into rectangles and have $D \subset \cup A_m \times B_m = \cup I_n$. Clearly we must have $[0, 1] \subseteq \cup (A_m \cap B_m)$ as $(x, x) \in A_m \times B_m$ iff $x \in A_m \cap B_m$. Keeping in mind that $A_m, B_m \in \mathcal{B}_{[0,1]}$ we thus have that $\sum \mu(A_m \cup B_m) \geq 1$ which implies that, for some m we must have $\mu(A_m), \mu(B_m) \geq \mu(A_m \cap B_m) > 0$ but then B_m has more than countably many elements making $\nu(B_m) = \infty$ so $\mu \times \nu(A_m \times B_m) = \mu(A_m)\nu(B_m) = \infty$. Thus, in the definition of $\mu \times \nu$ the numbers we take an infimum over is the singleton $\{\infty\}$ making the measure of D infinite and thus all the integrals are unequal. $\square$

Problem 2.47. Let $X = Y$ be an uncountable linearly ordered set such that for each $x \in X$, $\{y \in X : y < x\}$ is countable. (Example: the set of countable ordinals.) Let $\mathcal{M} = \mathcal{N}$ be the σ -algebra of countable or co-countable sets, and let $\mu = \nu$ be defined on $\mathcal{M}$ by $\mu(A) = 0$ if A is countable and $\mu(A) = 1$ if A is co-countable. Let $E = \{(x, y) \in X \times X : y < x\}$. Then E_x and E^y are measurable for all x, y , but $\iint \chi_E d\mu d\nu$ and $\iint \chi_E d\nu d\mu$ exist but are not equal. (If one believes in the continuum hypothesis, one can take $X = [0, 1]$ [with a nonstandard ordering] and thus obtain a set $E \subset [0, 1]^2$ such that E_x is countable and E^y is co-countable [in particular, Borel] for all x, y , but E is not Lebesgue measurable.)

Solution. Clearly $E_x = \{y \in X : y < x\}$ and $E^y = \{x \in X : x > y\}$. By definition we see that E_x is always countable and E^y always uncountable having a countable complement $E_y \cup \{y\}$, so these are always measurable. Thus,

$$\begin{aligned} \int \int \chi_E d\nu d\mu &= \int \nu(E_x) d\mu(x) = \int 0 d\mu(x) = 0 \\ \int \int \chi_E d\mu d\nu &= \int \mu(E^y) \nu(y) = \int 1 d\nu(y) = \nu(Y) = 1 \text{ as } Y \text{ is measurable} \end{aligned}$$

So these integrals are unequal. $\square$

Problem 2.48. Let $X = Y = \mathbb{N}$, $\mathcal{M} = \mathcal{N} = \mathcal{P}(\mathbb{N})$, $\mu = \nu =$ counting measure. Define $f(m, n) = 1$ if $m = n$, $f(m, n) = -1$ if $m = n + 1$, and $f(m, n) = 0$ otherwise. Then $\int f d\mu d\nu$ and $\int f d\nu d\mu$ exist and are unequal.

Solution. Firstly we see that $f(k, k) = 1$ and $f(k, k - 1) = -1$. Also, f is measurable (all functions are) as $\mathcal{P}(\mathbb{N}) \otimes \mathcal{P}(\mathbb{N}) = \mathcal{M}(\mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N})) = \mathcal{P}(\mathbb{N} \times \mathbb{N})$. Integration with respect to these counting measures is just summation, thus we can write,

$$\begin{aligned} \int \int f d\mu d\nu &= \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} f(m, n) = 1 + (1 - 1) + (1 - 1) + \dots = 1 \\ \int \int f d\nu d\mu &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} f(m, n) = (1 - 1) + (1 - 1) + \dots = 0 \end{aligned}$$

$\square$

Problem 2.49. Prove Theorem 2.39 by using Theorem 2.37 and Proposition 2.12 together with the following lemmas.

- If $E \in \mathcal{M} \otimes \mathcal{N}$ (typo in the book) and $\mu \times \nu(E) = 0$, then $\nu(E_x) = 0$ for a.e. x and $\mu(E^y) = 0$ for a.e. y .

- b. If f is $\mathcal{L}$ -measurable and $f = 0$ λ -a.e., then f_x and f_y are integrable for a.e. x and y , and $\int f_x d\nu = \int f_y d\mu = 0$ for a.e. x and y . (Here the completeness of μ and ν is needed.)

Solution. We first prove the lemmas.

- a. From the assumptions we have that

$$0 = \mu \times \nu(E) = \int \mu(E^y) \nu(y) = \int \nu(E_x) \mu(x)$$

thus, $\nu(E_x) = 0$ for a.e. x and similarly for y .

- b. As f is $\mathcal{L} = \overline{\mathcal{M} \otimes \mathcal{N}}$ measurable, by proposition 2.12, there exists a $\mathcal{M} \otimes \mathcal{N}$ measurable function g such that $f = g$ λ -a.e. which clearly implies that $f = g$ $\mu \times \nu$ -a.e. as well. Now, if f is 0 λ a.e. then so is g and thus we also similarly have that $g = 0$ $\mu \times \nu$ -a.e. We see that g_x and g_y are $\mathcal{N}$ and $\mathcal{M}$ measurable respectively. Suppose we had $f = g$ outside N with $\mu \times \nu(N) = 0$ then by part(a) we see that $\mu(N^y) = 0$ for a.e. y and $\nu(N_x) = 0$ for a.e. x . As, $f_x = g_x$ outside N_x which is null for a.e. x , we have that $f_x = g_x$ a.e. for a.e. x . In the context of the theorem, $\mathcal{N}$ is complete which makes f_x measurable if $f_x = g_x$ a.e. As g is 0 $\mu \times \nu$ -a.e. we see that $g \in L^1(\mu \times \nu)$ allowing us to use Tonelli's theorem to say that,

$$0 = \int g d(\mu \times \nu) = \int \int g_x(y) d\nu(y) d\mu(x) \implies \int g_x d\nu = 0 \text{ for a.e. } x$$

Combined with the fact that $f_x = g_x$ a.e. (implying $\int f_x = \int g_x$) for a.e. x we see that $\int f_x d\nu = 0$ for a.e. x . and thus f_x is integrable for a.e. x as well. The same can be shown for y as well following the same procedure but with different variables.

Now we prove the theorem. The claims concerning the measurability of f_x and f_y are evident from the proof of the second lemma. Now we will prove the rest assuming the same setup as used above, proving the other implications of the first case where $f \in L^+(\lambda)$ first. We can clearly choose g such that $g \geq 0$ as well (make it 0 on the null set where it isn't equal to f , i.e. consider $g\chi_{N^c}$ which is also measurable following the same notation as used above) and thus $g \in L^+(\mu \times \nu)$ making $x \mapsto \int g_x d\nu$ measurable by Tonelli's theorem. From the second lemma's proof we see that $f_x = g_x$ a.e for a.e. x making $\int f_x d\nu = \int g_x d\nu$ a.e. and as $\mathcal{M}$ is complete we must have that $x \mapsto \int f_x d\nu$ is measurable, being equal a.e. to a measurable function. Now, as $\int g_x d\nu = \int f_x d\nu$ a.e we also see that $\int \int g d\nu d\mu = \int \int f d\nu d\mu$. Using Tonelli's theorem again with the fact that the integrals w.r.t the completion and the underlying measure are equal for functions which are measurable w.r.t the underlying measure (this is easily proven) we find that,

$$\int f d\lambda = \int g d\lambda = \int g d(\mu \times \nu) = \int \int g d\nu d\mu = \int f d\nu d\mu$$

The other order for integration can be dealt with similarly. Now we prove the rest of the implications for part(b). If $f \in L^1(\lambda)$ then as $f = g$ a.e. we see that,

$$\int |g| d(\mu \times \nu) = \int |g| d(\lambda) = \int |f| d(\lambda) < \infty$$

and thus, $g \in L^1(\mu \times \nu)$. Tonelli's theorem tells us that $g_x \in L^1(\nu)$ for a.e. x and that the a.e. defined function $x \mapsto \int g_x d\nu$ is integrable (and also measurable outside a null set, this can be easily proven.) As $\mathcal{N}$ is complete we see that it is measurable (we can ignore it on a null set, if this is hard to understand just email me.) Now, clearly we have that $\int f_x = \int g_x$ for a.e. x making $x \mapsto \int f_x d\nu$ a measurable map. Thus, by Tonelli's theorem and the previous line we can say that

$$\int \int f_x(y) d\nu(y) d\mu(x) = \int \int g_x(y) d\nu(y) d\mu(x) = \int g d(\mu \times \nu) = \int f d\lambda$$

so we are done. $\square$

Problem 2.50. Suppose $(X, \mathcal{M}, \mu)$ is a σ -finite measure space and $f \in L^+(X)$. Let

$$G_f = \{(x, y) \in X \times [0, \infty] : y \leq f(x)\}.$$

Then G_f is $\mathcal{M} \times \mathcal{B}_{\mathbb{R}}$ -measurable and $\mu \times m(G_f) = \int f d\mu$; the same is also true if the inequality $y \leq f(x)$ in the definition of G_f is replaced by $y < f(x)$. (To show measurability of G_f , note that the map $(x, y) \mapsto f(x) - y$ is the composition of $(x, y) \mapsto (f(x), y)$ and $(z, y) \mapsto z - y$.) This is the definitive statement of the familiar theorem from calculus, “the integral of a function is the area under its graph.”

Solution. G_f is measurable as the composition of measurable maps is measurable, as hinted in the problem statement. Now, Theorem 2.36 tells us that, $\mu \times m(G_f) = \int m((G_f)_x) d\mu(x)$. Now, clearly $(G_f)_x = [0, f(x)]$ which has lebesgue measure $f(x)$ so we are done. If the inequality was strict then when $f(x) = 0$ the section would be an empty set, which also has $0 = f(x)$ and for $f(x) > 0$ the one sided open interval $[0, f(x))$ still has lebesgue measure $f(x)$. $\square$

Problem 2.51. Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be arbitrary measure spaces (not necessarily σ -finite).

- If $f : X \rightarrow \mathbb{C}$ is $\mathcal{M}$ -measurable, $g : Y \rightarrow \mathbb{C}$ is $\mathcal{N}$ -measurable, and $h(x, y) = f(x)g(y)$, then h is $\mathcal{M} \otimes \mathcal{N}$ -measurable.
- If $f \in L^1(\mu)$ and $g \in L^1(\nu)$, then $h \in L^1(\mu \times \nu)$ and $\int h d(\mu \times \nu) = (\int f d\mu)(\int g d\nu)$.

Solution. This is obvious for characteristic functions and we can extend it to simple functions by linearity. Part (a) is trivial as the composition of measurable maps is measurable so we prove part (b). By Theorem 2.10 there exists measurable simple functions ϕ_n, ψ_n , which tend to f, g such that $|\phi_n| \nearrow |f|$ and $|\psi_n| \nearrow |g|$. By the monotone convergence theorem we see that,

$$\int |fg| = \lim \int |\psi_n \phi_n| = \lim \left(\int |\psi_n| \int |\phi_n| \right) = \int |f| \int |g| < \infty \implies fg \in L^1(\mu \times \nu)$$

Now, by DCT (we use it on the extreme equalities) we have that,

$$\int fg = \lim \int \phi_n \psi_n = \lim \int \phi_n \int \psi_n = \left(\lim \int \phi_n \right) \left(\lim \int \psi_n \right) = \int f \int g$$

$\square$

Problem 2.52. The Fubini-Tonelli theorem is valid when $(X, \mathcal{M}, \mu)$ is an arbitrary measure space and Y is a countable set, $\mathcal{N} = \mathcal{P}(Y)$, and ν is counting measure on Y . (Cf. Theorems 2.15 and 2.25.)

Solution. By the countable additivity of the integral for any nonnegative measurable function f we have that,

$$\int_{X \times Y} f d\mu(\mu \times \nu) = \sum_{y \in Y} \int_{X \times \{y\}} f(x, y) d\mu(x) d\nu(y)$$

Now, any measurable subset of $X \times \{y\}$ clearly has form $E \times \{y\}$ where E is $\mathcal{M}$ measurable and we see that $\mu \times \nu(E \times \{y\}) = \mu(E)\nu(\{y\}) = \mu(E)$. From the definition of the integral it is clear that

$$\int_{X \times \{y\}} f(x, y) d\mu(x) d\nu(y) = \int_X f(x, y) d\mu(x)$$

The map $y \mapsto \int f(x, y) d\mu(x)$ is also trivially measurable as the sigma algebra on Y is the whole power set. As ν is the counting measure we thus have,

$$\int \int f(x, y) d\mu(x) d\nu(y) = \sum_{y \in Y} \int f(x, y) d\mu(x) = \int f d(\mu \times \nu)$$

Again, by the additivity of the integral here we see that

$$\int \int f(x, y) d\nu(y) d\mu(x) = \int \sum_{y \in Y} f(x, y) d\mu(x) = \sum_{y \in Y} \int f(x, y) d\mu(x)$$

which is $\int f d(\mu \times \nu)$ as we saw before. We also see that the map $x \mapsto \int f(x, y) d\nu(y)$ is measurable as,

$$\int f(x, y) d\nu(y) = \sum_{y \in Y} f(x, y)$$

as its the limit of a sequence of measurable functions $f^{y_1} + \dots + f^{y_n}$ (here we enumerate Y as $\{y_n : n \in \mathbb{N}\}$). We have proven Tonelli's theorem for this case and that of Fubini can be proven from this as in the book, or we can also reenact the same steps while using DCT instead of the additivity of the integral (which was just a corollary of MCT in this case). $\square$

Problem 2.53. Fill in the details of the proof of Theorem 2.41.

Solution. Fix $\epsilon > 0$. Take a sequence of simple measurable functions ϕ_n such that $|\phi_n| \nearrow |f|$ and by DCT, for large enough n we have $\int |\phi_n - f| < \epsilon$, we write $\phi := \phi_n$. Suppose $\phi = \sum_1^m a_j \chi_{E_j}$ where $a_j \neq 0$ and E_j are disjoint, then we see that

$$|a_j| \mu(E_j) = \int_{E_j} |\phi| \leq \int_{E_j} |f| < \infty$$

Implying that $\mu(E_j) < \infty$. By theorem 2.41c, for each E_j we can find a disjoint(disjuncy is needed here by the way) collection $\{R_{j,i}\}_{i=1}^{k_j}$ such that $m(E_j \Delta \bigcup_i R_{j,i}) < \epsilon$ and each $R_{j,i}$ is a product of intervals, we will call these cubes here on out. Define $\psi := \sum_{i,j} a_j \chi_{R_{j,i}}$, then we see that

$$\int |\phi - \psi| \leq \sum_{j=1}^m |a_j| \int \left| \chi_{E_j} - \sum_{i=1}^{k_j} \chi_{R_{j,i}} \right| \leq \sum_{j=1}^m |a_j| \cdot m \left(E_j \Delta \bigcup_{i=1}^{k_j} R_{j,i} \right) < \epsilon \cdot \sum_{j=1}^m |a_j|$$

Thus, we have that $\int |f - \psi| \leq \int |f - \phi| + \int |\phi - \psi| < \epsilon \cdot (1 + \sum_1^m |a_j|)$ which can be made arbitrarily small. But the fact that ψ is of the form described in the problem requires proof. It suffices to show that any finite union of cubes can be written as a disjoint union of such. We consider cubes whose sides (as defined in the book) are closed intervals, the rest follow similarly. Suppose, $A = \prod [a_j, b_j]$ and $B = \prod [c_j, d_j]$ then we clearly have that $A \cap B = \prod ([a_j, b_j] \cap [c_j, d_j])$ which is also a cube (it might be empty but that's a cube too.) We also find that

$$A^c = \bigcup_{f \in \{1,c\}^{\{1,\dots,n\}}} \prod_{j=1}^n [a_j, b_j]^{f(j)}$$

by extreme abuse of notation. Notice that this is a disjoint union. Also, $[a_j, b_j]^c = (-\infty, a_j) \cup (b_j, \infty)$ and this allows us to write any of the sets in the union above as a finite disjoint union of cubes as these sets are of form (after permuting the sides so it isn't as notationally strenuous to write)

$$((-\infty, a_1) \cup (b_1, \infty)) \times \dots \times ((-\infty, a_m) \cup (b_m, \infty)) \times [a_{m+1}, b_{m+1}] \times \dots [a_n, b_n]$$

which can be written as a disjoint union of cubes following this set theoretic identity

$$\prod_{i \in A} \bigcup_{j \in B} S_{i,j} = \bigcup_{f \in B^A} \prod_{i \in A} S_{i,f(j)}$$

as in our case, for fixed i , all the $S_{i,j}$ are disjoint. This shows that the set of all cubes is an elementary family and thus by Proposition 1.7 (page 23) we can say that the finite disjoint union of cubes is an algebra. This tells us that we can write the union of cubes $\cup_{i,j} R_{j,i}$ as a disjoint union of cubes and using this we can write ψ as a simple function where all the characteristic functions are those of cubes which are disjoint so we are done with the first claim. Again, by DCT for large enough m we must have that $\int |\psi \cdot \chi_{[-m,m]^n} - f| < 2 \cdot \epsilon$ starting with small enough $\epsilon > 0$. We clearly see that $\psi \chi_{[-m,m]^n}$ vanishes outside a bounded set and as it's defined using characteristic functions on disjoint cubes, we can approximate it on these cubes by a continuous function as we did in the case of the real line, while ignoring cubes of measure 0, which on the real line are just singletons, this allows us to get $\mathcal{O}(\epsilon^n)$ which can be made arbitrarily small. Then we can use the triangle inequality and conclude that $C_c(\mathbb{R}^n)$ is dense in $L^1(\mathbb{R}^n, \mathcal{L}^n, m)$ in the L^1 metric. $\square$

Problem 2.54. How much of Theorem 2.44 remains valid if T is not invertible?

Solution. Part (b) is true and we prove it below whereas part (a) is false, that is $f \circ T$ is not necessarily Lebesgue measurable, given f is. As T is not invertible we can find some invertible linear map S such that $ST(\mathbb{R}^n) = \mathbb{R}^k \times \{0\}^{n-k}$ where $k < n$ and this is measurable with measure 0. Thus, $m(S^{-1}ST(\mathbb{R}^n)) = |\det S^{-1}|m(ST(\mathbb{R}^n)) = 0$ showing that $T(\mathbb{R}^n)$ is a Lebesgue null set. Now, for any $E \in \mathcal{L}^n$ (any set even) we have that $T(E) \subseteq T(\mathbb{R}^n)$ proving part (b) by completeness. Now to disprove part (a), say $n = 2$ and consider the indicator function χ of the Lebesgue measurable set $N \times \{0\}$ where N is not Lebesgue measurable on the real line and let $T : (x, y) \mapsto (x, 0)$ then, $T^{-1}(f^{-1}(0.9, 1.1)) = N \times \mathbb{R}$. Now, if $N \times \mathbb{R}$ was in $\mathcal{L}^2$ then its indicator would be measurable and by Theorem 2.39 we would have that f^y is measurable on $\mathbb{R}$ for a.e. y which is false as these are all indicators of N on $\mathbb{R}$ so the set was not measurable to begin with. $\square$

Problem 2.55. Let $E = [0, 1] \times [0, 1]$. Investigate the existence and equality of $\int_E f \, dm^2$, $\int_0^1 \int_0^1 f(x, y) \, dx \, dy$, and $\int_0^1 \int_0^1 f(x, y) \, dy \, dx$ for the following f :

- a. $f(x, y) = (x^2 - y^2)(x^2 + y^2)^{-2}$.
- b. $f(x, y) = (1 - xy)^{-a}$ ($a > 0$).
- c. $f(x, y) = (x - \frac{1}{2})^{-3}$ if $0 < y < |x - \frac{1}{2}|$, $f(x, y) = 0$ otherwise.

Solution.

- a. We exclude $\{(0, 0)\}$ which has measure 0 and this does not alter anything we care about. As f is now continuous, using Theorem 2.39 we can work in $\mathcal{B}_{\mathbb{R}^2}$ and the positive and negative parts evaluate to ∞ after some basic calculus paired with Fubini's theorem and hence the integral does not exist however the iterated integrals evaluate to $\pi/4$ and $-\pi/4$ respectively.
- b. Removing $\{(1, 1)\}$ we can say that this is again continuous and following the same arguments above, revert back to $\mathcal{B}_{\mathbb{R}^n}$. As this is nonnegative, all the integrals exist and are equal.
- c. It's clearly Borel, we follow the same arguments above. The first iterated integral exists and equals 0 whereas the other does not. The integral of the positive and negative parts are both infinite and thus the integral over the product measure does not exist.

□

Problem 2.56. If f is Lebesgue integrable on $(0, a)$ and $g(x) = \int_x^a t^{-1} f(t) \, dt$, then g is integrable on $(0, a)$ and $\int_0^a g(x) \, dx = \int_0^a f(x) \, dx$.

Solution. Consider the function $H(t, x) := \chi_{t > x} \cdot f(t)t^{-1}$ on $(0, a)^2$. H is clearly measurable w.r.t $m \times m$ and by Tonelli's theorem we have that

$$\begin{aligned} \int_{(0, a)^2} |H| \, d(m \times m) &= \int_0^a \int_0^a |H(t, x)| \, dm(t) \, dm(x) \\ &= \int_0^a \int_0^t \frac{|f(t)|}{t} \, dm(x) \, dm(t) = \int_0^a |f(t)| \, dm(t) < \infty \end{aligned}$$

Making H integrable. Now, by Fubini's theorem we can say that $x \mapsto \int_0^a H(x, t) \, dm(t) = \int_x^a f(t)/t \, dm(t) = g(x)$ is measurable and

$$\int_0^a f(t) \, dm(t) = \int_{(0, a)^2} H \, d(m \times m) = \int_0^a \int_0^a H(t, x) \, dm(t) \, dm(x) = \int_0^a g(x) \, dm(x)$$

□

Problem 2.57. Show that $\int_0^\infty e^{-sx} x^{-1} \sin x \, dx = \arctan(s^{-1})$ for $s > 0$ by integrating $e^{-sxy} \sin x$ with respect to x and y . (It may be useful to recall that $\tan(\frac{\pi}{2} - \theta) = (\tan \theta)^{-1}$. Cf. Exercise 31d.)

Solution. Consider $f(x, y) = se^{-sxy} \sin(x)$ on $(0, \infty) \times (1, \infty)$, this is integrable as, by tonelli's we have

$$\int |f| = \int_0^\infty \int_1^\infty se^{-sxy} |\sin(x)| dy dx = \int_0^\infty e^{-sx} |\sin(x)/x| dx \leq \int_0^\infty e^{-sx} = s^{-1} < \infty$$

Now using fubini's theorem we see that,

$$\begin{aligned} \int_0^\infty e^{-sx} x^{-1} \sin(x) dx &= \int_0^\infty \int_1^\infty f(x, y) dx dy = \int_1^\infty \int_0^\infty se^{-sxy} \sin(x) dx dy \\ &= \int_1^\infty \frac{s}{1 + (sy)^2} dy = \pi/2 - \arctan(s) = \arctan(s^{-1}) \end{aligned}$$

□

Problem 2.58. Show that $\int e^{-sx} x^{-1} \sin^2 x dx = \frac{1}{4} \log(1 + 4s^{-2})$ for $s > 0$ by integrating $e^{-sx} \sin 2xy$ with respect to x and y .

Solution. Using the same setup as before and the identity $\sin(x)^2 = (1 - \cos(2x))/2$ we just need to evaluate integrals of form $\int e^{ax} \cos(bx) dx$ this time instead of using sine there. □

Problem 2.59. Let $f(x) = x^{-1} \sin x$.

- Show that $\int_0^\infty |f(x)| dx = \infty$.
- Show that $\lim_{b \rightarrow \infty} \int_0^b f(x) dx = \frac{1}{2}\pi$ by integrating $e^{-xy} \sin x$ with respect to x and y . (In view of part (a), some care is needed in passing to the limit as $b \rightarrow \infty$.)

Solution.

- (a) it can be shown with elementary methods that

$$\int_0^n |\sin(x)/x| dx \gg H_n$$

which suffices.

- (b) Consider $f(x, y) = e^{-xy} \sin(x)$. For any $b > 0$ we see that,

$$\int_0^b \int_0^\infty |f(x, y)| dy dx \leq \int_0^b \int_0^\infty xe^{-xy} dy dx = b < \infty$$

so its integrable on the domain used for the integral and by fubini's we have,

$$\begin{aligned} \int_0^b \frac{\sin x}{x} dx &= \int_0^b \int_0^\infty f(x, y) dy dx = \int_0^\infty \int_0^b f(x, y) dx dy \\ &= \lim_{b \rightarrow \infty} \int_0^\infty \underbrace{-\frac{e^{-by} \sin(b)y + e^{-by} \cos(b) - 1}{1 + y^2}}_{g_b(y)} dy \end{aligned}$$

We see that $g_b(y) \rightarrow (1 + y^2)^{-1}$ as $b \rightarrow \infty$ and $|g_b(y)| \leq 3(1 + y^2)^{-1}$ which is integrable allowing us to use DCT (by passing to sequences) to conclude that,

$$\int_0^\infty \frac{\sin x}{x} dx = \int_0^\infty \frac{1}{1 + y^2} dy = \frac{\pi}{2}$$

□

Problem 2.60. $\Gamma(x)\Gamma(y)/\Gamma(x+y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt$ for $x, y > 0$. (Recall that Γ was defined in §2.3. Write $\Gamma(x)\Gamma(y)$ as a double integral and use the argument of the exponential as a new variable of integration.)

Solution. Here, $x > y$ denotes the set $\{(x, y) \in \mathbb{R}^2 : x > y\}$. Let $f(x, y) = x^{\alpha-1}y^{\beta-1}e^{-(x+y)}$. Then, by [Problem 2.51](#) we see that

$$\Gamma(\alpha)\Gamma(\beta) = \int_{(0,\infty)^2} f d(m \times m)$$

and by Theorem 2.47, using the invertible linear map $G : (x, y) \mapsto (x-y, y)$ we see that,

$$\int_{G(\{x>y\})=(0,\infty)^2} f d(m \times m) = \int_{x>y} f(x-y, y) \cdot \underbrace{|\det G|}_1 d(m \times m)$$

and by Tonelli's we have,

$$\begin{aligned} \int_{x>y} f(x-y, y) &= \int_0^\infty \int_0^x (x-y)^{\alpha-1} y^{\beta-1} e^{-x} dy dx \\ &= \int_0^\infty \int_0^x \left(1 - \frac{y}{x}\right)^{\alpha-1} \left(\frac{y}{x}\right)^{\beta-1} x^{\alpha+\beta-1} e^{-x} dy dx = \int_0^\infty x^{\alpha+\beta-1} e^{-x} \int_0^1 v^{\beta-1} (1-v)^{\alpha-1} dv \end{aligned}$$

by a change of variable $v = y/x$. We thus have that

$$\Gamma(\alpha)\Gamma(\beta) = \Gamma(\alpha+\beta) \int_0^1 t^{\alpha-1}(1-t)^{\beta-1} dt$$

proving the claim. □

Problem 2.61. If f is continuous on $[0, \infty)$, for $\alpha > 0$ and $x \geq 0$ let

$$I_\alpha f(x) = \frac{1}{\Gamma(\alpha)} \int_0^x (x-t)^{\alpha-1} f(t) dt.$$

$I_\alpha f$ is called the α th **fractional integral** of f .

- $I_{\alpha+\beta}f = I_\alpha(I_\beta f)$ for all $\alpha, \beta > 0$. (Use Exercise 60.)
- If $n \in \mathbb{N}$, $I_n f$ is an n th-order antiderivative of f .

Solution.

- By expanding we have that,

$$I_\alpha(I_\beta f)(x) = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x \int_0^t (x-t)^{\alpha-1} (t-y)^{\beta-1} f(y) dy dt$$

Heuristically speaking we integrate over the region $x > t > y > 0$. Using the invertible linear map that switches the coordinates and Theorem 2.47 and Tonelli's theorem we see that this is equal to

$$\frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x \int_y^x (x-t)^{\alpha-1} (t-y)^{\beta-1} f(y) dt dy$$

$$= \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^x f(y)(x-y)^{\alpha+\beta-1} \int_y^x \left(\frac{x-t}{x-y}\right)^{\alpha-1} \left(\frac{t-y}{x-y}\right)^{\beta-1} \frac{dt}{x-y} dy$$

which is $I_{\alpha+\beta}f(x)$ after a change of variable $v = (t-y)/(x-y)$ and an application of the previous problem.

b. This follows from part (a) by repeatedly applying I_1 which is the antiderivative. □

Problem 2.62. The measure σ on S^{n-1} is invariant under rotations.

Solution. Let T be a rotation. Given any borel set $E \subseteq S^{n-1}$, let $E_1 = \Phi^{-1}((0, 1] \times E)$ as in the text, then we defined $\sigma(E) = n \cdot m(E_1)$ so it suffices to show that $m(E_1) = m(T(E)_1)$. For any $rx' \in E_1$ in usual notation (same as the text) we find that $T(rx') = rT(x')$ with $T(x') \in S^{n-1}$ as well due to T being a rotation ($\|Tv\| = \|v\|$ for all v), thus we have that $T(E_1) = \{rT(x') : r \in (0, 1], x' \in E\} = \Phi^{-1}((0, 1] \times T(E_1))$ giving us the equality $T(E_1) = T(E)_1$ and by Theorem 2.44 we have $m(T(E_1)) = |\det T| m(E_1) = m(E_1)$ as $|\det T| = 1$ here. □

Problem 2.63. The technique used to prove Proposition 2.54 can also be used to integrate any polynomial over S^{n-1} . In fact, suppose $f(x) = \prod_1^n x_j^{\alpha_j}$ ($\alpha_j \in \mathbb{N} \cup \{0\}$) is a monomial. Then $\int f d\sigma = 0$ if any α_j is odd, and if all α_j 's are even,

$$\int f d\sigma = \frac{2\Gamma(\beta_1) \dots \Gamma(\beta_n)}{\Gamma(\beta_1 + \dots + \beta_n)} \quad \text{where } \beta_j = \frac{\alpha_j + 1}{2}.$$

Solution.

$$\int_{\mathbb{R}^n} f(x) e^{-\|x\|^2} dm(x) = \int_0^\infty r^{n-1} \int_{S^{n-1}} f(rx') e^{-r^2} d\sigma(x') dm(r)$$

notice that $f(rx') = r^{\sum \alpha_i} f(x')$, this makes the above equal

$$\int_0^\infty r^{n-1+\sum \alpha_i} e^{-r^2} \int_{S^{n-1}} f(x') d\sigma(x') dm(r) = \int f d\sigma \int_0^\infty r^{n-1+\sum \alpha_i} e^{-r^2} dr$$

Now, by a change of variable we see that,

$$\int_0^\infty r^{n-1+\sum \alpha_i} e^{-r^2} dr = \frac{1}{2} \int_0^\infty r^{\frac{n+\sum \alpha_i}{2}-1} e^{-r} dr = \frac{1}{2} \Gamma\left(\frac{n+\alpha_1+\dots+\alpha_n}{2}\right) = \frac{1}{2} \Gamma(\beta_1 + \dots + \beta_n)$$

And by fubini's theorem, unless some α_j is odd we have,

$$\int_{\mathbb{R}^n} f(x) e^{-\|x\|^2} dm(x) = \prod_{j=1}^n \int_{-\infty}^\infty x_j^{\alpha_j} e^{-x_j^2} dx_j = \prod_{j=1}^n 2 \cdot \frac{1}{2} \Gamma\left(\frac{\alpha_j+1}{2}\right) = \Gamma(\beta_1) \dots \Gamma(\beta_n)$$

now rearranging these proves the claim in this case. If some α_j are odd then one of the integrals $\int_{-\infty}^\infty x_j^{\alpha_j} e^{-x_j^2} dx_j$ would be 0 making $\int f d\sigma = 0$ as $\Gamma(\sum \beta_i) > 0$ here, this is clear from the definition of the gamma function. □

Problem 2.64. For which real values of a and b is $|x|^a |\log |x||^b$ integrable over $\{x \in \mathbb{R}^n : |x| < \frac{1}{2}\}$? Over $\{x \in \mathbb{R}^n : |x| > 2\}$?

Solution. By corollary 2.51 we have, after attaching a suitable characteristic function,

$$\int_{B(0,1/2)} \|x\|^a |\log(\|x\|)|^b dm(x) = \sigma(S^{n-1}) \int_0^{1/2} r^{a+n-1} |\log(r)|^b dr$$

It is now trivial to check which a, b work. In the other case the integral is over $(2, \infty)$. $\square$

Problem 2.65. Define $G : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $G(r, \phi_1, \dots, \phi_{n-2}, \theta) = (x_1, \dots, x_n)$ where

$$\begin{aligned} x_1 &= r \cos \phi_1, & x_2 &= r \sin \phi_1 \cos \phi_2, & x_3 &= r \sin \phi_1 \sin \phi_2 \cos \phi_3, \dots, \\ x_{n-1} &= r \sin \phi_1 \dots \sin \phi_{n-2} \cos \theta, & x_n &= r \sin \phi_1 \dots \sin \phi_{n-2} \sin \theta. \end{aligned}$$

- G maps $\mathbb{R}^n$ onto $\mathbb{R}^n$, and $|G(r, \phi_1, \dots, \theta)| = |r|$.
- $\det D_{(r, \phi_1, \dots, \phi_{n-2}, \theta)} G = r^{n-1} \sin^{n-2} \phi_1 \sin^{n-3} \phi_2 \dots \sin \phi_{n-2}$.
- Let $\Omega = (0, \infty) \times (0, \pi)^{n-2} \times (0, 2\pi)$. Then $G|_{\Omega}$ is a diffeomorphism and $m(\mathbb{R}^n \setminus G(\Omega)) = 0$.
- Let $F(\phi_1, \dots, \phi_{n-2}, \theta) = G(1, \phi_1, \dots, \phi_{n-2}, \theta)$ and $\Omega' = (0, \pi)^{n-2} \times (0, 2\pi)$. Then $(F|_{\Omega'})^{-1}$ defines a coordinate system on S^{n-1} except on a σ -null set, and the measure σ is given in these coordinates by

$$d\sigma(\phi_1, \dots, \phi_{n-2}, \theta) = \sin^{n-2} \phi_1 \sin^{n-3} \phi_2 \dots \sin \phi_{n-2} d\phi_1 \dots d\phi_{n-2} d\theta.$$

Solution.

- Applying the identity $\sin^2(\theta) + \cos^2(\theta) = 1$ collapses $x_1^2 + \dots + x_n^2$ to r^2 proving the claim.
- Consider the transformations $J(r, \phi_1, \dots, \phi_{n-2}, \theta) = (r \cos \phi_1, r \sin \phi_1, \dots, \phi_2, \dots, \phi_{n-2}, \theta)$ and $G(x_1, w, \phi_2, \dots, \phi_{n-2}, \theta) = (x_1, G(w, \phi_2, \dots, \phi_{n-2}, \theta))$, then we see that $G = H \circ J$ and by the multivariate chain rule we must have $DG = DH \cdot DJ$, we can calculate DJ easily and DG by induction on the dimension, this proves the claim.
- The variables associated to the m -th dimension are subscripted by m . This is trivial for $n = 2$ as $G_2(\Omega_2) = \mathbb{R}^2 \setminus \{(x, 0) : x \geq 0\}$ and this is also clearly a diffeomorphism. Now, we assume it to be true for $n - 1$, that is G_{n-1} is a diffeomorphism and $G_{n-1}(\Omega_{n-1}) = U_{n-1}$ where $m(\mathbb{R}^{n-1} \setminus U_{n-1}) = 0$. We also see that $J_n(\Omega_n) = \mathbb{R} \times (0, \infty) \times (0, \pi)^{n-3} \times (0, 2\pi) = \mathbb{R} \times \Omega_{n-1}$ and this is a diffeomorphism as we only alter the first two coordinates and this corresponds to G_2 . By the definition of H_n and the assumption we see that $G_n(\Omega_n) = H_n(J_n(\Omega_n)) = \mathbb{R} \times U_{n-1}$ and this is also a diffeomorphism as the first coordinate is unaltered and the rest corresponds to G_{n-1} . Thus we have that, $\mathbb{R}^n \setminus G_n(\Omega_n) = \mathbb{R}^n \setminus (\mathbb{R} \times U_{n-1}) = \mathbb{R} \times (\mathbb{R}^{n-1} \setminus U_{n-1})$ and G_n being a composition of diffeomorphisms, is one itself. By our convention of $0 \cdot \infty = 0$ we see that this is of measure 0 (one could also use the fact that this has measure $\int_{\mathbb{R}} m_{n-1}(\mathbb{R}^{n-1} \setminus U_{n-1}) dm_1 = \int 0 dm_1 = 0$).
- By definition we see that $G_n(r, \omega) = rF_n(\omega)$, and as we had $|G_n(r, \omega)| = r$ we see that $F_n(\omega) = G_n(1, \omega)$ maps Ω'_n into S^{n-1} . As G_n was diffeomorphic on $\Omega_n = (0, \infty) \times \Omega'_n$ and acted injectively we must have F_n act diffeomorphically on Ω'_n as

its still C^1 (it shares partial derivatives with G_n clearly.) Thus, $F_n|_{\Omega'_n}^{-1}$ forms a valid coordinate system on $F_n(\Omega'_n) \subseteq S^{n-1}$. As $G_n(r, \omega) = rF_n(\omega)$ we see that all non-origin points outside a null set can be written uniquely in that form (radius times direction vector on the sphere) and as the radius can be arbitrary, the non-origin points we can not reach must be unreachable due to their direction not being in $F_n(\Omega'_n)$. Hence, $(\mathbb{R}^n \setminus U_n) \setminus \{0\} = \{rx' : r \in (0, \infty), x' \in S^{n-1} \setminus F_n(\Omega'_n)\}$. We thus see that, as $\chi_{(\mathbb{R}^n \setminus U_n) \setminus \{0\}}(rx') = \chi_{S^{n-1} \setminus F_n(\Omega'_n)}(x')$ and by polar integration, the measure of the $(\mathbb{R}^n \setminus G_n(\Omega_n)) \setminus \{0\}$ is

$$\int_0^\infty \int_{S^n} \chi_{S^{n-1} \setminus F_n(\Omega_n)}(x') r^{n-1} d\sigma(x') dm(r) = \sigma(S^{n-1} \setminus F_n(\Omega_n)) \int_0^\infty r^{n-1} dm(r)$$

and we must have the spherical measure be zero here for the whole thing to be zero which was proved in part (c). Now, we see that for borel sets $E \subseteq S^n$,

$$\begin{aligned} \int_0^1 \int_{S^{n-1}} \chi_E(x') r^{n-1} d\sigma(x') dm(r) &= \int_{\mathbb{R}^n} \chi_{E_1} dm = \int_{G(\Omega_n)} \chi_{E_1} dm \\ &= \int_{\Omega_n} \underbrace{\chi_{E_1}(G(r, \omega))}_{\chi_{(0,1]}(r) \cdot \chi_E(F(\omega))} |\det D_{(r, \omega)} G| dm((r, \omega)) \\ &= \int_0^\infty \chi_{(0,1]}(r) r^{n-1} \int_{\Omega'_n} \chi_E(F(\omega)) \sin^{n-2} \phi_1 \sin^{n-3} \phi_2 \cdots \sin \phi_{n-2} dm_{n-1}(\omega) dm(r) \end{aligned}$$

where we used all the facts developed above freely. Now, after some simplifications we have

$$\sigma(E) = \int_{F^{-1}|_{\Omega'_n}^{-1}(E)} \sin^{n-2} \phi_1 \sin^{n-3} \phi_2 \cdots \sin \phi_{n-2} dm((\phi_1, \phi_2, \dots, \theta))$$

so we are done. □

Chapter 3 : Signed Measures and Differentiation

Problem 3.1. Prove Proposition 3.1.

Solution. Wlog suppose that ν omits the value $-\infty$, then if E_n is a sequence increasing sets we have that,

$$\nu\left(\bigcup_{n=1}^\infty E_n\right) = \nu(E_1) + \sum_{n \geq 2} \nu(E_n \setminus E_{n-1}) = \lim_{m \rightarrow \infty} \nu(E_1) + \sum_{n \leq m} \nu(E_n \setminus E_{n-1}) = \lim_{m \rightarrow \infty} \nu(E_m)$$

And if it was decreasing with $\nu(E_1)$ finite then, $E_1 \setminus E_n$ is increasing and we have

$$\nu(E_1) - \nu\left(\bigcap E_n\right) = \nu\left(E_1 \setminus \bigcap E_n\right) = \nu\left(\bigcup (E_1 \setminus E_n)\right) = \lim_{m \rightarrow \infty} \nu(E_1 \setminus E_m) = \nu(E_1) - \lim_{m \rightarrow \infty} \nu(E_m)$$

after which we can cancel out the finite term of $\nu(E_1)$. □

Problem 3.2. If ν is a signed measure, E is ν -null iff $|\nu|(E) = 0$. Also, if ν and μ are signed measures, $\nu \perp \mu$ iff $|\nu| \perp \mu$ iff $\nu^+ \perp \mu$ and $\nu^- \perp \mu$.

Solution. If $|\nu|(E) = 0$ then $\nu^+(E) = \nu^-(E) = 0$ and as these are positive measures, by monotonicity see that $\nu^+(F) = \nu^-(F) = 0$ for any measurable $F \subseteq E$ making E ν -null. Now, if E was ν -null then $|\nu|(E) > 0$ would imply that $\nu^+(E)$ or $\nu^-(E)$ is strictly greater than 0 and if we assume the first to be as described then $\nu(E \cap P) = \nu^+(E \cap P) > 0$ which contradicts nullity of E w.r.t ν as $E \cap P \subseteq E$. The other case leads to a contradiction similarly, thus we must have $|\nu|(E) = 0$.

As a measurable set E is $|\nu|$ -null iff $|\nu|(E) = 0$ stemming from positivity we see that $\nu \perp \mu$ iff $|\nu| \perp \mu$ by the previous claim. We also see that $|\nu| \perp \mu$ iff $\nu^+, \nu^- \perp \mu$ as $|\nu|(E) = 0$ iff $\nu^-(E), \nu^+(E) = 0$, which proves the equivalence. $\square$

Problem 3.3. Let ν be a signed measure on $(X, \mathcal{M})$.

- $L^1(\nu) = L^1(|\nu|)$.
- If $f \in L^1(\nu)$, $|\int f d\nu| \leq \int |f| d|\nu|$.
- If $E \in \mathcal{M}$, $|\nu|(E) = \sup\{|\int_E f d\nu| : |f| \leq 1\}$.

Solution.

- As $L^1(\nu) := L^1(\nu^+) \cap L^1(\nu^-)$ we find that, $f \in L^1(\nu)$ implies $\int |f| d\nu^+, \int |f| d\nu^- < \infty$ and thus, $\int |f| d|\nu| = \int |f| d\nu^+ + \int |f| d\nu^- < \infty$ where we have used the following lemma: If μ_1, μ_2 are positive measures on the same sigma algebra then so is their sum and if $f \in L^+$ then $\int f d(\mu_1 + \mu_2) = \int f d\mu_1 + \int f d\mu_2$. For a proof, one can look at simple functions first where its trivial and then routinely extend to L^+ by MCT and approximations using simple functions. Now, if we had $f \in L^1(|\nu|)$ then as $\int f d|\nu| = \int |f| d\nu^+ + \int |f| d\nu^- < \infty$, both of these terms are individually finite which puts f in $L^1(\nu^+)$ and $L^1(\nu^-)$ proving the claim.

b.

$$\begin{aligned} \left| \int f d\nu \right| &= \left| \int f d\nu^+ - \int f d\nu^- \right| \\ &\leq \left| \int f d\nu^+ \right| + \left| \int f d\nu^- \right| \leq \int |f| d\nu^+ + \int |f| d\nu^- = \int |f| d|\nu| \end{aligned}$$

- Consider the hahn decomposition $P \cup N$ is usual notation leading upto the jordan decomposition, then letting $f = \chi_P - \chi_N$ we see that $\int f d\nu = \nu(P \cap E) - \nu(N \cap E) = \nu^+(E) + \nu^-(E) = |\nu|(E)$ so we have one direction of the inequality ($\leq$) and for the other direction, by part (b) we can say that $|f| \leq 1$ implies $|\int f d\nu| \leq \int 1 d|\nu| = |\nu|(E)$. so we are done.

$\square$

Problem 3.4. If ν is a signed measure and λ, μ are positive measures such that $\nu = \lambda - \mu$, then $\lambda \geq \nu^+$ and $\mu \geq \nu^-$.

Solution. Consider the hahn decomposition $P \cup N$ in usual notation, leading upto the jordan decomposition. Then we have $\nu^+(E) = \nu(E \cap P) = \lambda(E \cap P) - \mu(E \cap P) \leq \lambda(E \cap P) \leq \lambda(E)$ and the other inequality follows similarly. $\square$

Problem 3.5. If ν_1, ν_2 are signed measures that both omit the value $+\infty$ or $-\infty$, then $|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$. (Use Exercise 4.)

Solution. It can be easily shown that in this case $\nu_1 + \nu_2$ is also a signed measure. As all ν_1, ν_2 are finite, we can rearrange to get $\nu_1 + \nu_2 = \nu_1^+ - \nu_1^- + \nu_2^+ - \nu_2^- = (\nu_1^+ + \nu_2^+) - (\nu_1^- + \nu_2^-)$ which, by the previous problem implies $(\nu_1 + \nu_2)^+ \leq \nu_1^+ + \nu_2^+$ and $(\nu_1 + \nu_2)^- \leq \nu_1^- + \nu_2^-$ implying $|\nu_1 + \nu_2| \leq (\nu_1^+ + \nu_2^+) + (\nu_1^- + \nu_2^-) = (\nu_1^+ + \nu_1^-) + (\nu_2^+ + \nu_2^-) \leq |\nu_1| + |\nu_2|$. $\square$

Problem 3.6. Suppose $\nu(E) = \int f d\mu$ where μ is a positive measure and f is an extended μ -integrable function. Describe the Hahn decompositions of ν and the positive, negative, and total variations of ν in terms of f and μ .

Solution. Let $P = \{f > 0\}$ and $N = \{f \leq 0\}$, then $P \cup N$ is clearly a hahn decomposition and we see that $\nu^+(E) = \int_{E \cap P} f d\mu = \int_E f^+ d\mu$ and similarly $\nu^-(E) = \int_E f^- d\mu$ and thus, $|\nu|(E) = \int_E |f| d\mu$. $\square$

Problem 3.7. Suppose that ν is a signed measure on $(X, \mathcal{M})$ and $E \in \mathcal{M}$.

- $\nu^+(E) = \sup\{\nu(F) : F \in \mathcal{M}, F \subset E\}$ and $\nu^-(E) = -\inf\{\nu(F) : F \in \mathcal{M}, F \subset E\}$.
- $|\nu|(E) = \sup\{\sum_1^n |\nu(E_j)| : n \in \mathbb{N}, E_1, \dots, E_n \text{ are disjoint, and } \bigcup_1^n E_j = E\}$.

Solution.

- $F \subseteq E$ implies that $\nu(F) \leq \nu^+(F) \leq \nu^+(E)$ and $\nu(E \cap P) = \nu^+(E)$ with $E \cap P \subseteq E$ proving one part, the other follows similarly.
- Considering a hahn decomposition $P \cup N$ leading to a jordan decomposition, we see that $|\nu|(E) = \nu^+(E) + \nu^-(E) = |\nu(E \cap P)| + |\nu(E \cap N)|$ as $\nu^+(E) = \nu(E \cap P) \geq 0$ and $\nu^-(E) = -\nu(E \cap N) \geq 0$ proving one direction of the equality. For the other, it can be easily proved that $|\nu(E)| \leq |\nu|(E)$ and thus, $\bigcup_1^n E_j = E$ (disjoint union) implies $\sum_1^n |\nu(E_j)| \leq \sum_1^n |\nu|(E_j) = |\nu|(E)$ so we are done.

$\square$

Problem 3.8. $\nu \ll \mu$ iff $|\nu| \ll \mu$ iff $\nu^+ \ll \mu$ and $\nu^- \ll \mu$.

Solution. If $\nu \ll \mu$ then $\mu(E) = 0 \implies \nu(E) = 0$ and as μ is taken positive we must have that $\mu(E \cap P), \mu(E \cap N) = 0$ where $P \cup N$ is the hahn decomposition for ν , this implies that $\nu(E \cap P), \nu(E \cap N) = 0$ so $\nu^\pm(E) = 0$ and we have that $|\nu|(E) = 0$ as well proving $|\nu| \ll \mu$. As $|\nu|(E) \geq |\nu(E)|$ we have that $|\nu| \ll \mu$ implies $\nu \ll \mu$ and hence $\nu \ll \mu$ iff $|\nu| \ll \mu$. If $\nu^+, \nu^- \ll \mu$ then $\mu(E) = 0$ clearly implies $|\nu|(E) = 0$ and if we have $|\nu| \ll \mu$ then $\mu(E) = 0$ implies $\nu^+(E) + \nu^-(E) = 0$ and thus $\nu^+(E) = \nu^-(E) = 0$ proving $\nu^+, \nu^- \ll \mu$ and thus all these conditions are equivalent. $\square$

Problem 3.9. Suppose $\{\nu_j\}$ is a sequence of positive measures. If $\nu_j \perp \mu$ for all j , then $\sum_1^\infty \nu_j \perp \mu$; and if $\nu_j \ll \mu$ for all j , then $\sum_1^\infty \nu_j \ll \mu$.

Solution. Say E_j is ν_j -null and E_j^c is μ -null for all j , then we see that, let $E = \bigcap E_j$, then we clearly see that E is $\sum_1^\infty \nu_j$ -null and we claim that, $E^c = \bigcup E_j^c$ is μ -null. For any

measurable $F \subseteq E^c$, we can write $F = \bigcup_{j=1}^{\infty} (F \cap (E_j^c \setminus \bigcup_{i \neq j}^{\infty} E_i^c))$ is a disjoint union and

$$\mu(F) = \sum_{j=1}^{\infty} \mu(F \cap (E_j^c \setminus \bigcup_{i \neq j}^{\infty} E_i^c)) = 0$$

as $(F \cap (E_j^c \setminus \bigcup_{i \neq j}^{\infty} E_i^c)) \subseteq E_j^c$. The absolute continuity result is trivial. $\square$

Problem 3.10. Theorem 3.5 may fail when ν is not finite. (Consider $d\nu(x) = dx/x$ and $d\mu(x) = dx$ on $(0, 1)$, or $\nu =$ counting measure and $\mu(E) = \sum_{n \in E} 2^{-n}$ on $\mathbb{N}$.)

Solution. We choose the second counterexample. Clearly we have, $\nu \ll \mu$ but for any $\delta > 0$ we see that there exists a E_δ with $\mu(E_\delta) < \delta$ but $\nu(E_\delta) \geq 1$ which just reads “ E is nonempty” and as the sum $\sum_{n \in \mathbb{N}} 2^{-n}$ converges absolutely we can find such a E_δ always. $\square$

Problem 3.11. Let μ be a positive measure. A collection of functions $\{f_\alpha\}_{\alpha \in A} \subset L^1(\mu)$ is called uniformly integrable if for every $\epsilon > 0$ there exists $\delta > 0$ such that $|\int_E f_\alpha d\mu| < \epsilon$ for all $\alpha \in A$ whenever $\mu(E) < \delta$.

- Any finite subset of $L^1(\mu)$ is uniformly integrable.
- If $\{f_n\}$ is a sequence in $L^1(\mu)$ that converges in the L^1 metric to $f \in L^1(\mu)$, then $\{f_n\}$ is uniformly integrable.

Solution.

- Consider $A = \{1, \dots, n\}$; for any $\epsilon > 0$, as $f_j d\mu \ll \mu$ where $f_j d\mu$ are finite measures, we can find $\delta_j > 0$ such that $|\int_E f_j d\mu| < \epsilon$ for all $j \in A$ whenever $\mu(E) < \delta_j$ so taking $\delta = \min_j \delta_j$ suffices.
- As f_n converges in L^1 , it is Cauchy in this metric too so for any $\epsilon > 0$ we can find N such that $m, n \geq N \implies |\int_E f_m - \int_E f_n| \leq \int |f_m - f_n| < \epsilon$ and by the previous problem, as $\{f_1, \dots, f_N\}$ are uniformly integrable, we can find $\delta > 0$ such that, $|\int_E f_j| < \epsilon < 2\epsilon$ for $1 \leq j \leq N$ whenever $\mu(E) < \delta$ and for $n > N$,

$$\left| \int_E f_n d\mu \right| \leq \left| \int_E f_N d\mu \right| + \left| \int_E f_n - f_N d\mu \right| < \epsilon + \epsilon = 2\epsilon$$

so we are done. $\square$

Problem 3.12. For $j = 1, 2$, let μ_j, ν_j be σ -finite measures on $(X_j, \mathcal{M}_j)$ such that $\nu_j \ll \mu_j$. Then $\nu_1 \times \nu_2 \ll \mu_1 \times \mu_2$ and

$$\frac{d(\nu_1 \times \nu_2)}{d(\mu_1 \times \mu_2)}(x_1, x_2) = \frac{d\nu_1}{d\mu_1}(x_1) \frac{d\nu_2}{d\mu_2}(x_2).$$

Solution. Suppose $\mu_1 \times \mu_2(E) = 0$ then we see that $0 = \int \mu_1(E_x) d\mu_2(x)$ implying $\mu_1(E_x) = 0$ (this in turn implies $\nu_1(E_x) = 0$ here) for μ_2 -a.e. x so suppose its true for all x outside N with $\mu_2(N) = 0$ which implies $\nu_2(N) = 0$ so we have that $\nu_1 \times \nu_2(E) = \int \nu_1(E_x) d\nu_2(x) = 0$ as $\nu_1(E_x) = 0$ for ν_2 -a.e. x thus proving $\nu_1 \times \nu_2 \ll \mu_1 \times \mu_2$. Clearly

these product measures are sigma finite too, so we can find a radon nikodym derivative f i.e. $d(\nu_1 \times \nu_2) = f d(\mu_1 \times \mu_2)$ and also f_1, f_2 such that $f_j d\mu_j = \nu_j$ for $j = 1, 2$ and by fubini's we see that,

$$\begin{aligned} \int \chi_E(x, y) f_1(x) f_2(y) d(\mu_1 \times \mu_2(x, y)) &= \int \int f_1(x) \chi_{E_x}(y) f_2(y) d\mu_2(y) d\mu_1(x) \\ &= \int \nu_2(E_x) f_1(x) d\mu_1(x) = \int \nu_2(E_x) d\nu_1(x) = \nu_1 \times \nu_2(E) = \int \chi_E f(x, y) d(\mu_1 \times \mu_2(x, y)) \end{aligned}$$

which proves that $f(x, y) = f_1(x) f_2(y)$ for $\mu_1 \times \mu_2$ a.e. (x, y) so we have the equality treating them as equivalence classes. $\square$

Problem 3.13. Let $X = [0, 1]$, $\mathcal{M} = \mathcal{B}_{[0,1]}$, m = Lebesgue measure, and μ = counting measure on $\mathcal{M}$.

- $m \ll \mu$ but $dm \neq f d\mu$ for any f .
- μ has no Lebesgue decomposition with respect to m .

Solution.

- As $\mu(E) = 0 \implies E = \emptyset$, $m \ll \mu$ holds. If we had $dm = f d\mu$ then we would have $m(\{x\}) = f(x) = 0$ for all x which is absurd as it implies $dm = 0$.
- If we had $\mu = \rho + \lambda$ where $\lambda \perp m$ and $\rho \ll m$, notice how for all singletons we have $m(\{x\}) = 0 \implies \rho(\{x\}) = 0$ and thus $1 = \mu(\{x\}) = 0 + \lambda(\{x\}) = \lambda(\{x\})$ and we can find a partition $L \cup M$ where $\lambda(M) = m(L) = 0$. Now, we see that for any E we have $\lambda(E) = \lambda(E \cap L) = \lambda(E \cap L) + m(E \cap L) = \mu(E \cap L) \geq 0$ as $m(E \cap L) \leq m(L) = 0$ proving that λ is positive. Then, we must have that $\lambda(M) = 0 \implies \lambda(C) = 0$ where $C \subseteq M$ is countably infinite, we can choose such a C because M is required to be uncountable ($m(M) = 1$.) From this we see that $\infty = \mu(C) = \rho(C) + \lambda(C) = 0 + 0$ as $m(C) = 0$ forces $\rho(C) = 0$ so we have a contradiction and the hypothesis was false to begin with.

$\square$

Problem 3.14. If ν is an arbitrary signed measure and μ is a σ -finite measure on $(X, \mathcal{M})$ such that $\nu \ll \mu$, there exists an extended μ -integrable function $f : X \rightarrow [-\infty, \infty]$ such that $d\nu = f d\mu$. Hints:

- It suffices to assume μ is finite and ν is positive.
- With these assumptions there exists $E \in \mathcal{M}$ that is σ -finite for ν and $\mu(E) \geq \mu(F)$ for all sets F that are σ -finite for ν .
- The radon-nikodym theorem applies on E . If $F \cap E = \emptyset$, then either $\nu(F) = \mu(F) = 0$ or $\mu(F) > 0$ and $|\nu(F)| = \infty$.

Solution. If this is true for positive ν and finite μ , then for sigma finite μ we can write $X = \bigcup_1^\infty E_n$ with E_n being μ -finite and mutually disjoint we have $d\nu_n = f_n d\mu_n$ (we safely set $f_n = 0$ outside E_n) where $\mu_n(E) := \mu(E \cap E_n)$ and $\nu_n(E) = \nu(E \cap E_n)$ as μ_n is finite and clearly $\nu_n \ll \mu_n$ still holds. Then, with $f = \sum_1^\infty f_n$ we would have

$d\nu = f d\mu$. If ν was signed on top of that, then using [Problem 3.8](#) which states $\nu \ll \mu \implies \nu^+, \nu^- \ll \mu$ we can find f^+, f^- with $d\nu^- = f^- d\mu, d\nu^+ = f^+ d\mu$ we have $d\nu = d\nu^+ - d\nu^- = (f^+ - f^-) d\mu$ and $f^+ - f^-$ is clearly extended integrable as ν , by definition can only assume atmost one of the values $+\infty, -\infty$. We now prove it for this case. Consider the set $\{\mu(H) : H \in \mathcal{M}, H \text{ is } \nu \text{ } \sigma\text{-finite}\}$, let s be its supremum, then $s < \infty$ as μ is finite and we can find a ν σ -finite sequence of sets F_n such that $\mu(F_n) \rightarrow s$, then we can clearly see that $\mu(\cup F_n) = s$ as $\cup F_n$ is clearly still ν σ -finite and we can show both sides of the equality easily. Setting $E = \cup F_n$ gives us a set as described in hint (b). The radon nikodym theorem applies on E clearly. If $F \cap E = \emptyset$ then if we have $\nu(F) > 0$ we must have $\mu(F) > 0$ as $\nu \ll \mu$ which forces $\nu(F) = \infty$ as otherwise the existence of $E \cup F$ contradicts the maximality of E and if we have $\mu(F) > 0$ then again we similarly have $\nu(F) = \infty$ such as to not contradict the maximality of E so we have proved the claim in hint (c). There are now two cases, first one being when $\nu(E^c) = \mu(E^c) = 0$, in this case we can apply radon-nikodym to get a derivative f on E and extend it to the rest of the set by setting it to be zero outside this, then we see that,

$$\nu(K) = \nu(K \cap E) + \underbrace{\nu(K \cap E^c)}_{\leq \nu(E^c)=0} = \int_{K \cap E} f d\mu = \int_K f d\mu$$

as ν was assumed to be positive and $\mu(E^c) = 0$ so we have $d\nu = f d\mu$ in general so we are done. In the other case where $\mu(E^c) > 0$ and $\nu(E^c) = \infty$ we can set f to be ∞ on E^c , then we see that $\nu(E \cap K) = \int_{E \cap K} f d\mu$ as usual and either $\nu(E^c \cap K) = \mu(E^c \cap K) = 0$ or $\nu(E^c \cap K) = \infty$ and $\mu(E^c \cap K) > 0$ so we see that $\int_{E^c \cap K} f d\mu$ and $\nu(E^c \cap K)$ are either both infinite or both 0 by part (c) of the hint as $E^c \cap K$ is disjoint from E , hence we have the equality,

$$\nu(K) = \nu(E \cap K) + \nu(E^c \cap K) = \int_{E \cap K} f d\mu + \int_{E^c \cap K} f d\mu = \int_K f d\mu$$

so we are done. □

Problem 3.15. A measure μ on $(X, \mathcal{M})$ is called decomposable if there is a family $\mathfrak{F} \subset \mathcal{M}$ with the following properties: (i) $\mu(F) < \infty$ for all $F \in \mathfrak{F}$; (ii) the members of $\mathfrak{F}$ are disjoint and their union is X ; (iii) if $\mu(E) < \infty$ then $\mu(E) = \sum_{F \in \mathfrak{F}} \mu(E \cap F)$; (iv) if $E \subset X$ and $E \cap F \in \mathcal{M}$ for all $F \in \mathfrak{F}$ then $E \in \mathcal{M}$.

- a. Every σ -finite measure is decomposable.
- b. If μ is decomposable and ν is any signed measure on $(X, \mathcal{M})$ such that $\nu \ll \mu$, there exists a measurable $f : X \rightarrow [-\infty, \infty]$ such that $\nu(E) = \int_E f d\mu$ for any E that is σ -finite for μ , and $|f| < \infty$ on any $F \in \mathfrak{F}$ that is σ -finite for ν . (Use Exercise 14 if ν is not σ -finite.)

Solution.

- a. It can be easily shown that $\mathfrak{F} = \{E_n : n \in \mathbb{N}\}$ works where E_n partition X while being μ -finite.
- b. Let $\mathfrak{F}$ be as described in the problem. As $\mu(F) < \infty$ for any $F \in \mathfrak{F}$ we can find a measurable f_F on F such that $\nu(E \cap F) = \int_{E \cap F} f_F d\mu$ for all $E \in \mathcal{M}$ by the previous problem and if F is σ -finite for ν we can clearly force $|f_F| < \infty$ (in

the proof presente in the book we can take all f_j 's which are zero outside A_j to be real valued everywhere as shown in case I which makes their sum real valued too as the A_j are disjoint.) as in the proof of the radon-nikodym theorem. We claim that $f := f_F$ on $F \in \mathfrak{F}$ is measurable and it has the properties needed. For any $E \in \mathcal{B}_{\mathbb{R}}$ we see that $f^{-1}(E) = \bigcup_{F \in \mathfrak{F}} f^{-1}(E) \cap F$ and each $f^{-1}(E) \cap F = f|_F^{-1}(E) \cap F = f_F^{-1}(E) \cap F \in \mathcal{M}$ as f_F is measurable which implies that $f^{-1}(E)$ by property (iv). Now, for any E that is σ -finite for μ , suppose $E = \bigcup E_n$ (disjoint union) with $\mu(E_n) < \infty$, then we see that $\mu(E_n) = \sum_{\mathfrak{F}} \mu(E_n \cap F) < \infty$ and it is well known that only atmost countably many $\mu(E_n \cap F)$ are nonzero, so we can find $\{F_m\} \subseteq \mathfrak{F}$ such that $\mu(E_n) = \sum_{m=1}^{\infty} \mu(E_n \cap F_m)$ and thus we also see that $\mu(E_n \setminus (\bigcup_m (E_n \cap F_m))) = 0$ which, by absolute continuity implies $\nu(E_n) = \nu(\bigcup_m (F_m \cap E_n)) = \sum_{m=1}^{\infty} \mu(E_n \cap F_m) = \sum_{m=1}^{\infty} \int_{E_n \cap F_m} f d\mu = \int_{E_n} f d\mu$ and thus, $\nu(E) = \sum_1^{\infty} \nu(E_n) = \sum_1^{\infty} \int_{E_n} f d\mu = \int_E f d\mu$.

□

Problem 3.16. Suppose that μ, ν are measures on $(X, \mathcal{M})$ with $\nu \ll \mu$, and let $\lambda = \mu + \nu$. If $f = d\nu/d\lambda$, then $0 \leq f < 1$ λ -a.e. and $d\nu/d\mu = f/(1 - f)$.

Solution. (note: we treat the derivatives as equivalence classes here.) Firstly, it is obvious that $f \geq 0$ λ -a.e. We have $\int_E d\nu = \nu(E) = \int_E f d(\mu + \nu) \leq \int_E d\nu + \int_E d\mu = \int_E d(\mu + \nu)$, it can then be easily shown that $f \leq 1$ λ -a.e. If we had $f = 1$ over E with $\lambda(E) = \nu(E) + \mu(E) > 0$ then we see that, $\nu(E) = \int_E f d\lambda = \lambda(E)$ which implies that $\mu(E) = 0$ but then as $\nu \ll \mu$ we have $\nu(E) = 0$ too contradicting non-nullity, this proves that $f < 1$ λ -a.e. too so we have proven the first claim. We clearly have $\nu \ll \mu \ll \lambda$ and $\lambda \ll \mu$. Thus, $g = d\mu/d\lambda$ exists and by Proposition 3.9 we see that $f = d\nu/d\lambda = (d\nu/d\mu)(d\mu/d\lambda) = g(d\nu/d\mu)$ and substituting these in we get $d\nu/d\mu = f/g = f/(1 - f)$ as $f + g = d\nu/d\lambda + d\mu/d\lambda = d(\mu + \nu)/d\lambda = d\lambda/d\lambda = 1$. □

Problem 3.17. Let $(X, \mathcal{M}, \mu)$ be a σ -finite measure space, $\mathcal{N}$ a sub- σ -algebra of $\mathcal{M}$, and $\nu = \mu|_{\mathcal{N}}$ is σ -finite (This condition is missing in the book statement, without which this is false: Consider the lebesgue measure on $\mathbb{R}$ as the measure space and take $\mathcal{N} = \{\emptyset, \mathbb{R}\}$, $f = \chi_{[0,1]}$.) If $f \in L^1(\mu)$, there exists $g \in L^1(\nu)$ (thus g is $\mathcal{N}$ -measurable) such that $\int_E f d\mu = \int_E g d\nu$ for all $E \in \mathcal{N}$; if g' is another such function then $g = g'$ ν -a.e. (In probability theory, g is called the conditional expectation of f on $\mathcal{N}$.)

Solution. As $f \in L^1$ we have that $f d\mu$ is a signed measure and it is clearly still a measure restricted to $\mathcal{N}$ where we see that $(f d\mu)|_{\mathcal{N}} \ll \nu = \mu|_{\mathcal{N}}$ and as $\mathcal{N}$ is sigma finite for ν we can apply Problem 3.14 to deduce existence of some g such that $(f d\mu)|_{\mathcal{N}} = g d\nu$ which implies $\int_E f d\mu = \int_E g d\nu$ for all $E \in \mathcal{N}$. Here, g is extended ν integrable by construction. To show that $g \in L^1(\nu)$ consider the fact that $(|f| d\mu)|_{\mathcal{N}} = |(f d\mu)|_{\mathcal{N}} = |g d\nu| = |g| d\nu$ and thus, $\int_X |g| d\nu = \int_X |f| d\mu < \infty \implies g \in L^1(\nu)$. The uniqueness follows from the fact that $\int_E g d\nu = \int_E g' d\nu$ for all $E \in \mathcal{N}$ implies $g = g'$ ν -a.e. □

Problem 3.18. Prove Proposition 3.13c.

Solution. Let g be as in the definition ($d\nu = g d\mu$ with μ being positive) of $|\nu|$ then we

see that $|f|d|\nu| = |f||g|d\mu$ and we also have $f d\nu = fg d\mu$ and we have that,

$$\left| \int f d\nu \right| = \left| \int fg d\mu \right| \leq \int |fg| d\mu = \int |f|d|\nu|$$

□

Problem 3.19. If ν, λ are complex measures and μ is a positive measure, then $\nu \perp \mu$ iff $|\nu| \perp |\mu|$, and $\nu \ll \lambda$ iff $|\nu| \ll \lambda$.

Solution. First, we prove $\nu \perp \mu \iff |\nu| \perp \mu$. If $\nu \perp \mu$, then by definition $\nu_r \perp \mu$ and $\nu_i \perp \mu$. By [Problem 3.2](#), this implies $|\nu_r| \perp \mu$ and $|\nu_i| \perp \mu$. Using [Problem 3.9](#), their sum is also mutually singular to μ , meaning $(|\nu_r| + |\nu_i|) \perp \mu$. Since total variation satisfies $|\nu| \leq |\nu_r| + |\nu_i|$, any set that is null for $|\nu_r| + |\nu_i|$ is also null for $|\nu|$. Therefore, $|\nu| \perp \mu$. Conversely, if $|\nu| \perp \mu$, since $|\nu_r| \leq |\nu|$ and $|\nu_i| \leq |\nu|$, we have $|\nu_r| \perp \mu$ and $|\nu_i| \perp \mu$. By [Problem 3.2](#), this implies $\nu_r \perp \mu$ and $\nu_i \perp \mu$, meaning $\nu \perp \mu$. Now, we prove $\nu \ll \lambda \iff |\nu| \ll \lambda$. By definition, absolute continuity with respect to a complex measure means absolute continuity with respect to its total variation, so $\nu \ll \lambda$ means $|\lambda|(E) = 0 \implies \nu(E) = 0$. If $|\nu| \ll \lambda$, then $|\lambda|(E) = 0 \implies |\nu|(E) = 0$. Since $|\nu(E)| \leq |\nu|(E)$, this immediately implies $\nu(E) = 0$, proving $\nu \ll \lambda$. Conversely, suppose $\nu \ll \lambda$ and let $|\lambda|(E) = 0$. Since $|\lambda|$ is a positive measure, for any measurable subset $F \subseteq E$, we also have $|\lambda|(F) = 0$, which implies $\nu(F) = 0$. The total variation can be equivalently defined as $|\nu|(E) = \sup \sum_{j=1}^n |\nu(E_j)|$ over all finite measurable partitions $\{E_j\}$ of E by [Problem 3.21](#). Since each $E_j \subseteq E$, we have $\nu(E_j) = 0$ for all j . Thus, the supremum is exactly 0, giving $|\nu|(E) = 0$. This proves $|\nu| \ll \lambda$. □

Problem 3.20. If ν is a complex measure on $(X, \mathcal{M})$ and $\nu(X) = |\nu|(X)$, then $\nu = |\nu|$.

Solution. We see that $\nu(X) = |\nu(X)| = |\nu(E) + \nu(X \setminus E)| \leq |\nu(E)| + |\nu(X \setminus E)| \leq |\nu|(E) + |\nu|(X \setminus E) = |\nu|(X) = \nu(X)$ so all the inequalities must be equalities, which implies that $|\nu(E)| = |\nu|(E)$ as $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{M}$. As we have $|\nu(E)| + |\nu(X \setminus E)| = |\nu(E) + \nu(X \setminus E)|$ which is real, those two complex numbers share the same argument (unless 0 but that case is trivial) and as $\nu(E) + \nu(X \setminus E) = \nu(X)$ is real too, they must have no complex part making them real which implies that, $\nu(E) = |\nu(E)| = |\nu|(E)$ proving the claim. □

Problem 3.21. Let ν be a complex measure on $(X, \mathcal{M})$. If $E \in \mathcal{M}$ define

$$\mu_1(E) = \sup \left\{ \sum_1^n |\nu(E_j)| : n \in \mathbb{N}, E_1, \dots, E_n \text{ disjoint}, E = \bigcup_1^n E_j \right\},$$

$$\mu_2(E) = \sup \left\{ \sum_1^\infty |\nu(E_j)| : E_1, E_2, \dots \text{ disjoint}, E = \bigcup_1^\infty E_j \right\},$$

$$\mu_3(E) = \sup \left\{ \left| \int_E f d\nu \right| : |f| \leq 1 \right\}.$$

Then $\mu_1 = \mu_2 = \mu_3 = |\nu|$. (First show that $\mu_1 \leq \mu_2 \leq \mu_3$. To see that $\mu_3 = |\nu|$, let $f = \overline{d\nu/d|\nu|}$ and apply Proposition 3.13. To see that $\mu_3 \leq \mu_1$, approximate f by simple functions.)

Solution. $\mu_1 \leq \mu_2$ is obvious and $\mu_2 \leq \mu_3$ holds as for any given E and any partition $\{E_n\}$ of E we can define

$$f = \sum_n \frac{\overline{\nu(E_n)}}{|\nu(E_n)|} \cdot \chi_{E_n}$$

where the sum is over all n for which $\nu(E_n) \neq 0$, then we clearly see that $|f| \leq 1$ and $\int_E f d\nu = \sum_1^\infty \int_{E_n} f d\nu = \sum_1^\infty |\nu(E_n)|$ as $|z| = \bar{z}z/|z|$ for nonzero z . Clearly $\mu_3(E) \leq \int_E d|\nu| = |\nu|(E)$ by Proposition 3.13c and with f as in the hint (this can be made to have absolute value atmost 1 everywhere after changing it on a null set, not affecting any of the integrals below) we see that,

$$\int_E \frac{\overline{d\nu}}{d|\nu|} d\nu = \int_E \frac{\overline{d\nu}}{d|\nu|} \frac{d\nu}{d|\nu|} d|\nu| = \int_E |f|^2 d|\nu| = \int_E d|\nu| = |\nu|(E)$$

as $|f| = 1$ $|\nu|$ -a.e. so we must have $\mu_3 = |\nu|$. Now, we can find a sequence of simple functions ϕ_n converging f with $|\phi_n| \leq |f| \leq 1$ and convergence can be made uniform here as f is bounded on E , this clearly implies that $\lim |\int_E \phi_n d\nu| = |\int_E f d\nu|$ and thus, as $|\int_E \phi_n d\nu| \leq \mu_1(E)$ due to linearity and the triangle inequality we must have $\lim_n |\int_E f d\nu| \leq \mu_1(E)$ too so we are done. $\square$

Problem 3.22. If $f \in L^1(\mathbb{R}^n)$, $f \neq 0$, there exist $C, R > 0$ such that $Hf(x) \geq C|x|^{-n}$ for $|x| > R$. Hence $m(\{x : Hf(x) > \alpha\}) \geq C'/\alpha$ when α is small, so the estimate in the maximal theorem is essentially sharp.

Solution. We are given that f is not 0 a.e. and thus, $\int_{\mathbb{R}} |f| d\mu > 0$ and using DCT, we see that there exists some $r > 0$ such that $K = \int_{B(0,r)} |f| d\mu > 0$. Then, for all x , as $B(0, r) \subseteq B(x, |x| + r)$ we have

$$Hf(x) \geq \frac{1}{m(B(x, |x| + r))} \int_{B(x, |x| + r)} |f| d\mu \geq \frac{1}{c \cdot (|x| + r)^n} \int_{B(0,r)} |f| d\mu \geq \frac{K}{(|x| + r)^n}$$

where $c = m(B(0, 1))$. Thus, we can find some $C > 0$ and $R > 0$ like in the statement. Hence, for small $\alpha > 0$ we have $\{x : Hf(x) > \alpha\} \supseteq \{x : C\alpha^{-1} \geq |x|^n > R^n\}$. From this we can deduce that, $m(\{x : Hf(x) > \alpha\}) \geq c \cdot (C\alpha^{-1} - R^n)$ which is atleast $C'\alpha^{-1}$ for some $C' > 0$ and small α . $\square$

Problem 3.23. A useful variant of the Hardy-Littlewood maximal function is $H^*f(x) = \sup\{\frac{1}{m(B)} \int_B |f(y)| dy : B \text{ is a ball and } x \in B\}$. Show that $Hf \leq H^*f \leq 2^n Hf$.

Solution. If $x \in B(p, r) = B$ then we must have that $B(p, r) \subseteq B(x, |x - p| + r)$, thus

$$\frac{1}{m(B)} \int_B |f| dm \leq \frac{c \cdot (|x - p| + r)^n}{c \cdot r^n} Hf(x) \leq 2^n Hf(x)$$

where $c = m(B(0, 1))$, as $|x - p| < r$ so we must have $H^*f \leq 2^n Hf$ and $H^*f \geq Hf$ is trivially true. $\square$

Problem 3.24. If $f \in L^1_{loc}$ and f is continuous at x , then x is in the Lebesgue set of f .

Solution. For all $\epsilon > 0$ we can find $\delta > 0$ such that $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$, thus for all $0 < \delta' < \delta$ we must have $m(B(x, \delta'))^{-1} \int_{B(x, \delta')} |f(x) - f(y)| dy \leq \epsilon$ so it must converge to 0 implying $x \in L_f$. $\square$

Problem 3.25. If E is a Borel set in $\mathbb{R}^n$, the density $D_E(x)$ of E at x is defined as $D_E(x) = \lim_{r \rightarrow 0} \frac{m(E \cap B(r, x))}{m(B(r, x))}$ whenever the limit exists.

- Show that $D_E(x) = 1$ for a.e. $x \in E$ and $D_E(x) = 0$ for a.e. $x \in E^c$.
- Find examples of E and x such that $D_E(x)$ is a given number $\alpha \in (0, 1)$, or such that $D_E(x)$ does not exist.

Solution.

- Consider the borel measure $\nu(F) := \mu(E \cap F)$, then we have a lebesgue decomposition $\nu = 0 + \chi_E dm$ and Theorem 3.22. we have that $D_E(x) = \chi_E(x)$ for m -a.e. x . from which the conclusion follows.
- Consider $E = \bigcup_{k=1}^{\infty} (B(0, 2^{-2k+1}) \setminus B(0, 2^{-2k}))$ and $x = 0$. Let $c = m(B(0, 1))$. Then we see that for $m \in \mathbb{N}$ we have

$$m(E \cap B(0, 2^{-2m+1})) = c \sum_{k=m}^{\infty} (2^{n(-2k+1)} - 2^{-2kn}) = c(2^n - 1) \frac{2^{-2mn}}{1 - 2^{-2n}}.$$

Dividing by $m(B(0, 2^{-2m+1})) = c2^{n(-2m+1)}$, the density along this sequence of radii is:

$$\frac{m(E \cap B(0, 2^{-2m+1}))}{m(B(0, 2^{-2m+1}))} = \frac{c(2^n - 1)2^{-2mn}/(1 - 2^{-2n})}{c2^{-2mn}2^n} = \frac{2^n}{2^n + 1}.$$

Now consider the sequence of radii $r = 2^{-2m}$. The ball $B(0, 2^{-2m})$ intersects E starting from index $k = m + 1$, so:

$$m(E \cap B(0, 2^{-2m})) = c \sum_{k=m+1}^{\infty} (2^{n(-2k+1)} - 2^{-2kn}) = c(2^n - 1) \frac{2^{-2n(m+1)}}{1 - 2^{-2n}}.$$

Dividing by $m(B(0, 2^{-2m})) = c2^{-2mn}$ yields:

$$\frac{m(E \cap B(0, 2^{-2m}))}{m(B(0, 2^{-2m}))} = \frac{c(2^n - 1)2^{-2mn}2^{-2n}/(1 - 2^{-2n})}{c2^{-2mn}} = \frac{1}{2^n + 1}.$$

Since $\frac{2^n}{2^n + 1} \neq \frac{1}{2^n + 1}$ for all $n \geq 1$, two subsequences tend to two different limits, hence $D_E(0)$ does not exist. To find an example where $D_E(x) = \alpha \in (0, 1)$, choose $x = 0$. If $n \geq 2$, define E as a sector of $\mathbb{R}^n$ originating at 0. Let $A \subset S^{n-1}$ be a measurable set on the unit sphere such that its surface measure is $\alpha \sigma(S^{n-1})$ which we can do by considering the function $f : [-1, 1] \rightarrow \mathbb{R}$ defined by $f(t) := \sigma(\{x \in S^{n-1} : x_1 \leq t\})$ which can be showed to be continuous via lower and upper continuity, after which we can use the intermediate value theorem. Define $E = \{y \in \mathbb{R}^n \setminus \{0\} : y/|y| \in A\}$. For any $r > 0$, $m(E \cap B(0, r)) = \alpha m(B(0, r))$, meaning $D_E(0) = \alpha$ identically for all $r > 0$. If $n = 1$, define $E = \bigcup_{k=1}^{\infty} \left(\left[\frac{1}{k} - \frac{\alpha}{k(k+1)}, \frac{1}{k} \right] \cup \left[-\frac{1}{k}, -\frac{1}{k} + \frac{\alpha}{k(k+1)} \right] \right)$. For $r = 1/k$, $m(E \cap (-r, r)) = 2 \sum_{j=k}^{\infty} \frac{\alpha}{j(j+1)} = \frac{2\alpha}{k} = \alpha(2r)$. Bounding the measure for $r \in (\frac{1}{k+1}, \frac{1}{k})$ shows the ratio to α , hence $D_E(0) = \alpha$.

□

Problem 3.26. If λ and μ are positive, mutually singular Borel measures on $\mathbb{R}^n$ and $\lambda + \mu$ is regular, then so are λ and μ .

Solution. Define $\nu = \lambda + \mu$ and let $\lambda(A) = \mu(A^c) = 0$. For compact K we see that $\lambda(K) + \mu(K) = \nu(K) < \infty$, which implies that $\lambda(K), \mu(K) < \infty$. First, we show outer regularity. If $\lambda(E) = \infty$, the condition $\lambda(E) = \inf\{\lambda(U) : U \supseteq E, U \text{ open}\}$ is trivially satisfied by any open $U \supseteq E$. Assume $\lambda(E) < \infty$. We wish to find an open $U \supseteq E$ with $\lambda(U \setminus E) < \epsilon$. We split E into $E \cap A^c$ and $E \cap A$. Notice that $\nu(E \cap A^c) = \lambda(E \cap A^c) + \mu(E \cap A^c) = \lambda(E \cap A^c) + 0 \leq \lambda(E) < \infty$. Since $\nu(E \cap A^c) < \infty$ and ν is regular, there exists an open $U_1 \supseteq (E \cap A^c)$ with $\nu(U_1 \setminus (E \cap A^c)) < \epsilon/2$, which implies $\lambda(U_1 \setminus (E \cap A^c)) \leq \nu(U_1 \setminus (E \cap A^c)) < \epsilon/2$. For the remaining piece $E \cap A$, we have $\lambda(E \cap A) \leq \lambda(A) = 0$. We can write $A = \bigcup_{k=1}^{\infty} A_k$ where $A_k = A \cap B(0, k)$. As each A_k is bounded, $\nu(A_k) < \infty$, so there exist open sets $V_k \supseteq A_k$ with $\nu(V_k \setminus A_k) < \epsilon/2^{k+1}$. This implies $\lambda(V_k) = \lambda(V_k \setminus A_k) + \lambda(A_k) < \epsilon/2^{k+1} + 0$. Taking $U_2 = \bigcup_{k=1}^{\infty} V_k$ gives an open set containing A (and thus $E \cap A$) with $\lambda(U_2) \leq \sum \lambda(V_k) < \epsilon/2$. Let $U = U_1 \cup U_2$. Then U is open, $U \supseteq E$, and $U \setminus E \subseteq (U_1 \setminus (E \cap A^c)) \cup U_2$. Thus, $\lambda(U \setminus E) \leq \lambda(U_1 \setminus (E \cap A^c)) + \lambda(U_2) < \epsilon/2 + \epsilon/2 = \epsilon$ so we have the second condition. The first condition is trivial. $\square$

Problem 3.27. Verify the assertions in Examples 3.25.

Solution. This is trivial. $\square$

Problem 3.28. If $F \in NBV$, let $G(x) = |\mu_F|((-\infty, x])$. Prove that $|\mu_F| = \mu_{T_F}$ by showing that $G = T_F$ via the following steps.

- From the definition of T_F , $T_F \leq G$.
- $|\mu_F(E)| \leq \mu_{T_F}(E)$ when E is an interval, and hence when E is a Borel set.
- $|\mu_F| \leq \mu_{T_F}$, and hence $G \leq T_F$.

Solution.

- For any $x \in \mathbb{R}$ and $x_0 < \dots < x_n = x$ we see that,

$$\sum_{k=1}^n |F(x_k) - F(x_{k-1})| = \sum_{k=1}^n |\mu_F((x_{k-1}, x_k])| \leq \sum_{k=1}^n |\mu_F|((x_{k-1}, x_k]) = |\mu_F|((x_0, x])$$

Which is clearly less than $G(x)$. This shows that $T_F \leq G$ as T_F is the supremum over all sums of this form.

- For open (we only need these) intervals, the inequality follows from the fact that $T_F(x) - T_F(y) \geq |F(x) - F(y)|$ for $x \geq y$ as we can take limits and say that, for $a < b$

$$\mu_{T_F}((a, b)) = T_F(b-) - T_F(a) \geq |F(b-) - F(a)| = |\mu_F((a, b))|$$

(keep in mind that all the one sided limits exist in this case.) We know by construction that $\mu_F = \mu_1 - \mu_2 + i(\mu_3 - \mu_4)$ for finite lebesgue stjelt measures μ_i as in chapter 1. Let $\mu_5 = \mu_{T_F}$, another lebesgue stjelt measure. Given borel E , by regularity we can find open sets $U_n^{(i)} \supseteq E$ such that $\mu_i(U_n^{(i)}) \rightarrow \mu_i(E)$ for all i . Notice that by the $\diamond$ theorem and monotonicity of μ_i , the sequence of open sets $V_n := \bigcap_i U_n^{(i)}$ have the same convergence property by monotonicity of measures and the sandwich property. As μ_i are all finite we see that $\mu_F(V_n) \rightarrow \mu_F(E)$. If $\bigcup I_n$ is a countable disjoint union of open intervals we see that $|\mu_F(\bigcup I_n)| \leq \mu_{T_F}(\bigcup I_n)$ by the triangle inequality and sigma additivity of measures. From this, we can conclude

that $|\mu_F(V_n)| \leq \mu_{T_F}(V_n)$ for all n and we have the desired inequality by taking the limit as $n \rightarrow \infty$

- c. This can be proven by the previous part and [Problem 3.21](#) directly (look at μ_1) in the hyperlinked problem. From this we thus have that $G \leq T_F$ by considering left open rays.

Now that we have $G = T_F$ we must have that $\mu_G = \mu_{T_F}$ and μ_G is none other than $|\mu_F|$. $\square$

Problem 3.29. If $F \in NBV$ is real-valued, then $\mu_F^+ = \mu_P$ and $\mu_F^- = \mu_N$ where P and N are the positive and negative variations of F .

Solution. It is easily seen that, the positive and negative variations are nonnegative for NBV functions as $F(-\infty) = T_F(-\infty) = 0$. Now, we clearly have $\mu_F = \mu_P - \mu_N$ (this is true for h-intervals and the sets where this holds forms a sigma algebra due to these being finite measures) which proves that $\mu_P \geq \mu_F^+$ and $\mu_N \geq \mu_F^-$ by [Problem 3.4](#). Now, from page 103 we can also see that

$$P(x) = \sup \left\{ \sum_{k=1}^n [F(x_k) - F(x_{k-1})]^+ : x_0 < \dots < x_n = x \right\} + 0$$

as $F \in NBV$, now notice that for $a < b$, we have

$$[F(b) - F(a)]^+ = [\mu_F((a, b))]^+ = [\mu_F^+((a, b)) - \mu_F^-((a, b))]^+ \leq [\mu_F^+((a, b))]^+ = \mu_F^+((a, b))$$

From this we can easily see that $\mu_P((-\infty, x]) = P(x) \leq \mu_F^+((-\infty, x])$. But earlier we proved that $\mu_P \geq \mu_F^+$ so we must have equality above and hence $\mu_P = \mu_F^+$ can be easily shown to hold. $\square$

Problem 3.30. Construct an increasing function on $\mathbb{R}$ whose set of discontinuities is $\mathbb{Q}$.

Solution. Consider an enumeration $\{q_n\}$ of the rationals. Define $f(x) := \sum_{q_n < x} 2^{-n}$ then it is obvious that for rationals r we have $f(q_n+) - f(q_n-) \geq 2^{-n}$ making it discontinuous on the rationals. Now, if x is an irrational, let $\delta > 0$ be such that $q_1, \dots, q_m \notin (x - \delta, x + \delta)$, we can do this because x is irrational. In this case, we see that for $y \in (x - \delta, x + \delta)$

$$|f(x) - f(y)| = \sum_{q_n \in [x - \delta, x + \delta)} \frac{1}{2^n} \leq \sum_{n > m} \frac{1}{2^n} = \frac{1}{2^{m+1}}$$

Thus we have continuity at the irrationals as this can be made arbitrarily small for larger m and small enough $\delta > 0$. $\square$

Problem 3.31. Let $F(x) = x^2 \sin(x^{-1})$ and $G(x) = x^2 \sin(x^{-2})$ for $x \neq 0$, and $F(0) = G(0) = 0$.

- a. F and G are differentiable everywhere (including $x = 0$).
b. $F \in BV([-1, 1])$, but $G \notin BV([-1, 1])$.

Solution. The first part is an exercise in calculus and the next follows from F' being bounded ($|F'| \leq 3$ to be a bit more specific) and $T_G(1) - T_G(n^{-1/2}) \gg \log n$ (this is not to be interpreted as $\infty - \infty$ but rather the total variation in the smaller interval $(n^{-1/2}, 1)$) $\square$

Problem 3.32. If $F_1, F_2, \dots, F \in NBV$ and $F_j \rightarrow F$ pointwise, then $T_F \leq \liminf T_{F_j}$.

Solution. For any $x \in \mathbb{R}$ and $\epsilon > 0$ consider some $x_0 < \dots < x_n = x$ such that,

$$\sum_{k=1}^n |F(x_k) - F(x_{k-1})| \geq T_F(x) - \epsilon$$

Then, by pointwise convergence we see that for large enough j ,

$$-\epsilon + \sum_{k=1}^n |F(x_k) - F(x_{k-1})| \leq \sum_{k=1}^n |F_j(x_k) - F_j(x_{k-1})| \leq T_{F_j}(x)$$

This shows that $T_F(x) - 2\epsilon \leq T_{F_j}(x)$ for large enough j and thus, $T_F(x) - 2\epsilon \leq \liminf T_{F_j}(x)$. As $\epsilon > 0$ was arbitrary we are done. $\square$

Problem 3.33. If F is increasing on $\mathbb{R}$, then $F(b) - F(a) \geq \int_a^b F'(t)dt$.

Solution. As we only care about things being defined a.e. we define $F_n(x) := n(F(x + 1/n) - F(x)) \geq 0$ and we see that, $F_n \rightarrow F'$ a.e. thus, by fatou's lemma we have

$$\int_a^b F' \leq \liminf \int_a^b F_n = \liminf_n n \int_a^b F(x + 1/n) - F(x) dm(x)$$

If we set $G(x) := \int_a^x F(t)dm(t)$ then we can pass to rieman integrability (monotone functions are rieman integrable) and see that the last expression above is

$$\liminf_n \frac{G(b + 1/n) - G(a + 1/n) - (G(b) - G(a))}{1/n} = G'(b) - G'(a) = F(b) - F(a)$$

As $G'(t) = F(t)$ by the fundamental theorem of calculus, when F is continuous at a, b . Now, as F is continuous a.e. the set of points where it is continuous must be dense and we can thus find sequences $a_n \searrow a < b \nearrow b_n$ where F is continuous and thus,

$$F(b) - F(a) \geq F(b-) - F(a+) = \lim_{n \rightarrow \infty} F(b_n) - F(a_n) \geq \lim \int_{[a_n, b_n]} F' = \int_{(a, b)} F' = \int_a^b F'$$

where the last two equalities are due to the fact that $A \mapsto \int_A F'$ is a measure here ((a_n, b_n) increases to (a, b) and $m(\{a, b\}) = 0$). $\square$

Problem 3.34. Suppose $F, G \in NBV$ and $-\infty < a < b < \infty$.

a. By adapting the proof of Theorem 3.36, show that

$$\int_{[a, b]} \frac{F(x) + F(x-)}{2} dG(x) + \int_{[a, b]} \frac{G(x) + G(x-)}{2} dF(x) = F(b)G(b) - F(a-)G(a-)$$

b. If there are no points in $[a, b]$ where F and G are both discontinuous, then

$$\int_{[a, b]} F dG + \int_{[a, b]} G dF = F(b)G(b) - F(a-)G(a-).$$

Solution.

- a. Following steps similar to those the proof of theorem 3.36, taking $\Omega = \{(x, y) : a \leq x \leq y \leq b\}$ we can see that,

$$\int_{[a,b]} FdG - F(a-)[G(b) - G(a-)] = G(b)[F(b) - F(a-)] - \int_{[a,b]} G(x-)dF(x)$$

We can also switch F, G and get another equation

$$\int_{[a,b]} GdF - G(a-)[F(b) - F(a-)] = F(b)[G(b) - G(a-)] - \int_{[a,b]} F(x-)dG(x)$$

rearranging and adding these we get,

$$\int_{[a,b]} (F(x) + F(x-))dG(x) + \int_{[a,b]} (G(x) + G(x-))dF(x) = 2(F(b)G(b) - F(a-)G(a-))$$

- b. If F is continuous at x then we see that

$$\mu_F(\{x\}) = \lim_n \mu_F((x + 1/n, x]) = \lim_n F(x) - F(x + 1/n) = 0$$

and similarly for G . As F, G are NBV we can write them as the difference of two increasing functions which only ever have countably many discontinuities and thus the set of discontinuities here form countable sets which we call D_F, D_G for F and G respectively. Then we can see that $\mu_F(D_G) = \sum_{x \in D_G} \mu_F(\{x\}) = \sum_{x \in D_G} 0 = 0$ because F is continuous on D_G and similarly $\mu_G(D_F) = 0$. Now, in the equation from the previous part we can write the LHS as,

$$\int_{[a,b] \setminus D_F} (F(x) + F(x-))dF(x) + \int_{[a,b] \setminus D_G} (G(x) + G(x-))dF(x)$$

as the integrands are continuous over their domain of integration, we see that $F(x) + F(x-) = 2F(x)$ and similarly for G . From this we can conclude that,

$$\int_{[a,b]} 2FdG + \int_{[a,b]} 2GdF = 2(F(b)G(b) - F(a-)G(a-))$$

from which we get the desired equality after cancelling the 2 out.

□

Problem 3.35. If F and G are absolutely continuous on $[a, b]$, then so is FG , and

$$\int_a^b (FG' + GF')(x)dx = F(b)G(b) - F(a)G(a).$$

Solution. It suffices to show that F being absolutely continuous implies F^2 is absolutely continuous. Indeed, if F is absolutely continuous then it is continuous too and thus is bounded on the compact set $[a, b]$, say $|F| \leq M$. If, $\epsilon > 0$ is arbitrary, we can find $\delta > 0$ with,

$$\sum_{i=1}^n (b_i - a_i) < \delta \implies \sum_{i=1}^n |F(b_i) - F(a_i)| < \frac{\epsilon}{M + 1} \text{ for } a \leq a_1 < b_1 < \dots < b_n \leq b$$

Then,

$$\sum_{i=1}^n |F(b_i)^2 - F(a_i)^2| \leq M \sum_{i=1}^n |F(b_i) - F(a_i)| \leq \frac{\epsilon M}{M+1} < \epsilon$$

so F^2 is also absolutely continuous. Now, it is clear that F, G being absolutely continuous makes their product absolutely continuous too. From Theorem 3.35c we see that $(FG)' \in L^1([a, b], m)$ and the same also holds for F', G' . As they all exist a.e., they exist together a.e. as well and we can say that $(FG)' = FG' + F'G$ a.e. and again by the cited theorem,

$$F(b)G(b) - F(a)G(a) = \int_a^b (FG)'(x)dx = \int_a^b (FG' + F'G)(x)dx$$

□

Problem 3.36. Let G be a continuous increasing (we need G to be strictly increasing, otherwise consider constant functions.) function on $[a, b]$ and let $G(a) = c$, $G(b) = d$.

- If $E \subset [c, d]$ is a Borel set, then $m(E) = \mu_G(G^{-1}(E))$. (First consider the case where E is an interval.)
- If f is a Borel measurable and integrable function on $[c, d]$, then $\int_c^d f(y)dy = \int_a^b f(G(x))dG(x)$. In particular, $\int_c^d f(y)dy = \int_a^b f(G(x))G'(x)dx$ if G is absolutely continuous.
- The validity of (b) may fail if G is merely right continuous rather than continuous.

Solution.

- The case where this is an interval follows as shown, for $c \leq x < y \leq d$ we just have $\mu_G(G^{-1}([x, y])) = \mu_G([G^{-1}(x), G^{-1}(y)]) = G \circ G^{-1}(y) - G \circ G^{-1}(x) = y - x = m([x, y])$. We can easily show that the collection of sets E for which this works is a sigma algebra as the inverse commutes with the complement and unions, we thus have that the borel sigma algebra over $[c, d]$ is contained in this one which suffices.
- If $f = \chi_E$ ($E \in \mathcal{B}_{[c, d]}$) then, $\int_a^b \chi_E(G(x))dG(x) = \int_a^b \chi_{G^{-1}(E)}dG(x) = \mu_G(G^{-1}(E)) = m(E) = \int_c^d \chi_E dm$ and we can conclude by linearity of the integral and MCT as usual. If G is NBV, we have that $dG = G'dm$ (can be proven from 3.35c) and by [Problem 2.14](#) we are done (split the integrable function into its two parts as usual.)
- In that case the range of f needn't be an interval in the first place. For example, take $f(x) := x\chi_{[0, 1/2)}(x) + (x+1)\chi_{[1/2, 1]}$ which is right continuous, increasing and maps $[0, 1]$ to $[0, 1/2) \cup [3/2, 2]$.

□

Problem 3.37. Suppose $F : \mathbb{R} \rightarrow \mathbb{C}$. There is a constant M such that $|F(x) - F(y)| \leq M|x - y|$ for all $x, y \in \mathbb{R}$ (that is, F is Lipschitz continuous) iff F is absolutely continuous and $|F'| \leq M$ a.e.

Solution. For the only if direction, absolute continuity follows directly from the definition by considering $\epsilon/M + 1$ instead of ϵ and for the derivative to exist a.e. (it is trivially bounded as needed when it does exist) we consider intervals $(n, n+1]$ for $n \in \mathbb{Z}$. Clearly

F is still absolutely continuous on this small interval and by Theorem 3.35c we again see that F' exists a.e. in $(n, n+1]$ and thus it exists a.e. in $\mathbb{R}$ too as the union of countably many disjoint null sets is still null. To prove the if part, for $x < y$ consider F on the interval $[x, y]$. By Theorem 3.35c we can write $F(x) - F(y) = \int_x^y F'(t)dt$ again and by hypothesis we have,

$$|F(x) - F(y)| = \left| \int_x^y F'(t)dt \right| \leq \int_x^y |F'(t)|dt \leq M(y - x) = M|x - y|$$

The case of $x > y$ is no different. □

Problem 3.38. If $f : [a, b] \rightarrow \mathbb{R}$, consider the graph of f as a subset of $\mathbb{C}$, namely, $\{t + if(t) : t \in [a, b]\}$. The length L of this graph is by definition the supremum of the lengths of all inscribed polygons.

- a. Let $F(t) = t + if(t)$; then L is the total variation of F on $[a, b]$.
- b. If f is absolutely continuous, $L = \int_a^b [1 + f'(t)^2]^{1/2} dt$.

Solution.

- a. This is trivial, just remember how $|x + iy| = \sqrt{x^2 + y^2}$.
- b. If f is absolutely continuous then so is F and by problem 3.35c it can be shown that $\mu_F = F'dm$ on this interval and thus, $\mu_{T_F} = |F'|dm$ by [Problem 3.28](#), this shows that

$$L = T_F(b) - T_F(a) = \mu_{T_F}((a, b]) = \int_{(a, b]} |F'(t)|dm(t) = \int_a^b [1 + f'(t)^2]^{1/2} dm(t)$$

as $F'(t) = 1 + i \cdot f'(t)$ (when it exists.) □

Problem 3.39. If $\{F_j\}$ is a sequence of nonnegative increasing functions on $[a, b]$ such that $F(x) = \sum_1^\infty F_j(x) < \infty$ for all $x \in [a, b]$, then $F'(x) = \sum_1^\infty F'_j(x)$ for a.e. $x \in [a, b]$. (It suffices to assume $F_j \in NBV$. consider the measures μ_{F_j} .)

Solution. Keep in mind the following lemma : If $P_1, P_2, \dots$ hold a.e. then $\bigwedge P_i$ does too, the proof is left as an exercise. We can consider the functions $G_j(x) = F_j(x+) - F_j(a)$ and $G(x) = F(x+) - F(a)$ instead. These have the same properties as F_j and F , and we extend them to make them NBV as done before (they were trivially in $BV([a, b])$ already). Note that $G = \sum G_j$ everywhere, which follows from an application of the monotone convergence theorem with respect to the counting measure. Consider the measures μ_G and μ_{G_j} , for which it holds that $\mu_G = \sum_j \mu_{G_j}$ since they agree on right-closed rays. Now, consider the Lebesgue-Radon-Nikodym decompositions $dG = g dm + d\lambda$ and $dG_j = g_j dm + d\lambda_j$. We have $\lambda_j \perp m$, and thus $\sum \lambda_j \perp m$ too by [Problem 3.9](#) (notice that the proof is still valid if ν_j are complex measures as well.) Since $\mu_G = \sum \mu_{G_j}$, we must have $dG = (\sum_j g_j) dm + \sum \lambda_j$. By the uniqueness criterion in the Lebesgue-Radon-Nikodym theorem, this implies $g = \sum g_j$ a.e. and $\lambda = \sum \lambda_j$. It is clear from the lebesgue differentiation theorem that $g_j = G'_j$ a.e. and $g = G'$ a.e., so the lemma at the beginning implies that $G' = \sum G'_j$ a.e. If we now show that $F' = G'$ and $F'_j = G'_j$ a.e., then we would

be done using the same lemmat. Suppose p and q are such that $p = q$ a.e. and both are differentiable a.e. We want to show that $p' = q'$ a.e. We can find a set $A \subseteq [a, b]$ where $p = q$ and both are differentiable, such that $m([a, b] \setminus A) = 0$ by the lemma. Notice that A must be dense in this interval, so for any $x \in A$, we can find a sequence $x_n \in A \setminus \{x\}$ that tends to x . By the sequential criterion for limits, we have

$$p'(x) = \lim_{n \rightarrow \infty} \frac{p(x_n) - p(x)}{x_n - x} = \lim_{n \rightarrow \infty} \frac{q(x_n) - q(x)}{x_n - x} = q'(x)$$

for all $x \in A$, so we are done. $\square$

Problem 3.40. Let F denote the Cantor function on $[0, 1]$ (see §1.5), and set $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 1$. Let $\{[a_n, b_n]\}$ be an enumeration of the closed subintervals of $[0, 1]$ with rational endpoints, and let $F_n(x) = F((x - a_n)/(b_n - a_n))$. Then $G = \sum_1^\infty 2^{-n} F_n$ is continuous and strictly increasing on $[0, 1]$, and $G' = 0$ a.e. (Use exercise 39)

Solution. F_n are clearly increasing, continuous and nonnegative and we also see that $G = \sum F_n \leq 1 < \infty$ as $F_n \leq 1$ for all n by definition. G is also continuous by weierstrass' M-test being the limit of an uniformly converging sequence of continuous functions. Given any $0 \leq x < y \leq 1$ we can find some interval with rational endpoints $[a_n, b_n] \subset (x, y)$ and we see that $G(y) - G(x) \geq 2^{-n}(F_n(y) - F_n(x)) = 2^{-n}(1 - 0) > 0$ by definition again, making G strictly increasing. We see that F is constant on intervals outside the cantor set and thus has derivative 0 here and as the cantor set has lebesgue measure 0 we can show that $F' = 0$ a.e. by standard procedures. Now, by Problem 3.39 we see that $G'(x) = \sum 2^{-n} F'_n(x)$ a.e. and as $F'_n = 0$ a.e. too (can be easily shown), we must have $G' = 0$ a.e. $\square$

Problem 3.41. Let $A \subset [0, 1]$ be a Borel set such that $0 < m(A \cap I) < m(I)$ for every subinterval I of $[0, 1]$.

- Let $F(x) = m([0, x] \cap A)$. Then F is absolutely continuous and strictly increasing on $[0, 1]$, but $F' = 0$ on a set of positive measure.
- Let $G(x) = m([0, x] \cap A) - m([0, x] \setminus A)$. Then G is absolutely continuous on $[0, 1]$, but G is not monotone on any subinterval of $[0, 1]$.

Solution.

- Clearly F is lipschitz continuous with constant 1 and thus its absolutely continuous by Problem 3.37 and its strictly increasing as $F(y) - F(x) = m((x, y] \cap A) > 0$ for any $x < y$ by definition. As F is absolutely continuous, by theorem 3.35c we see that $F' \geq 0$ exists a.e, is integrable and $F(x) = F(x) - F(0) = \int_0^x F'(t) dm(t)$. F can be extended to $\mathbb{R}$ by setting it equal to $0 = F(0)$ for $x < 0$ and $m(A) = F(1)$ for $x > 1$ and this is NBV so we can talk about μ_F . We also see that $\mu_F((-\infty, x]) = (F' dm)((-\infty, x])$ for all x and thus, $\mu_F = F' dm$. If E is a set of positive measure and we have $\mu_F(E) = \int_E F' dm = 0$ then it must be true that $F' = 0$ a.e. in E which again furnishes a set of positive measure (same as E) on which $F' = 0$. Notice that the measure defined by $\nu(E) = m(E \cap A)$ is the same as μ_F as they agree on right closed rays and hence we just need to find some E such that $m(E) > 0$ whereas $\nu(E) = 0$ and the perfect (not a topological property!) candidate is A^c for which $m(A^c) = 1 - m(A) > 0$ whereas $\nu(A^c) = m(A^c \cap A) = 0$ so we are done.

- b. We can easily see that $m(A^c \cap I) = m(I) - m(A \cap I) \in (0, m(I))$ as well, if we let $H(x) = m([0, x] \cap A^c)$ then we see that H has the same properties as F from the previous part. Thus, G is also absolutely continuous being the linear combination of two absolutely continuous functions. Given any interval $[x, y] \subset [0, 1]$ ($x < y$) suppose for the sake of contradiction that WLOG. G is increasing in this interval, this implies that $G' \geq 0$ a.e. here but this can not be true as it implies $\mu_G = \mu_F - \mu_H$ has $[x, y]$ as a positive set. But we clearly see that for $V = A^c \cap [x, y] \subseteq [x, y]$ we have $\mu_F(V) = m(V \cap A) = 0$ whereas $\mu_H(V) = m(V \cap A^c) > 0$ and thus, $\mu_G(V) < 0$ which is a contradiction and thus we are done.

□

Problem 3.42. A function $F : (a, b) \rightarrow \mathbb{R}$ ($-\infty \leq a < b \leq \infty$) is called convex if $F(\lambda s + (1 - \lambda)t) \leq \lambda F(s) + (1 - \lambda)F(t)$ for all $s, t \in (a, b)$ and $\lambda \in (0, 1)$.

- a. F is convex iff for all $s, t, s', t' \in (a, b)$ such that $s \leq s' < t'$ and $s < t \leq t'$,

$$\frac{F(t) - F(s)}{t - s} \leq \frac{F(t') - F(s')}{t' - s'}$$

- b. F is convex iff F is absolutely continuous on every compact subinterval of (a, b) and F' is increasing (on the set where it is defined).
- c. If F is convex and $t_0 \in (a, b)$, there exists $\beta \in \mathbb{R}$ such that $F(t) - F(t_0) \geq \beta(t - t_0)$ for all $t \in (a, b)$.
- d. (Jensen's Inequality) If $(X, \mathcal{M}, \mu)$ is a measure space with $\mu(X) = 1$, $g : X \rightarrow (a, b)$ is in $L^1(\mu)$, and F is convex on (a, b) , then

$$F\left(\int g \, d\mu\right) \leq \int F \circ g \, d\mu.$$

(let $t_0 = \int g \, d\mu$ and $t = g(x)$ in (c), and integrate.)

Solution.

- a. This is a standard result from introductory real analysis, it can be proved by just manipulating the inequalities.
- b. If F is convex then we see that on any compact subinterval $[c, d]$, for $x, y \in [c, d]$ if F' exists on x, y ($x < y$) then, for $h > 0$ small enough, by part (a)

$$\frac{F(x+h) - F(x)}{(x+h) - x} \leq \frac{F(y) - F(x)}{y - x} \leq \frac{F(y+h) - F(y)}{(y+h) - y}$$

where $x < x+h < y < y+h$ and as $h \rightarrow 0$ from above, as the derivatives exist at x, y we get that $F'(x) \leq F'(y)$. It is another standard result that convex functions are continuous and we can also see that if, $c = s' < t'$ and $c < t \leq t'$ then,

$$\frac{F(t) - F(c)}{t - c} \leq \frac{F(t') - F(c)}{t' - c}$$

Thus, $G(x) := (F(x) - F(c))/(x - c)$ is increasing, similarly we can show that $H(x) = (F(d) - F(x))/(d - x)$ is decreasing. This, implies that $G(c+), H(d-)$ both

exist and we can show that these bound all slopes using an argument similar to the one we used to show that F' is increasing on the set its defined. We thus have, for all x, y such that $c \leq x < y \leq d$ we have

$$G(c+) \leq \frac{F(y) - F(x)}{y - x} \leq H(d-)$$

and thus F is lipschitz with constant atmost $\max\{|G(c+)|, |H(d-)|\}$ so it is absolutely continuous by a previous problem. We now prove the if direction. By hypothesis, for any such s, t, s', t' as in part (a), we work in the compact subinterval $[s, t]$ where the inequality in part (a) reduces to the following by theorem 3.35c,

$$\frac{1}{t - s} \int_s^t F'(t) dt \leq \frac{1}{t' - s'} \int_{s'}^{t'} F'(t) dt$$

where $s \leq s' < t'$ and $s < t \leq t'$. This is trivial if $s' > t$ so we assume otherwise that $s' \leq t$, where this inequality is true as,

$$\frac{1}{t - s} \int_s^t F' = \frac{s' - s}{t - s} \cdot \underbrace{\frac{1}{s' - s} \int_s^{s'} F'}_{\leq \frac{1}{t' - s'} \int_{s'}^{t'} F'} + \frac{t - s'}{t - s} \cdot \frac{1}{t - s'} \int_{s'}^t F' \leq \frac{1}{t - s'} \int_{s'}^t F'$$

The inequality in the underbrace is true as all the values appearing in the integralin the left are atmost those appearing in the right and we average these. It can be similarly shown that the expression on the right is atmost $\frac{1}{t' - s'} \int_{s'}^{t'} F'$ so we are done.

- c. Just like we did for c, d in part (b), it can be showed that the left and right derivatives exist at every point and we pick any $\beta \in [F'(t_0-), F'(t_0+)]$ (left and right derivatives and not the one sided limits of our derivative.) The conclusion follows just as we did in part (b), we show part of it.

$$F'(t_0+) = \lim_{h \searrow 0} \frac{F(t_0 + h) - F(t_0)}{h} \leq \frac{F(y) - F(t_0)}{y - t_0}$$

for any $y > t_0$ as $t_0 + h < y$ eventually.

- d. We see that $\int_X g d\mu \in (a, b)$ so from part (c), we can find some constant c such that,

$$F(t) - F\left(\int g d\mu\right) \geq c \cdot \left(t - \int g d\mu\right)$$

for all $t \in (a, b)$ and thus we can substitute $t = g(x)$ to get,

$$F \circ g(x) - F\left(\int g d\mu\right) \geq c \cdot \left(g(x) - \int g d\mu\right)$$

for all $x \in X$. Now, integratint both sides we see that,

$$\int F \circ g d\mu - F\left(\int g d\mu\right) \geq c \cdot \int g d\mu - \int g d\mu = 0$$

so we are done.

□